

Experimental apparatus for universal spin-motion control in multi-ion registers



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Introduction

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Should something go here?

1.1 Classical and Quantum Computing

two paragraphs

- Why we want to build a quantum computer. (Feynman, Shor, Grover)
- What constitutes a quantum computer? (Qubits, gates, circuits)

Quantum mechanics successfully describes the behaviour of sufficiently small things [**<empty citation>**]. The theory is useful in the description of sub-atomic particles, atoms, the molecule they compose, crystals, light, the vacuum, and many other parts of our reality. That is to say, given a choice, bet things obey quantum mechanics. From the 1980s? work began in earnest on marrying the fields of

quantum mechanics and information, with the intuition that if reality is described by QM, then surely QM is the efficient language in which to “compute” reality ¹. To discuss this idea, I will first devote a paragraph to the classical computers that have gone before.

We will define “compute” as modelling some physical system with another, disparate, physical system. This could be an analogue simulacra with continuous input and outputs, or it could be some digital model that mimics the system at an arbitrary precision. Apart from the engineering practicalities of realising the device, this analogue and digital distinction does not matter. The *Computer* enters when Turing proposed a class of these systems that not only mimic one specific system, but rather can model any physical system, provided enough time and memory. This remarkable result lends *computers* their other name: the Turing machines. Surprisingly few operations are required for a device to be a Turing machine (i.e. is Turing complete), and so the question of whether a Turing machine can be constructed was never up for debate (as this was likely answered at least XX years before Turing’s birth by probably some astrolabe XX). But now comes the practicality - although a physical device is a Turing machine, it may still not be feasible for an operator to “code” some desired behaviour ², or for a useful result to be returned in reasonable time ³. Regardless, highly advanced and operable computers have been constructed and have successfully modelled and transformed almost all aspects of modern life. However, there still remains vast regions of computational problems that fall into this category of unfeasible on current hardware. Worse still, the scaling of resources and time required for some real-world problems suggest that classical computing will never be able to solve them.

One category of problem that is often recognised to be unfeasible on classical computers, is that of modelling quantum systems. Fundamentally this is due to the concept of superposition (**I have seen interference being the often quoted feature, check my understanding of this**) – a quantum state is generally

¹Paraphrased, rather than the tired Feynman spiel [**<empty citation>**].

²play with esolangs for an hour to confirm this for yourself.

³factor a 256-bit number for **1e10** years to confirm this for yourself.

described by the superposition of all possible 2^n basis states, where n is the system dimension **Check this**. Each basis state will have some amplitude and phase, and with some normalising results in $2^n - 1$ complex numbers required to track and accurately model the generic state. This exponential scaling is not kind to a digital computer, whose own internal classical state (something about how they scale...). **Read deutsch a bit again to finalise the argument here**. Returning to the 80s?, it was thought that this detrimental scaling of quantum systems might offer a distinct advantage if we were to construct a Turing machine from quantum objects ⁴. This prompted the requirements given by DiVincenzo on the components to construct such a device: Some aggregation of fundamental quantum information carrying units, most commonly two-level systems called quantum-bits (or qubits), although higher dimensional systems such as qudits [**<empty citation>**], or continuous systems such as qumodes [**<empty citation>**] are areas of active research. We further require some way to prepare these quantum objects into a known initial state. Then, a set of unitary operations (quantum gates) are used to manipulate the state and perform the desired computation. Finally, some measurement of the state is required to extract information. The performance of a physical QC is then characterised by how faithfully each of these requirements is fulfilled in the error prone world. To realise a resource scaled quantum computer, we must recognise the double-edged sword that is superposition: this large state space in which to encode information, is increasingly sensitive to environmental noise [**<empty citation>**] due to this large state space. The loss of recoverable information due to noise is called decoherence, and is the major challenge to building a practical quantum computer [**<empty citation>**].

The physical origin and coupling of noise into the quantum system is highly dependent on the hardware implementation of the computer, although the effects of noise fall into a few classes, which we shall discuss in the next chapter. This thesis

⁴As an aside, the implications of quantum mechanics on information extend further than purely mimicking other dynamic systems – using and distributing quantum objects also allows for dominant strategies in cryptography [**<empty citation>**], and some competitive (although currently esoteric) games [**<empty citation>**].

is concerned practically on the trapped ion platform, and so a specific focus will be lent to it, and we shall now spend some time outlining what it consists of.

1.2 Trapped Ion Platform

two paragraphs

- Atoms are convenient qubits.
- Description and history of experimental trapped ion systems.

An effective quantum computer requires meticulous control over a collection of simple quantum systems. A good starting point is thus selecting that simple system to build our control around. Nature provides us with a bounty of choices in this regard: Atoms. By definition it is difficult to choose a simpler object, and according to our understanding of physics, atoms of the same species are identical to one another no matter where or when you are. We may use their electronic energy structure, which atomic physicists have scrutinized for over a century, to define our quantum bits, and we can use an assortment of electromagnetic frequencies to manipulate these internal states. This is the strategy of both the “neutral-atom”, and “trapped-ion” platforms, with the latter choosing to remove or add a few electrons to more easily manipulate the atoms position in space (although we note that the neutral-atom platform has made significant progress in optical traps, and I suspect this distinction could become a moot point relatively soon).

These atomic ions held in place through some combination of electric and/or magnetic fields. They are stored in ultra-high vacuum conditions, and physically distant ($>10\text{ }\mu\text{m}$) from any bulk material to limit the coupling of environmental noise to our fragile state. Lasers, radio, and microwave drives are then used to effect the electronic and motional states of the ions.

1.3 Scaling Trapped Ion Systems

- Why we need to scale up trapped ion systems (Steane) in register, and in depth.
- What are the technical challenges to scaling up trapped ion systems. Including motional control and hybrid systems.
- What are the new errors that appear when we scale up trapped ion systems. This is a main contribution of this thesis.

Useful circuits are proposed to be of order x qubits, x gates. increased complexity changes the relevant error processes

Errors increase with both space and time in QC, motivating hardware improvements, quantum error mitigation and correction techniques. Hardware challenges for scaling up trapped ion systems include number of qubits, addressing (control line problem), speed of operation, and motional control. Particularly, as the resource scales, new errors appear that are not present in small systems. These include spatially correlated noise, and crosstalk between qubits. This thesis focuses on the construction of a new experimental apparatus to enable the study of these errors, and the development of techniques to mitigate them.

1.4 Structure of this Thesis

This thesis is concerned with the construction, and characterisation of a new ion trap apparatus targetting spin and motional control of multi-ion chains; the errors that arise at this scale; and error mitigation through virtual distillation.

Chapter 2 provides motivation and theory required for the remainder of the thesis, including a background on randomised benchmarking and the relevant group theoretic results.

Then follows in chapter ?? a hardware description of apparatus constructed during my DPhil, followed by performance characterisations in chapter 4.

Chapter ?? presents a novel randomised benchmarking method for detecting and quantifying spatially correlated errors in multi-qubit devices, including experimental results on up-to an 8-ion chain.

The motional states of the ions as a computational resource are discussed in chapter ??, with relevant device characterisation including a demonstration of squeezing, numerical simulations supporting the expected leading errors for our device, and an outlook on future experiments on hybrid spin-motion computation. Chapter 7, presents an experimental implementation of virtual distillation, an error mitigation technique with practical application to computation involving both spin and motional degrees of freedom.

The thesis is concluded in chapter ?? with details on the implication on this work to the field, and an outlook of future work planned on the constructed apparatus.

2

Theory

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2.1 Trapping Ions

2.1.1 Trapping Potential

From Earnshaw's theorem, $\nabla^2 V = 0$, a stable stationary point in 3D cannot be realised using only static electric potentials, V , as if the potential is confining in two dimensions, it will be anti-confining in the third. Therefore, to achieve stable trapping, either an oscillating electric field (Paul trap [paul1990electromagnetic]), or a static magnetic field (Penning trap) is used.

Will include pseudopotential relating V_{RF} and Ω_{RF} to the radial mode frequencies ω_m (see Vera eqn 3.4),

$$\omega_r \propto \frac{V_{RF}}{M\Omega_{RF}}. \quad (2.1)$$

Further, will use James to give ion spacing and non-COM mode frequencies as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitati as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitation.

2.2 Light Matter Interactions

2.2.1 Rabi Flopping

Explain this is how we do single qubit gates.

2.2.2 Coupling to Motion

2.2.3 Geometric Phase Gates

2.3 Randomised Benchmarking

Some overview and motivation for RB. Need to cover:

- Twirling over a group

- Liouville representation of a CPTP map
- Irreps of the Clifford group
- Clifford approximates twirling over the unitary group (Haar measure)
- How Liouville representation is transformed under twirling
- The specific representation-theory tools you use later: irreps, multiplicity spaces, invariant sectors, Schur basis.
- RB protocol for single qubit Clifford group, generalisation to multi-qubit Clifford group.
- How to extract error per Clifford from decay rate

2.4 Noise channels

As will also be used in virtual distillation, may move to theory sections

A description of the noise channels we consider in this work. Explicit CPTP maps for these channels.

2.4.1 Why isn't everything white noise?

2.4.2 Unitary noise

Both continuous and probabilistic.

2.4.3 Depolarising noise

An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere. Get this from twirling over Clifford group!

2.4.4 Spatially correlated noise

Common noise An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere, but the same unitary is applied to multiple qubits. Can enforce some symmetry to this channel by twirling over the k -copy Clifford group. This is the channel we will use to model correlated noise in our experiments.

Entangling noise In SC ZZ entangling noise is common.

2.4.5 Can we make everything white noise?

Have a look at Balint paper, deep random circuits, etc.

2.5 v1

This chapter lays the theoretical groundwork for Chapter ???. It reviews standard Clifford randomised benchmarking (RB), develops the representation-theoretic machinery of group twirling in some generality before specialising it to the two-copy Clifford group, and surveys the classes of noise channel and prior characterisation strategies that motivate a symmetry-based approach to correlated-noise detection.

2.6 Standard Clifford randomised benchmarking

2.6.1 Motivation: characterising noise without full tomography

As quantum processors scale, fault tolerance and error mitigation become necessary to overcome decoherence and systematic control errors [1–3]. Errors that induce correlations across multiple qubits — variously termed spatial correlations or crosstalk errors [4] — require that mitigation and correction strategies be built on noise models that faithfully capture their correlated structure. Assuming a globally depolarising (white) noise model where correlations are in fact present can lead to systematic underestimation of logical error rates [5], or systematic bias in the outcome of an error-corrected observable [6].

Methods exist to simplify correlated noise channels into more favourable depolarising or white-noise forms, for example Pauli twirling and random circuit compilation [7–9]. However, the efficacy of such methods must still be characterised, verified, and optimised on physical devices with realistic noise, which is only possible with a scalable diagnostic for the correlated component of that noise in the first place.

Add a short paragraph situating this thesis chapter within the broader RB literature landscape (interleaved RB, cycle benchmarking, purity benchmarking, etc.), with a comparison table of scaling/expressiveness trade-offs. This context was compressed to a single sentence in the PRL and would benefit from a fuller literature review here.

2.6.2 The single-qubit Clifford RB protocol

Standard single-qubit Clifford RB proceeds by applying a sequence of $M - 1$ Cliffords, sampled uniformly at random from the single-qubit Clifford group, followed by a single inverting Clifford, C_{inv} , chosen so that the sequence implements the identity operation in the absence of errors [10, 11]. For each sequence length M , r independent random sequences are generated, and each is repeated s times to estimate the probability of the qubit returning to its target computational state. The target state is conventionally randomised between $|0\rangle$ and $|1\rangle$ from sequence to sequence, which reduces sensitivity to asymmetries in state preparation and measurement [12].

Circuit diagram: single-qubit Clifford RB sequence of length M , with per-step error box and final inverting Clifford C_{inv} . Schematic single-qubit Clifford RB circuit (reproduced/adapted from Fig. 1(a) of the manuscript). **Insert final figure once redrawn for thesis formatting; the manuscript version also shows k parallel copies, which belongs in Chapter ?? instead.**

The survival probability of a single qubit after m Cliffords follows a single exponential decay,

$$\mathcal{P}_0 = \frac{1}{2} (1 + (1 - \varepsilon_c)^m), \quad (2.2)$$

where ε_c is the “error per Clifford” (EPC), the single scalar figure of merit that standard RB is able to extract. This compression of an entire error channel into one averaged parameter is precisely what gives RB its scalability, but it is also the limitation that motivates the correlated-noise-resolving protocol developed in Chapter ??: two qubits with identical single-qubit EPC can have entirely different amounts of correlated error between them, and Eq. (2.2) cannot distinguish these cases.

2.6.3 State-preparation and measurement (SPAM) errors

RB’s scalability partly derives from its insensitivity to SPAM errors up to an overall scaling of the exponential decay. Modelling independent bit-flip type state-preparation error on a single qubit with probability $\varepsilon_{sp}^{(i)}/2$, the prepared state is

$$\rho_{sp}^{(i)} = \left(1 - \frac{\varepsilon_{sp}^{(i)}}{2}\right) |0\rangle\langle 0| + \frac{\varepsilon_{sp}^{(i)}}{2} |1\rangle\langle 1|, \quad (2.3)$$

equivalently expressed as a probabilistic X channel acting on an arbitrary input state,

$$\rho \mapsto \mathcal{E}_{sp}^{(i)}(\rho) = \left(1 - \frac{\varepsilon_{sp}^{(i)}}{2}\right) \rho + \frac{\varepsilon_{sp}^{(i)}}{2} X\rho X. \quad (2.4)$$

In the Liouville picture this simply rescales the overlap of the initial state with each sub-representation of the twirled channel by a factor $A_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)})$ for every sub-representation in which qubit i participates non-trivially. This single-qubit treatment generalises directly to the two-qubit setting used in Chapter ??: correlated SPAM errors, such as a correlated spin-flip channel, leave certain sub-representations (in particular the fully symmetric sector) invariant, and so can mimic a genuine correlated error signature if left unaccounted for in a fit.

Expand this subsection with the general n -qubit SPAM formalism, and a short discussion of readout-error mitigation strategies compared against this multiplicative-scaling picture. The manuscript SM only develops the 2-qubit special case, which is deferred to Chapter ??.

2.7 Twirling and representation theory

2.7.1 General twirling formalism

RB owes its practicality to the process of *twirling*: averaging a general noise channel over a chosen group of unitaries. Consider a general CPTP error channel Λ , expressed in Liouville (transfer-matrix) form acting on a vectorised density matrix. Twirling Λ over a discrete group $\mathcal{G}$ produces

$$\Lambda \mapsto \tilde{\Lambda} = \frac{1}{|\mathcal{G}|} \sum_{G \in \mathcal{G}} G \Lambda G^\dagger, \quad (2.5)$$

where G denotes the vectorised representation of a group element. By construction, the twirled channel commutes with every representation element,

$$G \tilde{\Lambda} G^\dagger = \tilde{\Lambda} \quad \forall G \in \mathcal{G}. \quad (2.6)$$

The Hilbert space on which $\mathcal{G}$ acts decomposes into a direct sum of irreducible representations (irreps),

$$\mathcal{H} = \bigoplus_{\lambda} (V_{\lambda} \otimes \mathbb{C}^{m_{\lambda}}), \quad (2.7)$$

where V_{λ} is the irrep space and $\mathbb{C}^{m_{\lambda}}$ is a multiplicity space of dimension m_{λ} . Schur's lemma then constrains any operator satisfying Eq. (2.6) to be block diagonal in this decomposition,

$$\tilde{\Lambda} = \bigoplus_{\lambda} (I_{\lambda} \otimes \mathcal{A}_{\lambda}), \quad (2.8)$$

where I_{λ} is the identity on V_{λ} and $\mathcal{A}_{\lambda}$ is an $m_{\lambda} \times m_{\lambda}$ matrix acting only within the multiplicity space. Physically, Eq. (2.8) is the statement that twirling collapses an otherwise arbitrarily complex error channel down to a comparatively small number of scalar decay rates (diagonal terms of $\mathcal{A}_{\lambda}$) and couplings (off-diagonal terms), one set per irrep. This is the general mechanism underlying the scalability of every RB variant, and the choice of twirling group $\mathcal{G}$ determines which physical information survives the averaging and which is discarded.

Add a short worked example here with a simple, well-known group (e.g. the single-qubit Clifford group acting on one qubit, recovering

Sub-representation	irrep	Basis operators
Identity	1	II
Left qubit	3	XI, YI, ZI
Right qubit	3	IX, IY, IZ
Trace	1	$(XX + YY + ZZ)/\sqrt{3}$
Diagonal traceless	2	$(XX + YY - 2ZZ)/\sqrt{6}, (XX - YY)/\sqrt{2}$
Symmetric off-diagonal	3'	$(XY + YX)/\sqrt{2}, (XZ + ZX)/\sqrt{2}, (YZ + ZY)/\sqrt{2}$
Antisymmetric	3	$(XY - YX)/\sqrt{2}, (XZ - ZX)/\sqrt{2}, (YZ - ZY)/\sqrt{2}$

Table 2.1: Sub-representations of the $Cl^{\otimes 2}$ group, adapted from Eq. 56 of Ref. [14]. Each row gives the irrep label, its dimension, and a spanning set of two-qubit Pauli-basis operators.

standard RB's single decay parameter) to build intuition before jumping to the two-copy case below. This would also be a natural place to discuss the trade-off between expressiveness and scalability more formally, referencing Pauli-twirling as an alternative choice of $\mathcal{G}$.

2.7.2 The two-copy Clifford group and its irreducible representations

To characterise correlated noise between a pair of qubits, we are interested in the group generated by applying the *same* randomly sampled single-qubit Clifford simultaneously to both qubits: the diagonal action of the Clifford group on two copies, denoted $Cl^{\otimes 2}$. The representation theory of this group has been analysed in detail elsewhere [13, 14]; the key results are summarised here.

The general 16-dimensional two-qubit Liouville space decomposes under $Cl^{\otimes 2}$ into four irreducible representations, of dimensions $\{1, 2, 3, 3'\}$ (the label $3'$ distinguishes the second, inequivalent, three-dimensional irrep from the first), with respective multiplicities $\{2, 1, 3, 1\}$. Table 2.1 lists the six sub-representations that follow from these irreps and multiplicities, together with a spanning set of Pauli-basis operators for each.

Insert the derivation connecting Tables 2.1 and 2.2 explicitly (i.e. show the character-table computation step by step, rather than quoting the result). The manuscript SM states this result without full derivation;

irrep	e	(12)	(12)(34)	(123)	(1234)
Identity/Trace	1	1	1	1	1
Left/Right/Antisymmetric	3	0	-1	1	1
Diagonal traceless	2	-1	2	0	0
Symmetric off-diagonal	3	0	-1	-1	-1

Table 2.2: Character table for the irreps of Table 2.1, listed against representative conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle.

Coupling	irrep	Mechanism
$m_{id \rightarrow tr}$	1	Decays (non-unital)
$m_{L \rightarrow R}, m_{R \rightarrow L}$	3	Qubit swap, correlated decays (non-unital)
$m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R}$	3	Entanglement

Table 2.3: Physical mechanisms capable of coupling sub-representation sectors of the twirled $Cl^{\otimes 2}$ error channel.

a thesis chapter is the appropriate place to walk through it, likely via Young-tableaux / permutation-representation arguments.

Given the block-diagonal structure of Eq. (2.8), and further imposing that the twirled channel be trace preserving, a general twirled two-qubit channel is described by 13 independent scalars: six diagonal decay rates, $\vec{\alpha} = \{\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as}\}$, one per sub-representation, and seven off-diagonal coupling terms, $\vec{m} = \{m_{id \rightarrow tr}, m_{L \rightarrow R}, m_{R \rightarrow L}, m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R}\}$, describing transitions between multiplicity spaces of the same irrep. Table 2.3 associates each coupling term with a physical error mechanism capable of generating it.

This decomposition is a purely group-theoretic statement: it is agnostic to any particular physical error model and simply enumerates the possible subspaces that a two-qubit error channel can populate once diagonal Clifford twirling has been applied. Section 2.8 below specialises this general structure to specific, physically motivated error channels, and Chapter ?? uses the resulting scalar decay rates to build the observable population dynamics measured experimentally.

Add a subsection discussing the multiplicity structure in more depth — i.e. what it physically means for an irrep to appear with multiplicity > 1 , and how this connects to the existence of the coupling terms $\vec{m}$. Also worth adding a small worked example reducing a simple noise channel (e.g. pure local depolarisation) through Eq. (2.8) by hand for pedagogical value.

2.8 Noise models

Having established the general block-diagonal structure available to any twirled two-qubit channel, this section reviews the specific classes of physical error mechanism relevant to the results in Chapter ??, and how each maps onto the sub-representations of Table 2.1.

2.8.1 Local depolarising noise

Independent noise acting separately on each qubit is, without loss of generality (following twirling), well modelled by a depolarising channel,

$$\rho^{(1q)} \mapsto \mathcal{E}_{dp,a}(\rho^{(1q)}) = (1 - \varepsilon_a)\rho^{(1q)} + \varepsilon_a \frac{I}{2} \text{Tr}(\rho^{(1q)}), \quad (2.9)$$

where the subscript $a \in \{L, R\}$ labels the qubit and ε_a is its depolarising error per Clifford. In Liouville form, restricted to the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, independent depolarisation on both qubits gives

$$\Lambda_{dp} = \text{diag}^{\mathcal{P}} \left(1, (1 - \varepsilon_L)I_3, (1 - \varepsilon_R)I_3, (1 - \varepsilon_L)(1 - \varepsilon_R)I_9 \right). \quad (2.10)$$

Local depolarising noise generates no couplings between sub-representations: it is diagonal in the Schur basis introduced in Section 2.7.2, and drives the two-qubit state toward the fully mixed state.

2.8.2 Correlated (“common”) single-qubit rotation noise

The central noise model of interest is correlated random-unitary noise, in which the *same* randomly sampled single-qubit rotation error is applied to both qubits, here termed *common noise*. This channel is

$$\rho \mapsto \mathcal{E}_{U \otimes U}^{\mathcal{A}}(\rho) = \sum_{a \in \mathcal{A}} p_a (U_a \otimes U_a) \rho (U_a \otimes U_a)^\dagger, \quad (2.11)$$

where $\mathcal{A}$ is an ensemble of unitaries applied with probabilities p_a ($\sum_a p_a = 1$), and each $U_a(\theta_a, \hat{n}_a) = e^{-i(\theta_a/2)\hat{n}_a \cdot \hat{\sigma}}$ is a rotation of angle θ_a about axis $\hat{n}_a$. A subset $\mathcal{A}' \subseteq \mathcal{A}$ of this class produces genuinely non-separable two-qubit channels, i.e. $\mathcal{E}_{U \otimes U}^{\mathcal{A}'} \neq \mathcal{E}_U^{(L)} \otimes \mathcal{E}_U^{(R)}$, and it is precisely this subset that constitutes spatially correlated noise.

Physically, this class captures correlated but non-entangling control and environmental fluctuations, which are particularly relevant to trapped-ion and neutral-atom qubits subject to common fluctuations of qubit drive phase, amplitude, and detuning; magnetic field noise and gradients; beam pointing and intensity drift; AC Stark-shift fluctuations; motional-state-dependent over-rotations; and gate-duration calibration errors.

Twirling Eq. (2.11) over $Cl^{\otimes 2}$ reveals that only the fully symmetric “trace” sub-representation is exactly preserved by common single-qubit rotations, $\alpha_{tr} = 1$, regardless of the error angle. To second order in a small rotation angle θ , the remaining scalar decay rates for a single common rotation are

$$\alpha_L = \alpha_R = \alpha_{as} = 1 - \theta^2/3, \quad \alpha_{tr} = 1, \quad \alpha_{dt} = \alpha_{so} = 1 - \theta^2, \quad (2.12)$$

so that, to leading order, the dynamics depend only on the magnitude of the error angle and not on its axis. Averaging over the ensemble $\mathcal{A}$ gives $\alpha_i^{(\mathcal{A})} = \sum_a p_a \alpha_a$, leading to the convenient definition of a correlated error per Clifford,

$$\varepsilon_{\text{corr}} = \frac{1}{3} \sum_{a \in \mathcal{A}} p_a \theta_a^2. \quad (2.13)$$

Add the full (non-small-angle) expressions here for completeness, i.e. $\alpha_L = \alpha_R = \alpha_{as} = (1 + 2 \cos \theta)/3$ and the corresponding α_{dt} , α_{so} including the anisotropic term n_{an} , together with a short discussion of when the small-angle approximation breaks down. Also worth adding the tensor-contraction derivation ($s_{tr}, s_{id}, s_{swap}, s_{diag}$) that produces these α_i from first principles, since it is a reusable technique beyond this specific channel.

2.8.3 Entangling coherent errors: ZZ-type noise

A distinct and experimentally important source of two-qubit correlation is coherent entangling coupling, exemplified in superconducting-qubit architectures by always-on ZZ crosstalk [fors_comprehensive_2024, alghadeer_crosstalk_2025, 15], described by the Hamiltonian

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z. \quad (2.14)$$

Unlike common single-qubit rotation noise, H_{ZZ} couples the two multiplicity copies of the 3-dimensional irrep (the “left”, “right”, and “antisymmetric” subrepresentations of Table 2.1), and so cannot in general be described by a diagonal Liouville matrix in the Schur basis. A more tractable, stochastic variant of this error,

$$\rho \mapsto \mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ} (Z \otimes Z)\rho(Z \otimes Z), \quad (2.15)$$

does remain diagonal, with

$$\alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} = 1 - 4p_{ZZ}/3, \quad \alpha_{tr} = \alpha_{dt} = 1. \quad (2.16)$$

This channel is included here for completeness and generality; it is not probed experimentally in this thesis (see Chapter ??, Outlook).

Expand this subsection substantially if ZZ-error characterisation numerics or a proposed experiment are added elsewhere in the thesis. As it stands this is a very compressed treatment inherited directly from the manuscript SM; a thesis-length treatment might derive the full coherent (non-stochastic) block-diagonal Liouville matrix, rather than only the stochastic special case.

3

Ion Trap Apparatus

“We never experiment with just one electron or atom or (small) molecule.” –
Schrödinger 1952

To keep him happy we will try to load at least two.

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After 70 years of ion confinement development (since the quadrupole mass spectrometer in the 1950s [16]), and the myriad use-cases from time standards, to

analytic chemistry, to quantum computing, there are naturally many systems and these apparatus chapters in existence today. As such, we structure this chapter in the following way: Following on from the standard requirements of an ion trap system, provided in the introduction section 2.1, we discuss the unique motivation and design specifications of our system; We then present an abstract schematic of the system in its current form, highlighting the constituent subsystems, and the optical and electrical signal paths; The remaining sections of the chapter contain the technical detail, perhaps more relevant to the practicing experimental student, and the hardware enthusiastic.

3.1 Specification

Ion-trap experiments are exceptionally versatile in their application, however in the vast search space of quantum-information applications, we want a system particularly tailored for three avenues of study:

1. Error channels on multi-qubit chains.
2. Continuous variable quantum computing and simulation utilising the spin and motional degrees of freedom of the confined ions.
3. Fast two-qubit entangling gates via phase stabilised optical standing waves.

These latter two research directions were explored within our group on a previous apparatus, leading to state-of-the-art two-qubit entanglement rates on the ion trap platform [17, 18], creation of highly non-classical motional states [19, 20], and a simulation of Z2 gauge theories encoded via the spin and motional degrees of freedom of a two-ion chain [21, 22]. However, hard limitations of both scaling these demonstrations to longer ion chains, and of interaction strength due to available optical intensity, were often restrictive. To solve both these issues, a new apparatus has been designed, constructed, and characterised, motivated by these applications. Colloquially the system is named “*FastGates*”, owing to its experimental objectives,

and we shall refer to it as such throughout this chapter.

These objectives give the following specific control requirements:

- Storage of medium length (5-10) qubit registers, section 3.2.
- Individual internal state control of each qubit, section 3.6.5.
- Stable motional modes for storing quantum information, section 3.2.1.
- Dual high numerical aperture (NA) optical access for high-intensity standing waves, section 3.4.

As the lifetime of such an apparatus extends past a single PhD generation, this thesis serves as a point-in-time description of the system, and where applicable, the planned future extensions will be highlighted as such. An abstract schematic of the current configuration is shown in figure 3.1 with blocks resembling either electronic, mechanical, or optical subsystems.

3.2 The Ion Trap

Recently, the micro-fabricated surface style linear Paul trap has gained popularity due to the maturity of chip fabrication technologies [24] and the potential route to scalability this offers. In the surface trap, the 3D radial and axial electrodes of a “macro” trap are effectively projected onto a 2D surface. The stable confining point of such a trap is typically designed to be on the order of $50 - 100 \mu\text{m}$ above the chip surface. The maturity of chip fabrication has enabled the manufacturing of complex multi-zone devices with many DC electrodes, amenable to ion shuttling [25, 26], and hence furthering progress towards Quantum CCD type architectures [27]. However, this surface style geometry typically comes with two costs: the depth of the trapping potential is often greatly reduced due to anharmonicity [28], and the close proximity of the surface to the ion can be a large contributor to motional heating rates [29–32].

Our trap, provided by the *National Physical Laboratory* in the UK, brings together the advantages of chip fabrication as well as the low heating rates and high trapping

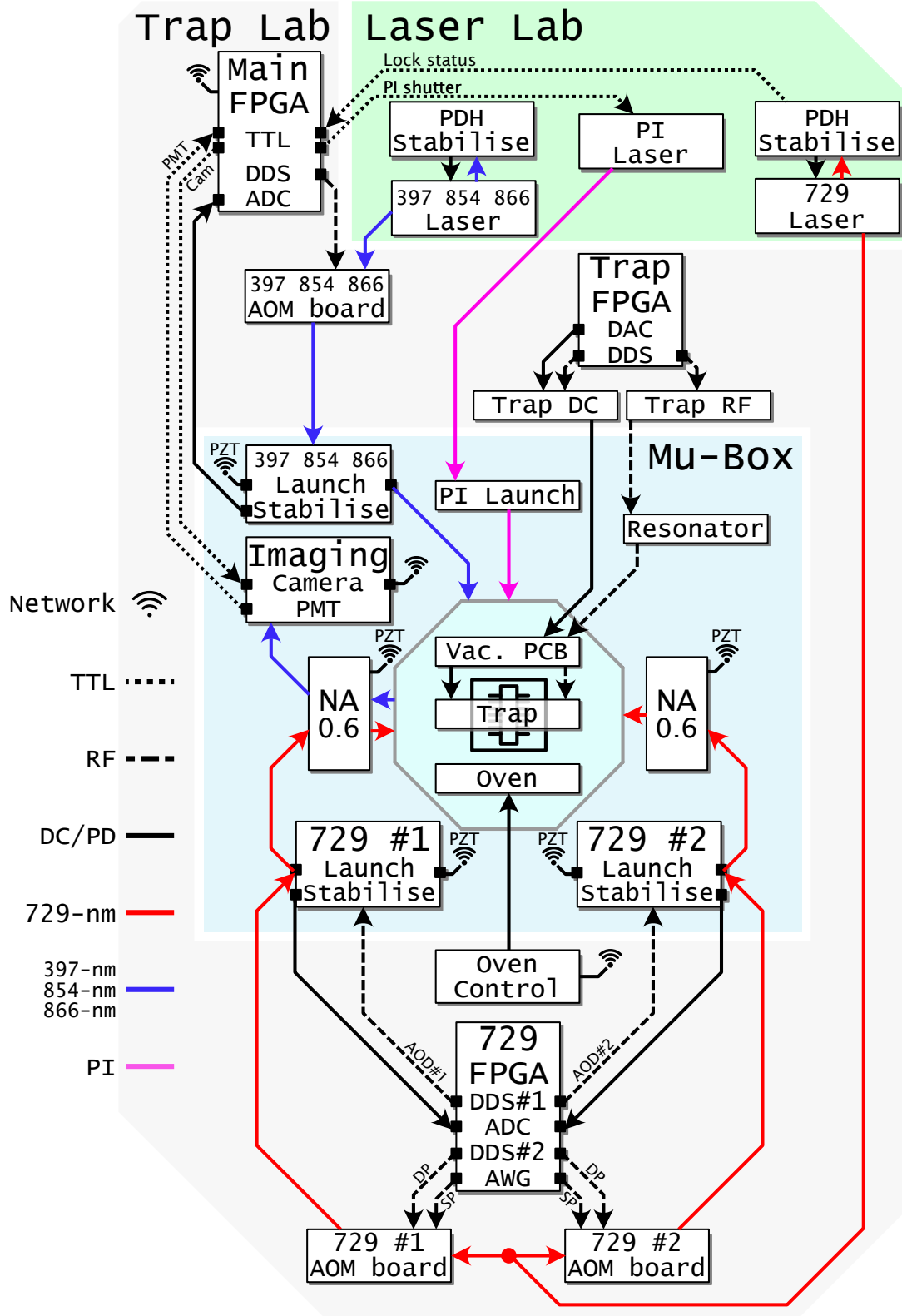


Figure 3.1: Block diagram schematic of the *FastGates* apparatus. Blocks resemble discrete subsystems, which shall be discussed within this chapter. Lines resemble relevant optical (free-space or in-fibre), RF, DC, and communication paths between subsystems. Block location signifies whether the device is in-vacuum, within the magnetic shielded (MuMetal) box, within the “trap-lab”, or within the “laser-lab”.

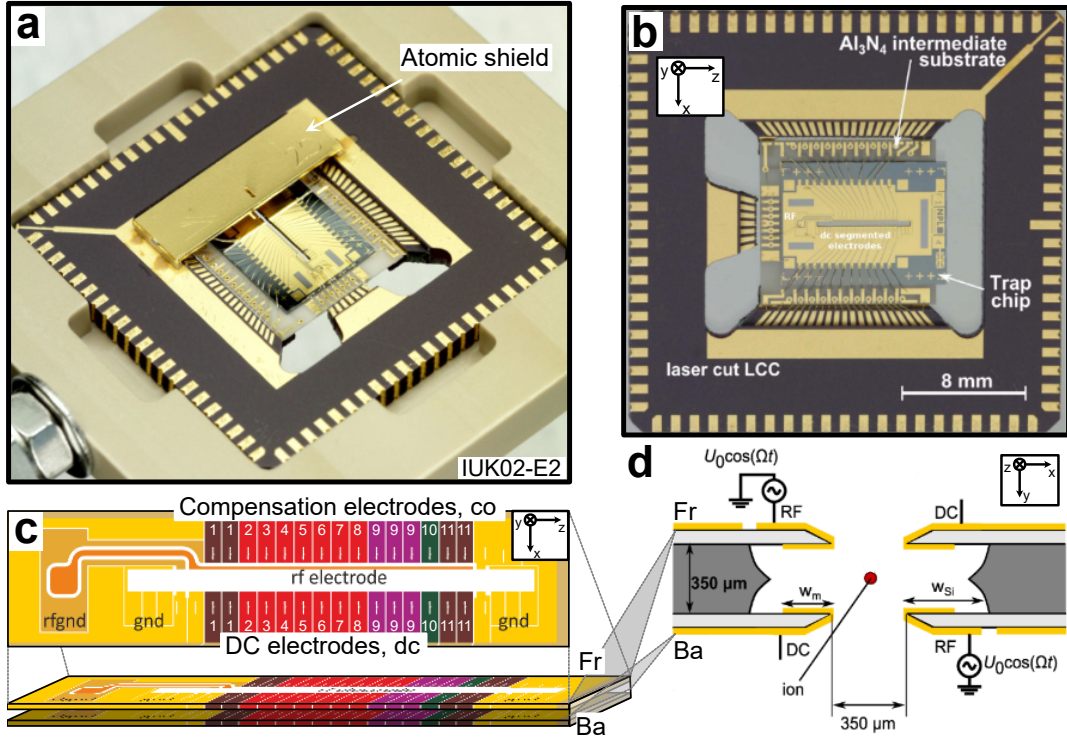


Figure 3.2: Schematics of the *NPL* trap chip used in this experiment. **a)** A photograph of the trap chip used in the *FastGates* experiment before being placed under vacuum. The top “atomic shield” can be seen here. **b)** A front view of the trap chip. Axial confinement is provided by a subset of the segmented DC electrodes. Radial confinement is provided by the RF rails. The trap chip and leadless chip carrier (LCC) are highlighted. **c)** A schematic of the electrode geometries and labelling for the front-face of the trap chip. A similar arrangement exists for the back-face however with the RF rail on the diagonally opposite electrode. **d)** A slice view of the trap. The red point here represents the ions, the gold are the RF and DC electrodes, and the grey is the silica substrate. It can be seen that the ion is $350 \times \sqrt{2}/2 \simeq 250 \mu\text{m}$ away from the nearest electrode. The numerical aperture of the ion plane due to the electrodes is ~ 0.71 . Figures b) and d) adapted from Ref. [23], other panels adapted from private communication with the *NPL* team.

depths of a macro 3D style trap through utilising a micro-fabricated 3D trap geometry with greater ion-surface distances than typical planar traps. Details on its design and characterisations can be found in Ref. [23, 33, 34]. Figure 3.2 shows the electrode geometry of the trap and the relevant length scales. The trap package consists of an outer leadless chip carrier (LCC) with outer pads routing DC, RF, and ground connections to an intermediate Al₃N₄ substrate. The connection to the front and back surfaces of the trap chip is made through vias and wire bonds to the intermediate substrate [35]. The trap chip is a monolithic double-sided

silicon device with gold plated electrodes on the front (Fr) and back (Ba) surfaces. These two layers are shown schematically in the bottom panel of (c). The electrode schematic for the front surface is shown, and the electrode indexing convention is highlighted. Electrodes recessed from the ion due to the RF rail are named “compensation”-electrodes (co), while electrodes on the opposite side provide the strong DC potential required for axial confinement¹. We typically trap ions centered on electrode dc5. Panel (d) shows a slice view of the trapping region, highlighting the ion position between the front and back surface electrodes. The dual RF rails and DC electrodes are shown in gold, and the undercut silicon bulk is highlighted in dark grey.

The next two subsections describe the source of RF and DC voltages to the trap electrodes. The physical mounting of the trap in the vacuum system is discussed in section 3.4, and the optical access to the trap is discussed in section 3.4.

3.2.1 Trap RF Chain

To utilise radial motional modes for low error quantum gates, we require the radio frequency (RF) field that produces the desired trapping pseudopotential to be both frequency and amplitude stable. Furthermore, as the natural time scale for two-qubit gate durations is $\tau = 2\pi/\omega_m$ (see section 2.2.3), where ω_m is the motional mode angular frequency, then greater motional mode frequency is desired for fast entanglement. From equation 2.1, it can be seen that radial motional mode frequency is proportional to RF voltage, and inversely proportional to the frequency. The *NPL* trap used in the *FastGates* apparatus is rated for a max peak-to-peak voltage of 400 Vpp on the RF electrodes. To target high motional mode frequencies, saturating this maximum voltage is desirable, however, for safe operation, typically the trap is provided a maximum RF voltage $V_{RF} = 360$ Vpp. The RF chain which provides this signal is schematically shown as:

¹Note: Compensation electrodes co1 and co11 on both front and back surfaces are grounded on the trap package, and so are not available for use. There is a continuity fault on front dc11 and so this electrode is also not available for use.

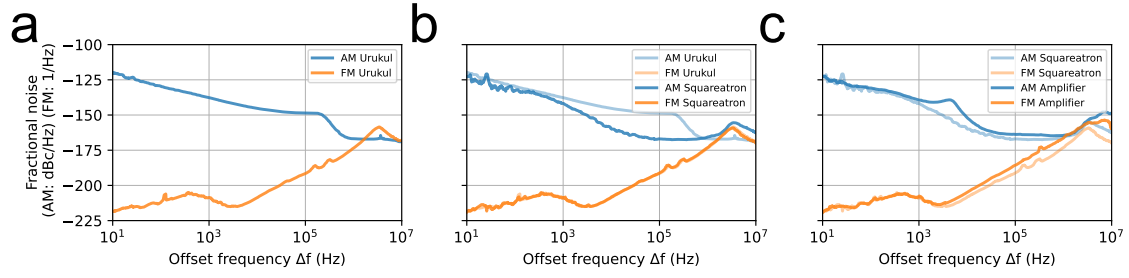


Figure 3.3: Characterisation of the RF chain. **a)** SSB spectra for AM and PM, given as fractional noise, measured after the *Urukul*. **b)** Measured after the *Squareatron*. **c)** Measured after the amplifier.

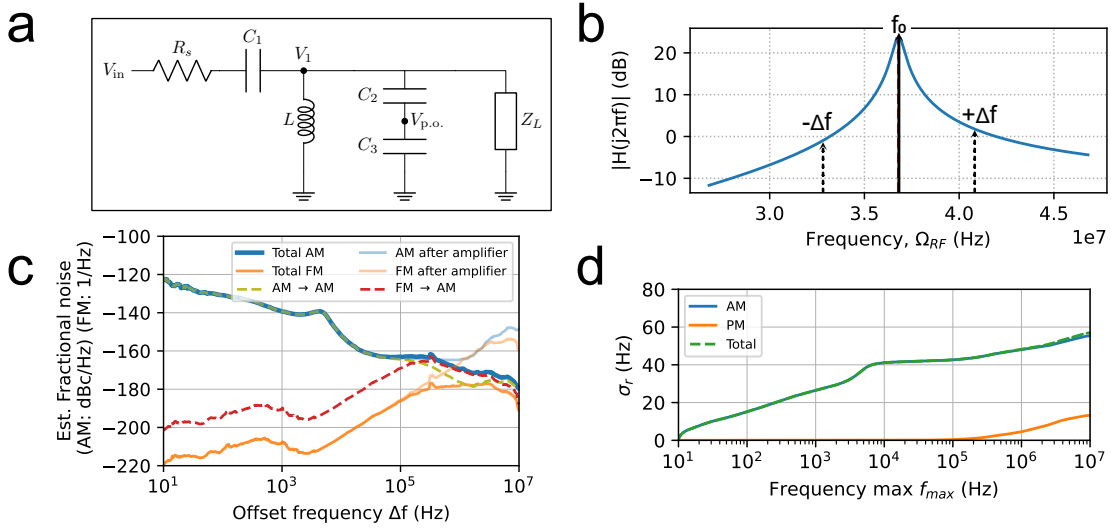
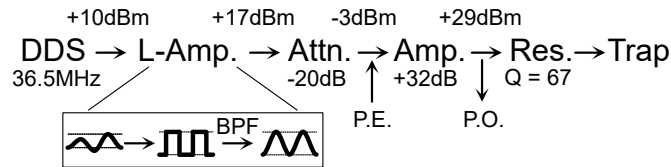


Figure 3.4: Expected SSB after the resonator, and estimated mode frequency noise. **a)** Resonator circuit diagram. **b)** Resonator lineshape, given by magnitude of the transfer function $|H(f)|$. **c)** Expected AM and FM fractional noise spectra after the resonator. **d)** Integrated RMS noise of radial motional mode frequency due to the RF chain.



and embedded within the schematic 3.1, beginning with the *Trap FPGA*. Each element of the chain is designed to provide a low-noise, high-voltage RF signal to the trap electrodes, and is discussed in turn below.

The frequency source is a direct-digital-synthesiser (DDS), named *Urukul* [36],

part of the *Artiq Sinara* [37] hardware ecosystem. *Urukul* is used extensively in our lab for driving acousto-optic modulators (AOMs) and here used as the source drive of our trap RF. The *Urukul* is operated at maximum output power, +10 dBm, and with a frequency ~ 35 MHz (set by resonator, discussed below). As can be seen in equation 2.1, amplitude and frequency instability will directly lead to radial mode instability. Expanding for small amplitude, dV_{RF} , and drive frequency, $d\Omega_{RF}$, errors we find the motional mode frequency, ω'_r , is related to the ideal frequency, ω_r , by

$$\omega'_r = \omega_r(1 + dV_{RF}/V_{RF} - d\Omega_{RF}/\Omega_{RF}), \quad (3.1)$$

where we can see both errors affect the motional mode frequency linearly. The instability of the oscillator can limit motional mode coherence time (see section 4.3.4 for characterisation with the ion). We measure noise originating from both amplitude modulation (AM) and frequency modulation (FM) across the RF chain using a phase noise analyser². This was done to verify low-noise operation of the individual devices, and to estimate expected motional mode errors due to the inherent noise on the RF chain.

We quote FM as the fractional frequency stability power spectral density (PSD),

$$S_y(f) = \frac{f^2}{f_0^2} S_\phi(f), \quad (3.2)$$

where $S_\phi(f)$ is the phase noise PSD (which is the Fourier transform of the Autocorrelation function) f is the offset frequency, and f_0 is the carrier frequency. $S_y(f)$ has units of $\left[\frac{1}{\text{Hz}}\right]$. AM noise is quoted as the single sideband (SSB) power, S_α , in a 1 Hz bandwidth at an offset frequency, f , from the carrier, normalised to the total power of the carrier, and is here given in units of $\left[\frac{\text{dBc}}{\text{Hz}}\right]$.

Figure 3.3, panel (a) displays the *Urukul* AM and FM spectra. The *Urukul* is phase referenced to the master FPGA board (*Kasli*), which is in turn phase locked to an external 10 MHz atomic clock³. It can be seen that AM noise dominates up ~ 1 MHz, suggesting that this will be the limiting error source for motional mode stability. The AM noise may be reduced by using a non-linear voltage clamp. We

²Rhode and Schwartz FSWP PNA

³**ToDo: Abaqus atomic clock PN**

use an “ultra-low noise limiting amplifier” named *Squareatron* developed previously in the group, and further refined by *Oxford Ionics*. *Squareatron* takes an input RF signal and converts to a stable amplitude square-wave output which is subsequently bandpassed around the desired RF frequency to recover the sinusoidal signal, but with significantly reduced AM noise. The AM and FM after the *Squareatron* is shown in panel (b). It can be seen that between 1 kHz and 1 MHz, AM is considerably suppressed. There appears to be an increase in AM at higher frequencies **ToDo: explore this further**. The *Squareatron* has a 17 dBm output power, which is attenuated by -20 dB before an amplification stage to reach the desired power for the trap. Highlighted in the above chain schematic is the option to input a parametric excitation signal (P.E.), which we shall discuss further in section 4.2.3. For the following measurements, the P.E. branch was removed. AM and FM introduced by the low-noise amplifier⁴, can be seen in panel (c). Most notably an increase in AM is seen between 1 kHz and 10 kHz.

To impedance match the $50\ \Omega$ line from the amplifier to the small capacitance of the trap electrodes and vacuum feedthrough, a resonant matching circuit is used. This “resonator” has three main effects on our system:

1. The source $50\ \Omega$ is matched to the (almost) purely capacitive load.
2. The voltage step up provides the necessary amplitude for high motional mode frequencies.
3. The input RF is bandpassed around the desired trapping frequency.

The latter two points are practically useful, especially the bandpass effect of the resonator attenuating noise on the RF line, however, changes to the resonators Q factor and central frequency due to thermal and mechanical noise can be a significant source of low frequency motional mode instability. Opting for good passive stability, a PCB based design⁵ with good thermal management was seen as

⁴Mini-Circuits ZHL-1-2W-S+

⁵This style PCB resonator has been used around the Oxford Ion Trapping Group for some time, and is based originally on a design by Tom Harty

preferable to large helical resonators, which are often used in ion trap experiments for their high Q factors. Aiming for a moderate Q factor of ~ 70 allows for a good voltage step up, and some bandpass filtering, without being overly sensitive to thermal and mechanical noise. The circuit diagram of the constructed resonator⁶ is shown in figure 3.4 (a). The inductor, $L \sim 0.8 \mu\text{H}$, is hand-wound copper wire with a toroidal iron powder core⁷, and the capacitors are low tempco C0G rated ceramic, $C = 5.4 \text{ pF}$. The vacuum can feedthrough and trap were measured to present an $\sim 18 \text{ pF}$ load. The trap itself presents a $C_T = 4 \text{ pF}$ [35], the cable between the resonator and the vacuum feedthrough contributes $\sim 4 \text{ pF}$, and the remaining presented capacitance is expected to originate from the feedthrough and in-vacuum RF line. In vacuum, the RF line is a *Kapton* coated 0.4 mm diameter single strand copper wire, which connects to an in-vacuum distribution PCB. The RF connection to the trap LCC is made by *fuzz button* vias.

Measuring the motional mode frequency of a trapped ion while varying the trap RF frequency (see section 4.3.3), yields a resonator quality factor $Q = 57.88(6)$, and central frequency $34.659(1) \text{ MHz}$. The effect of the resonator on the AM and FM noise can be calculated through the transfer function, $H(f)$, of the resonator circuit. Capacitors ($C2$ and $C3$) are used to provide a pick-off (p.o.) voltage which can be monitored, and if required, used in a closed feedback loop. We however have not attempted this as of yet, and shall ignore the effect of the pick-off capacitors for this analysis. The transfer function of the simplified circuit is then given by,

$$H(f) = \frac{V_{out}}{V_{in}} = \frac{Z_p}{Z_s + Z_p}, \quad (3.3)$$

where

$$\begin{aligned} Z_p &= \left(\frac{1}{j2\pi f L} + \frac{1}{Z_L} \right)^{-1}, \\ Z_s &= R_s + \frac{1}{j2\pi f C_1}. \end{aligned} \quad (3.4)$$

Given the approximate capacitance and inductance values of the resonator and trap, the analytic $|H(f)|$ is plotted in figure 3.4 panel (b), and shows acceptable

⁶Credit to Emily Hirsch for construction of the recent resonator.

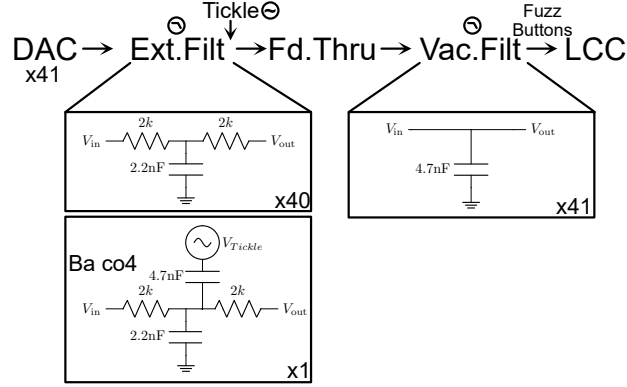
⁷Micrometals T80-6

agreement with the measured quality factor and central frequency (here $Q = 69$ and $f_0 = 36.8\text{MHz}$) with the discrepancy of quality factor expected to originate from excess resistance in the inductor not accounted for in the model. Final AM and FM spectra are calculated through considering both upper ($+\Delta f$) and lower sidebands ($-\Delta f$), as shown in panel (b), following reference [38]. We find that for both AM $\rightarrow$ AM and FM $\rightarrow$ FM (where " $\rightarrow$ " represents the transform from input noise type to output noise type) the input PSD are low pass filtered with a cut-off frequency of ~ 0.25 MHz. We also consider the cross term, FM $\rightarrow$ AM, which here we expect to be non-negligible at higher offset frequencies due to the assymetry of the resonator lineshape. Panel (c) shows the expected fractional noise after the resonator, with total FM noise (orange line), total AM noise (blue line), and contributions from AM $\rightarrow$ AM (dashed green), FM $\rightarrow$ AM (dashed red). Integrating the AM and FM fractional noise spectra between some chosen range of offset frequencies, f_{min} and f_{max} , gives the expected fractional RMS noise on the radial mode frequency of the ions. Assuming a mode frequency of $w_r = 3$ MHz the RMS, σ_r , is shown in figure 3.4 panel (d) as a function of f_{max} , with $f_{min} = 10$ Hz. It can be seen that the expected noise is dominated by the AM contribution, and in the frequency range considered, corresponds to a mode frequency instability of $\sigma_r = 60$ Hz. This analysis excludes any noise less than 10 Hz, and environmental noise coupled in by the resonator, which may be a significant contribution at low frequencies due to thermal and mechanical noise. The motional modes will also be subject to noise from other coupling mechanisms, such as voltage noise on the DC electrodes, or fluctuating stray electric fields, which we do not consider here.

Here we have characterised both AM and FM noise on the RF chain and estimated the expected contribution to motional mode instability. Further characterisations of the motional mode stability using the ion as a sensor is discussed in section 4.3.3.

3.2.2 Trap DC Voltages

Static fields are required for reshaping the trapping psuedo-potential. This is useful for setting desired mode frequency splittings, for shuttling ions, and for compensating stray electric fields. As shown in figure 3.2, the *NPL* trap has 40 independently controllable DC electrodes. A further DC bias voltage can be applied directly to the silicon bulk. DC Voltage is supplied to the LCC through the following chain:



and embedded within the schematic 3.1. Two multi-channel DACs, known as *Fastino*, part of the *Artiq Sinara* hardware ecosystem, provide the initial 41 DC channels. The Fastino outputs are filtered by a passive cascaded RC low pass filters with resistances, $R = 2k$, and capacitances $C_1 = 2.2 \text{ nF}$ and $C_2 = 4.7 \text{ nF}$ ⁸ yielding a -3 dB “cut-off” frequency of 7.4 kHz . The low pass filter circuit is shown in the above schematic, with the first stage on an external to vacuum PCB (Ext. Filt), and the final shunt capacitor on an in-vacuum PCB (Vac. Filt)⁹ within close proximity of the trap chip to mitigate RF pickup on the DC lines. The feedthrough to the vacuum system has 51 pins¹⁰ with non-DC pins used to provide ground connections. On the external filter board, one channel, corresponding to electrode *Ba co4*, has a capacitively coupled in AC signal. This originates from a DDS (*Urukul*) and is used to resonantly excite the motional modes of the ion chain for characterisation and calibration purposes (see section 4.3.3). The in-vacuum PCB is connected to the trap chip via a custom PEEK interposer with embedded *Fuzz buttons* vias¹¹.

⁸For in-vacuum use. 1206 NPO 4.7nF 100V, SRT 1206A472JBCB

⁹Vacuum compatible PCB using Rogers RT6002. Filter boards were designed by Mariella Minder.

¹⁰*Allectra* 51-way Micro-D

¹¹*Custom Interconnects* Au/BeCu 0.025" diameter x 0.104" length Fuzz Button

The ion is confined in the “operation” zone, seen in figure 3.2 (c). The axial trapping is provided by electrodes *Ba dc5* and *Fr dc5*, where *Ba* and *Fr* correspond to the back and front plane of the trap respectively. The DAC can provide up to 10 V to these electrodes to produce an axial frequency of 1.6 MHz.

3.2.3 System Grounding

Thinking of removing this section. Poor ground design (or no design at all). RF signal and return in vacuum may be important.

3.3 Magnetic Field

Due to the fragility of the internal states of the ion (these are *literally* state-of-the-art magnetic field sensors), we must take great care in isolating the ion from a noisy environment. We do not have access to magnetic field insensitive transitions in $^{40}\text{Ca}^+$, therefore uncontrolled Zeeman shifts of the internal atomic energy levels must be suppressed through enforcing a stable external magnetic field. This magnetic field stability is required for long spin coherence times (see section 4.2.5), and for low error single- and two-qubit gates. A permanent magnet array of Samarium Cobalt¹², $\text{Sm}_2\text{Co}_{17}$, is used in a Helmholtz configuration to create a stable magnetic field of $\simeq 5 \text{ G}$ ¹³. Samarium Cobalt was chosen for its low temperature coefficient of remenance, $-0.03\%/K$.

The ion is shielded from unwanted external magnetic fields by a custom designed MuMetal enclosure¹⁴, with an internal dimension of $650 \times 650 \times 650 \text{ mm}$. The shielding consists of two layers of MuMetal shielding with internal shield thickness of 1.5mm and external shield thickness of 2.0mm. The factory quoted shielding factor at the center of the box is $\times 546$ for DC fields. Spin coherence time comparisons with and without magnetic shielding are shown in section 4.2.5.

¹²*Bunting Magnets* E643SM and E644SM.

¹³Designed and constructed by Sophie Howarth.

¹⁴*Magnetic Shields* custom design.

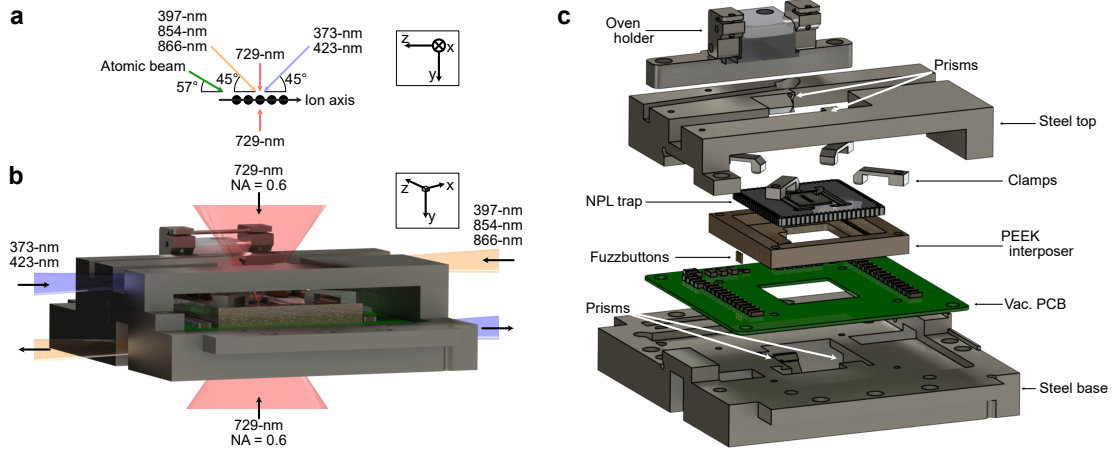


Figure 3.5: a) Laser and atomic beam geometries with respect to the ion chain. b) Compact trap housing with dual NA=0.6 access. The size of the high NA objectives restricts access for other beams, and so in-vacuum mirrors are used to redirect low-NA beams (373, 397, 423, 854, 866-nm) to the trap center. c) Exploded view of the trap housing with components labelled.

3.4 Vacuum System and Beam Geometry

Ultra High Vacuum (UHV) is required to prevent ion loss. However, UHV equipment is often bulky, and puts constraints on the access and visibility of the ion chain. Here, we describe the custom vacuum system, allowing for multiple beam access geometries including dual high numerical aperture (NA) objectives required for single addressed optical standing waves. The vacuum system was designed by Sebastian Saner, Chris Ballance, and Mariella Minder, and constructed by Sebastian Saner, Fabian Pokorny, and myself.

A residual pressure of $< 10^{-11}$ mbar is desired. For this strict requirement, care must be taken in selecting in-vacuum materials, and thorough cleaning and baking procedures must be followed. A summary of tactics that were useful in the construction of the vacuum system can be found in [39, 40].

The system consists of a 6" spherical-octagonal experimental chamber¹⁵ connected to a spherical chamber with ion pump¹⁶, ion gauge¹⁷ and Titanium sublimator

¹⁵Kimball MCF600-SphOct-F2C8

¹⁶Agilent VacIon Plus 20 Pump

¹⁷Agilent UHV-24P Ion gauge

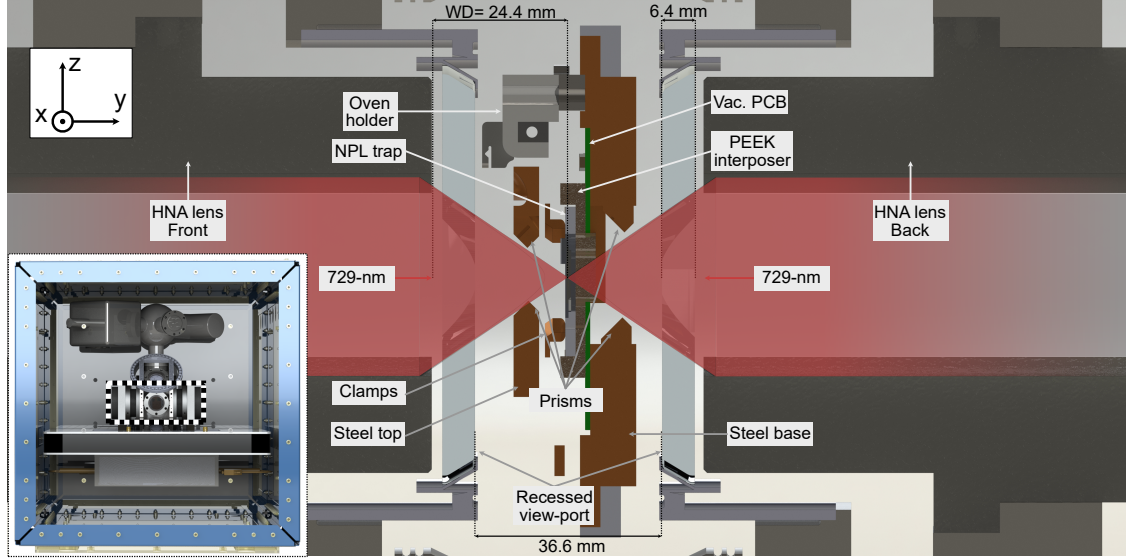


Figure 3.6: Cross section of the vacuum chamber and dual high NA objective sandwich. Lower right inset shows the vacuum chamber housed in the MuMetal enclosure, with dashed lines indicating the cross section region. Custom recessed CF-100 viewports are used to provide optical access for the high NA objectives (working distance = 24.4 mm).

pump (TSP)¹⁸ attached. After the final system bake, a He leak test was performed and no external leaks were found. We do observe a gradual pressure increase that we suspect is either due to the in-vacuum optics, the optics epoxy adhesive¹⁹, or the PCB solder. Regardless, the ion pump and TSP maintain UHV at the desired $< 10^{-11}$ mbar (the TSP is fired every 4 weeks at 41 A for 60 seconds to provide a fresh Ti layer).

Figure 3.5 (a) shows the beam geometries with respect to the ion chain axis. The quadrupole 729-nm beam has high NA ($NA = 0.6$) access perpendicular to the ion chain axis, while 373-nm, 397-nm, 423-nm, 854-nm and 866-nm beams have low NA ($NA < 0.05$) access at an angle of 45° to the ion chain axis. The atomic beam from the oven is at an angle of $\sim 57^\circ$ to the ion chain axis. As such, there is a doppler shift of the atomic beam with respect to the 423-nm photoionisation laser, which is accounted for in the set laser frequency (given 12° angle between atomic and 423-nm beam, and 500 K mean temperature, expect +126 MHz frequency shift). Figure 3.5 (b) shows the custom trap housing including $NA=0.6$ access for the

¹⁸Scanwel custom housing

¹⁹EPO-TEK 353ND

729-nm beams, and in-vacuum mirrors (prisms) to redirect the low NA beams to the trap center. The low NA beams enter the housing via one of the four beam access channels parallel to the trap chip plane, and are reflected through the trap slit. The exiting beam is reflected by a second mirror out of the housing to prevent scattered light incident on the ion chain. Figure 3.5 (c) shows an exploded view of the trap housing with components labelled. The trap chip is interfaced to an in-vacuum PCB (described in section 3.2.2) via a custom PEEK interposer with embedded *Fuzzbutton* vias (one for each DC and RF channel). The trap chip is held in place by 4 steel clamp arms, which attach to a steel base plate which provides mechanical stability and houses the lower in-vacuum prisms. A top steel plate houses the upper prisms. The tube oven is mounted on two steel clamps which are electrically isolated from the base plate by a PEEK spacer. The trap housing is fastened to the vacuum can by 4 steel arms attached to alternating side ports of the octagonal chamber²⁰. Figure 3.6 shows a cross section of the vacuum chamber and recessed dual high NA objectives. Two recessed CF100 viewports²¹ are required to bring the custom objectives²² to the designed working distance of 24.4 mm from the ion chain.

3.5 Imaging System

To image the ion chain, and readout the qubit states, fluorescent scatter of incident 397-nm light is collected. Here we briefly describe the imaging system, while section 4.2.2 describes the qubit readout procedure and fidelity. The imaging system is designed to provide high spatial resolution of the ion chain, while also providing a large field of view to accommodate long ion chains. The imaging system must also be compatible with the high NA objectives used for single addressed optical standing waves. Figure 3.7 shows the optical path schematic of the imaging system, with both a high quantum-efficiency ($\simeq 70\%$ at 397 nm) CMOS camera²³ for spatially selective qubit readout on a multi-ion chain, and a photon multiplier tube

²⁰Using *Kimball Groove grabbers* P/N 53-482610

²¹*UK Atomic Energy Authority* P/N VPR100015

²²*Photon Gear, inc.* Custom 18WD Atom Imager

²³*Hamamatsu Orca-Fusion BT* C15440-20UP

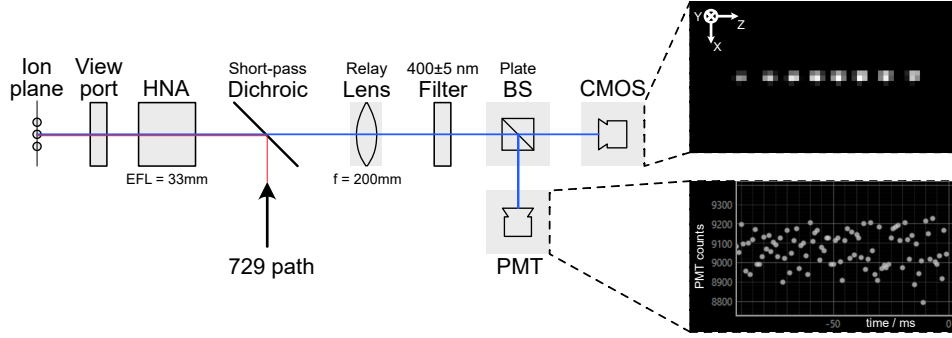


Figure 3.7: Optical path for fluorescent imaging of the ions using both a CMOS camera and a Photon-multiplier tube. Insets on right hand side show typical signal for: spatial resolution of eight ions in a linear chain (top), time resolution of detected photons from the ion chain (bottom).

(PMT)²⁴, for experiments requiring time resolved photon counting (e.g. micromotion compensation, section 4.2.3). Typical signals of both devices can be seen in right panels of figure 3.7. The optical path consists of a high numerical aperture objective²⁵ (effective focal length, $f = 33$ mm), $NA = 0.6$, a dichroic mirror²⁶ (transmission: 397 nm; reflection: 729 nm), a relay lens²⁷ ($f = 200$ mm), a 90 : 10 plate beamsplitter²⁸, and the camera and PMT.

The objective lens is custom designed to compensate for aberrations due to the recessed vacuum viewport, however, due to the large NA, accurate alignment of all degrees of freedom of the objective is critical. As such, the NA is attached to a 5-axis alignment stage²⁹, with z-axis focus controlled by a piezo micrometer³⁰ for fine adjustment of the ion focus with the MuMetal box closed (requires $< \mu\text{m}$ precision).

Given the chosen lens focal lengths, we expect a magnification of $\times 6$ from the ion plane to the imaging plane. With an expected ion separation of $5 \mu\text{m}$, and the camera pixel pitch of $6.5 \mu\text{m}$, we expect around 4.5 pixels between ions, which, if the system is well aligned, is more than sufficient for low cross-talk spatial readout of qubits.

²⁴Hamamatsu Photon counting head H10682-210

²⁵Photon Gear, inc. Custom 18WD Atom Imager

²⁶**Thorlabs TODO**

²⁷Thorlabs Achromat Doublet, ACT508-200-A-ML

²⁸**Thorlabs TODO**

²⁹Newport ULTRAlign M-562-XYZ, M-36, SM-13

³⁰Newport CONEX-TRA12CC

3.6 Ca^+ Laser Systems

Singly-charged group 2 elements are popular in ion trap experiments due to their single outer electron resulting in Hydrogen-like energy levels. In this thesis we use $^{40}\text{Ca}^+$.

$^{40}\text{Ca}^+$ has no nuclear spin giving the (relatively) simple level structure shown in figure 4.1 with the relevant laser transitions used in our apparatus are indicated. Details on the ion choice, and utility of the lasers is given in section 4.1, here we only describe the hardware and feedback control to provide this stabilised light.

The relevant wavelengths are the following: The qubit is defined by the quadrupole transition at 729-nm. Access to other excited levels, outside of our defined two level system, are crucial for ion trap quantum logic to enable qubit readout via state selective fluorescence (397-nm, and 866-nm transitions), and state preparation via optical pumping (866-nm, and 854-nm transitions). Access to two additional transitions in neutral calcium for isotope selective ionisation, 423-nm and 373-nm, are also required [41].

This section begins with a description of laser frequency and power stabilisation techniques common to multiple laser systems used in this thesis. Following this, the specific systems are described, starting with the diode lasers, which drive all but the 729-nm transition. Subsequently, the Ti:Sapph 729-nm source is discussed. Finally, the global and single addressed beam arrangements for the 729-nm system are described.

3.6.1 Frequency Stabilisation

All lasers exhibit frequency noise due to unavoidable environmental coupling (e.g. temperature fluctuations, and acoustics), and to instabilities in control signals (e.g. voltage and current controls). For the lasers corresponding to dipole transitions used in this thesis, we are relatively lenient and require slow frequency excursions of < 1 MHz with respect to the qubit transition, and laser linewidths less than the natural linewidth of the dipole transitions, which are often multiple MHz.

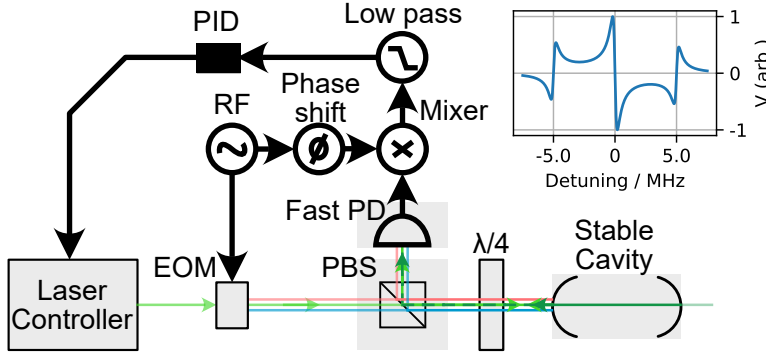


Figure 3.8: PDH optical and control schematic. Green optical path represents the carrier frequency, blue and orange lines represent the upper and lower sidebands respectively, generated by an EOM. The reflected light is directed onto a fast photodiode, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The signal is finally low pass filtered to recover the error signal, an example of which is seen in the upper right panel for a finesse $F \sim 4000$, a free spectral range of $FSR = 1.5$ GHz, and a modulation frequency of $f_{mod} = 5$ MHz.

However, for the quadrupole transition at 729-nm, we require significantly more stringent frequency stability, with laser linewidths of < 100 Hz, and negligible phase modulation of the laser light over a few MHz of the qubit transition to ensure spectator transitions are not excited during qubit operations. Although the order of magnitude of the required frequency stability is different for the dipole and quadrupole transitions, the concepts and techniques used to achieve this stability are similar. Generally, a closed loop consisting of measurement of the optical frequency of the laser and feedback to the laser control is required. In this work, stable external optical cavities are used for referencing the laser frequency. Specifically, we use the Pound-Drever-Hall (PDH) techniques to generate the error signal for the feedback loop. The PDH technique is well described in the literature [42–44], and so here we only give a brief overview of the technique to contextualise the required hardware introduced in the following sections.

The conceptual goal of the PDH technique is to measure the frequency difference between the laser and a resonant frequency of a stable optical cavity to generate an error signal, preferably linear over typical frequency excursions (while *locked*). This error signal can then be modified by some controller (typically PID), to directly infer

how to tune the laser controls to minimise this error signal, often termed *locking* the laser to the cavity. Further, the error signal should be insensitive to amplitude noise on the laser light, to avoid amplitude to frequency noise conversion due to the feedback loop (in contrast, side-of-fringe stabilisation is sensitive to amplitude drifts).

Practically this scheme consists of phase modulation of the incident laser light, here by electro-optical modulators (EOMs), to generate sidebands, which are used as phase references to compare the phase of the incoming laser field to that of the phase of the field inside the cavity. Figure 3.8 shows a schematic of the PDH scheme, highlighting the beam paths and closed loop. The phase modulated light is directed onto the cavity, with the total return signal being some combination of the reflected sidebands, carrier, and the cavity leakage field, with the latter providing the phase of the electric field inside the cavity. The reflected light is directed onto a fast photodetector, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The resulting signal is low pass filtered to remove high frequency components, yielding the final error signal, an example of which is seen in the upper right panel of Figure 3.8. The central region is anti-symmetric and linear, centered on the desired lock point. Using a tuned controller for feedback to the laser controller, this provides a stable lock of the laser to the cavity.

As the cavity is used as a frequency reference, both a high finesse and stable optical lengths are desirable. The gradient of the PDH error signal is proportional to finesse, leading to increased signal to noise with higher finesse (see Ref. [43] for more details). Optical length deviations are minimised through mounting the cavity in a vacuum chamber, and taking care in isolating it from temperature fluctuations and vibrations.

Particular details of the PDH hardware and control loops are given in the following sections for each laser system.

3.6.2 Intensity Stabilisation

The light-matter interactions considered in this thesis are all proportional either to intensity or amplitude of the optical electric field at the ion's position. Intensity fluctuations over time therefore lead to fluctuations in the interaction strength, which are detrimental to the fidelity of quantum operations. Intensity noise originates either from total beam power fluctuations, or due to beam pointing drift with respect to the ion position. The former is mitigated through a closed feedback loop between a pick-off photodiode signal before the ion monitoring the optical power, and the drive RF amplitude of an upstream AOM. For all beams, this optical pick off is one of the last elements before the beam enters the vacuum system to reduce the impact of lossy elements between the stabilise point and the ion position. The low-noise photodiodes used were designed by David Nadlinger (*QLNPD*), and the feedback system controller, known as *SUServo*, was written by Peter Drmota. Beam pointing drifts are mitigated through careful mechanical design of the optical paths for passive stability, and through automated intermittent beam pointing calibrations using the ion as a sensor (see section 4.2.3).

3.6.3 Diode Lasers

For transitions 397-nm, 854-nm, and 866-nm, we require light that is frequency stable to < 1 MHz level, well below the natural line widths of the dipole transitions, and power stable to < 1 μ W. All beam paths and relevant control loops for the diode laser systems are shown in figure 3.9. The laser diode heads³¹, controllers³², and splitterboards³³ (fig 3.9 (a)) are housed in a single 19" rack. The splitterboard distributes the light between two ion-trap experiments, the PDH lock cavity (fig 3.9 (c)), and a diagnostic wavemeter for checking wavelength and lock status³⁴. The splitterboard also houses an Electro-optic modulator (EOM) module for applying

³¹*Topica* diodes. 854/866: MDL DL pro; 397: MDL DL pro HP; 423: DL pro

³²*Topica* DLC Pro with PDH boards

³³Designed by *OITG Cavities* group

³⁴*HighFinesse* WS7

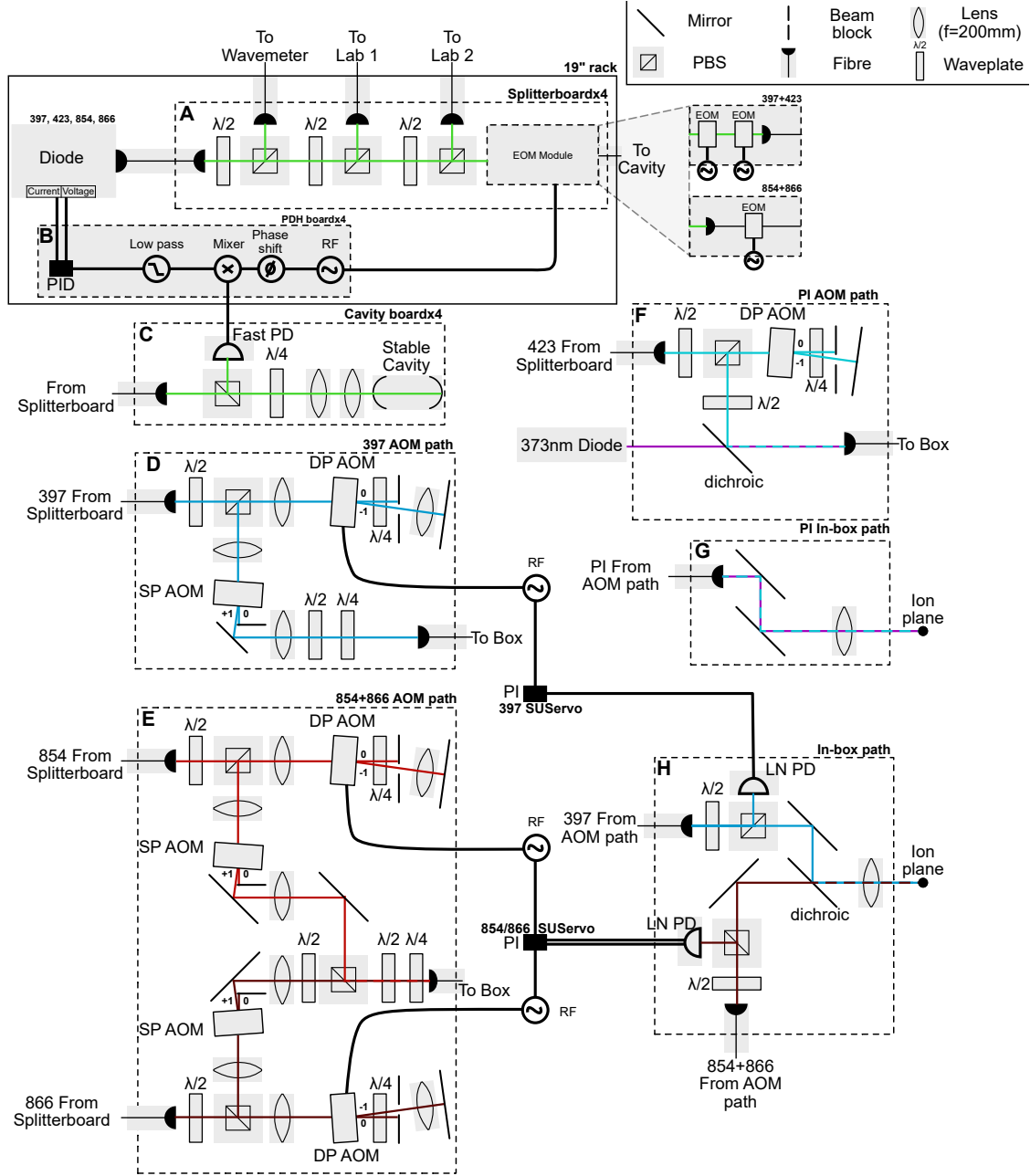


Figure 3.9: All beam paths for diode laser systems. Dotted boxes resemble modular optical paths with fibre and control lines in and out only. Frequency and optical power control loops are included.

both offset and PDH sidebands for an offset PDH lock. The offset sidebands may be arbitrarily set (within the GHz range of the EOM) to provide a continuous choice of laser lock frequencies between cavity modes, rather than maintaining a fixed offset between the laser output frequency and the discrete cavity mode frequencies. For 397-nm and 423-nm, resonant free space EOMs³⁵ are used, for 854-nm and 866-nm broadband fibre EOMs³⁶ are used for convenience. The path on the reference cavity board (fig 3.9 (c)) consists of a four-bore reference cavity³⁷ under vacuum and temperature control, two lens for matching the collimated input beam to the cavity mode, and a beamsplitter and $\lambda/4$ waveplate for directing the reflected cavity light to a fast photodiode³⁸. The beatnote due to the carrier and PDH sidebands are mixed down and low-passed by the *Toptica* PDH module to produce the signature antisymmetric PDH trace. Gains for PID controllers of both the diode current and voltage are set to stabilise the optical frequency.

The light from the “laser” laboratory is carried to the “trap” laboratory using 16 m polarisation maintaining fibres³⁹. In the “trap” laboratory we use AOMs in a double pass configuration for both frequency shifting and power stabilising the light (fig 3.9 (d) and (e)).

As space inside the MuMetal enclosure is constrained, it is advantageous to couple multiple beams in a single fibre before the enclosure. The second stage photo-ionisation light, at 373-nm⁴⁰, is combined with the 423-nm beam by a dichroic (fig 3.9 (f)) before being coupled to a 16 m fibre that takes the light directly into the MuMetal enclosure. The respective refractive indices of fused silica for 373-nm and 423-nm differ to the extent that the two beams were separated by more than one waist at the ion plane after initial rough alignment. To ensure the two beams overlap at the ion plane, care was taken for the beam propagation to be perpendicular to both the lens and vacuum viewport window, and a parabolic mirror outcoupler⁴¹

³⁵*Qubig* PM7-VIS-25 and PM9-VIS-2.25

³⁶*EOSPACE* PM-0s5-05-PFA-PFA-850

³⁷*Stable Laser Systems* SLS-6010-1-4bore

³⁸*Thorlabs* PDA10A2

³⁹*Schafter-Kirchoff ...*

⁴⁰*Toptica* iBEAM-SMART-375-S.

⁴¹*Thorlabs* RC02APC-F01

(rather than a lens) was used inside the MuMetal box.

The 854-nm and 866-nm light is combined after their respective double and single pass AOMs, and coupled into a 3 m PM⁴² fibre carrying the light into the enclosure. Light from both the PI and the 397/854/866-nm paths are directed onto the ions via the in-vacuum prisms, discussed in section 3.4. The final mirrors prior to the vacuum system are motorised⁴³, so that beam alignment to the ions can be optimised when the MuMetal enclosure is closed. Practically we find that these beams rarely need to be realigned (few months), unless there is a significant ambient temperature change of a few degrees.

3.6.4 Narrow Linewidth 729 Laser

Lasers are a key tool for creating the highly localised, strong electric field amplitudes and gradients needed to drive both carrier and sideband transitions of the trapped ion.

As shown in Figure 4.1, two sublevels within the $4S_{1/2}$ to $3D_{5/2}$ manifolds define the qubit. This is an electric quadrupole transition as $\Delta l = 2$. For the Calcium ion this transition is at 729-nm, and so a near resonance 729-nm laser is used to implement single- and multi-qubit gates (sections 4.2.4 and 4.4.2). This transition is also used after Doppler cooling for resolved sideband cooling to prepare the motional mode close to its ground state (as discussed in section 4.3.1). This transition is narrow linewidth due to the long lived $3D_{5/2}$ level, and so, for power efficiency, a narrow linewidth laser must be used. For the “fast entangling gates” use-case, high intensity electric-fields are required at the ion to drive sideband transitions, this will be discussed further in the single-addressing and two-qubit gate sections, but here it is sufficient to say we require >100 mW of light at the ion plane. Here we describe the 729-nm system consisting of a Ti:Sapph laser system pumped by an Nd:YAG 532-nm laser.

An *M2 Solstis* Ti:Sapph is pumped via 18 W of 532-nm light from a *Coherent*

⁴² **SK** ...

⁴³ *Newport Picomotor* piezo mirror mount P/N 8821

Verdi-V system to produce around 5 W of 729-nm light. The Ti:Sapph is engineered to operate with a stable < 50 kHz linewidth. Ti:Sapph crystals have broadband gain profiles which make them amenable in research environments to create frequency tunable laser systems. They also often have lower intrinsic phase noise than diode lasers at the same wavelength, which is favourable for our application. For single mode operation, the *Solstis* employs multiple intracavity frequency selective elements which consist of (in order of coarse frequency selectivity), a birefringent filter, a tunable Fabry-Pérot etalon, and the surrounding bow-tie cavity. To lock the tunable etalon to one longitudinal cavity mode, an actively stabilised “dither” phase-lock loop is used. This dither consists of periodically varying the etalon spacing with a frequency of around 20 kHz. This dither frequency directly phase modulates the light, which leads to the creation of sidebands which can interact with the ion causing errors in gates. This dither frequency (and harmonics of it), can be observed via composite pulse experiments on the ion, and are discussed in section 4.2.5.

Referencing the *Solstis* output with an external cavity and using PDH feedback can greatly suppress phase noise. Here an ultra high finesse cavity⁴⁴ is used, with a schematic of the 729-nm system shown in Figure 3.10. As opposed to the simpler controller implemented for the diode lasers, the 729-nm uses a cascaded PID lock with a fast feedback loop to the AOM situated after the *Solstis*, and an intermediate and slow feedback loop to piezos controlling the *Solstis* cavity length. With this system, a linewidth of $\ll 1$ kHz for the 729-nm light is expected. The electronics for this control loop are provided by *Stable Laser Systems* in the form of their *FPGA Servo* lock box. For an effective PDH lock, both short and long term stability of the reference cavity is required. To ensure the cavity is insensitive to the environment, it is made of an ultra low expansion material, and mechanically isolated from the environment by an acoustic dampening enclosure and a vibration isolation platform. The cavity is housed in a vacuum system to prevent refractive index fluctuations due to air, and to further reduce thermal coupling to the environment. The cavity is temperature stabilised at the zero crossing temperature of 30.6°C. Long term

⁴⁴*Stable Laser Systems* P/N 6020-4

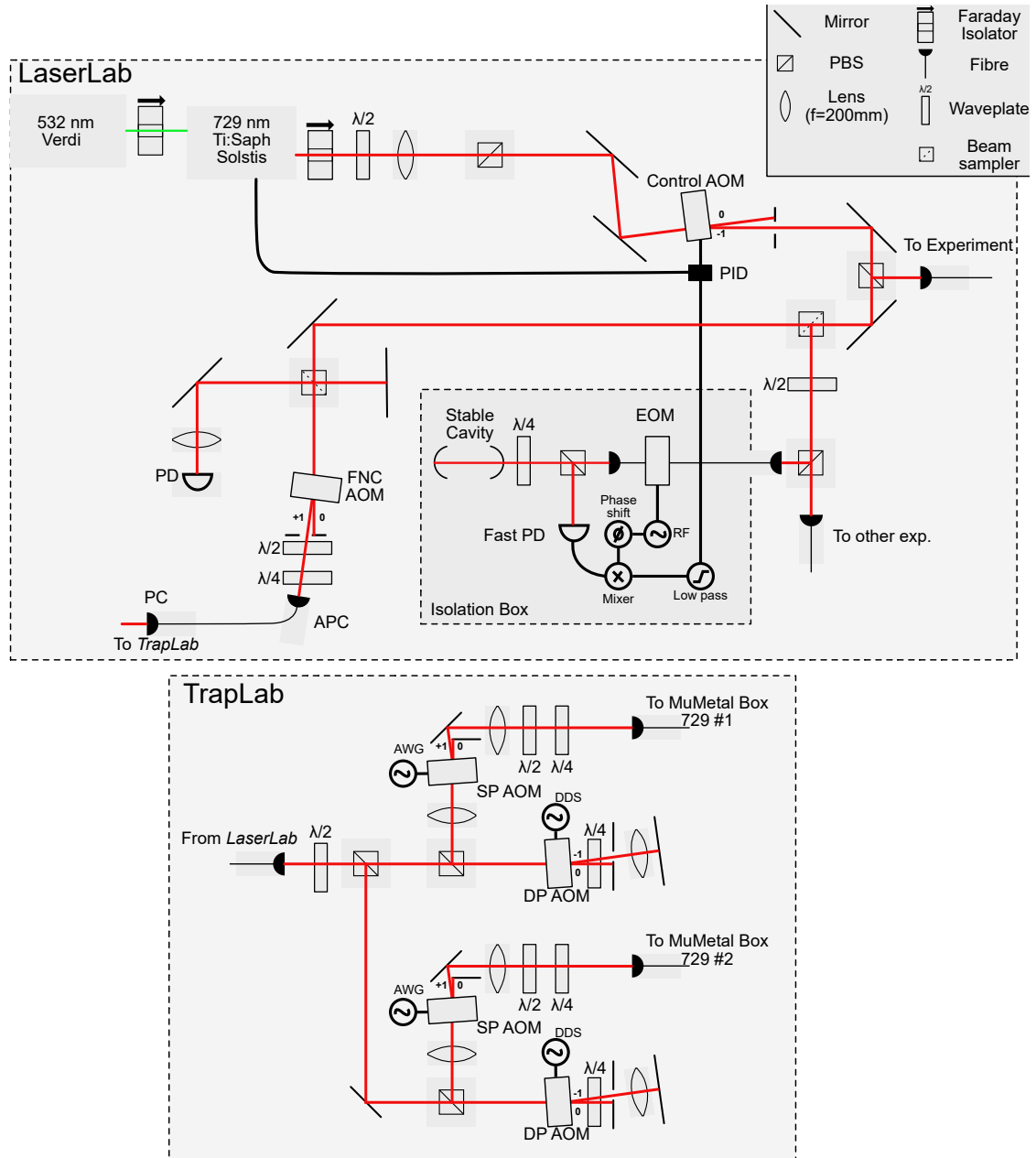


Figure 3.10: The 729-nm system. A Ti:Sapph laser tuned to 729-nm is pumped by a 532-nm source. Light is picked off at the first beam sampler to stabilise by PDH locking to a cavity. Light is distributed to two experiments, with the *FastGates* line including a fibre noise cancellation system.

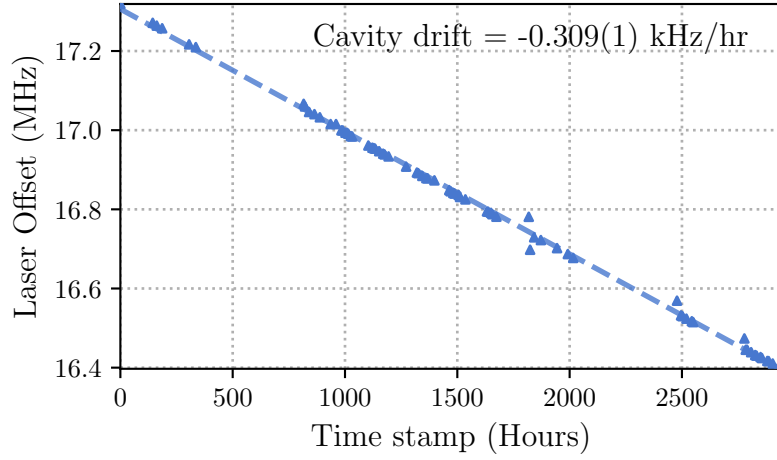


Figure 3.11: Cavity drift over 125 days measured by reference to the ion transition. The laser offset here is with respect to running all 729-nm AOMs at their central frequencies. Outliers are due to bad fits of the ion transition frequency, not due to changes in the cavity resonance.

cavity drifts of 309(1) Hz/hr are measured over 125 days, seen in figure 3.11. This measurement uses the ion as a frequency reference to probe the cavity frequency, with details given in section 4.2.3.

Figure 3.10 displays the other beam paths for our 729-nm system. Some light is picked off and sent to a wavemeter to continuously monitor the frequency. However, the majority is coupled to two output fibres for our experiment and another within the group. The 729-nm light is transported from a dedicated laser lab to a the trap apparatus lab by a 10 m single mode polarization maintaining fibre⁴⁵. The fibre introduces phase noise due to mechanical and thermal perturbations along the 10 m length. To remove this introduced noise, passive stabilisation in the form of thick foam tubing along the fibre length, as well as active stabilisation by a fibre-noise-cancellation (FNC) technique [45], is used⁴⁶. The efficacy of the FNC system is characterised in section 4.2.5 by measuring the spin coherence time of the ion.

⁴⁵ *SK ...*

⁴⁶FNC control consists of *Sinara Stabilizer* and *Pounder* mezzanine board with PID software developed by Ayush Agrawal.

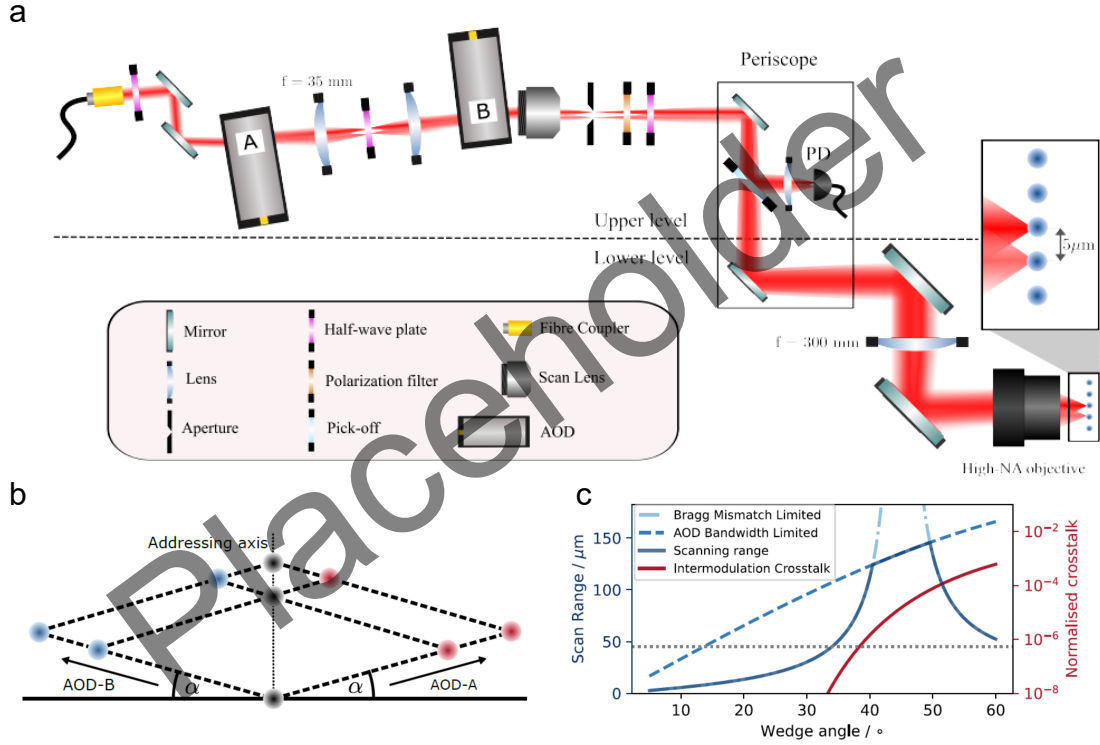


Figure 3.12: AOD setup

3.6.5 Single Addressing System

Single ion addressing is a requirement for executing arbitrary circuits on a register of ion-qubits. Furthermore, it is advantageous to focus the optical drive onto the ion, increasing the intensity of light, and so driving faster gates. A unique design feature of our system is the ability to produce an array of single ion addressing standing waves with individually contrrollable spatial phase, thus enabling any-to-any connectivity for fast entangling gates based on the scheme demonstrated in Ref. [18].

The design and construction of the single addressing system was led by Iver Øvergaard and, as of writing, has been in constant use in the experiment for around 1 year. Here we shall quote the design specification and relevant details of the single addressing system. In section 4.2.4 we present the characterisation data upon installation. Further technical information can be found in IRO's Transfer report [46], available upon request.

Multiple methods for single ion addressing have been developed in the ion trap community, including waveguide arrays [47, 48], micro-electronic-mechanical mirror arrays [49, 50], spatial-light-modulators [51], and acousto-optic deflectors [52, 53]. Each method has its own advantages and disadvantages in terms of speed, resolution, ease of integration, and scalability, and so choice of hardware is often a matter of experimental preference and constraints. We require:

- Addressing of up to two ions in a ~ 10 ion chain with low crosstalk to neighbouring ions.
- Fast reconfigureability of the beam position within a single experimental shot (i.e. between different gates within a single circuit execution).
- Good optical power efficiency to ensure high intensity electric fields at the ion.
- Flexibility in beam positioning to allow for fine tuning of the overlap of two counterpropagating beams used to create the standing wave.
- Individual phase control of each addressed beam to stabilise the standing wave phase at the ion plane.
- Beam position stability to avoid power fluctuations.
- A compact design to fit within the magnetic shielded region of the experiment.
- Good telecentricity (fixed beam k-vector direction across the scanned area) to avoid unwanted travelling wave component in a standing wave gate scheme.
- Zero frequency shift of the beam at the ion plane.

Crossed Acousto-Optic Deflectors (cAODs) are a popular choice for single ion addressing, offering fast rise times ($\sim 10 \mu\text{s}$), and good resolution over a large scanning area (~ 500 resolvable spots with 2° angular range) with zero frequency shift along the addressing axis. AODs operate through the same mechanism as AOMs, but are optimised for beam deflection rather than frequency modulation. As

an addressing system cAODs have poor scaling of power efficiency with the number of addressed spots, and so are not ideal for large scale systems. As we are interested in addressing ≤ 2 ions at a time, this is not a significant issue for our system.

Figure 5.16 (a) shows a schematic of the constructed addressing system. Due to space constraints within the magnetic shielded region, the optical elements are located on two height levels with a periscope steering the beam between them. The beam is delivered via single-mode polarisation maintaining fibre to the upper level where it is steered through the first AOD, with beam angle optimised for maximum diffraction efficiency into the -1^{st} order. Two lenses⁴⁷ in a 4f configuration re-image the deflection origin of the first AOD⁴⁸ onto the second AOD with the second AOD angled for maximum diffraction efficiency into the $+1^{st}$ order. The cAOD arrangement is termed “crossed” as the second AOD is rotated by some angle α with respect to the first, so that the deflection from the second AOD has some orthogonal component to that of the first. When the two AODs are driven with RF tones of the same frequency, f , the high efficiency diffracted order combinations are: $(0,0)$, $(-f,0)$, $(0,+f)$, $(-f,+f)$. The last combination of $(-f,+f)$ has zero overall frequency shift yet retains some frequency dependent deflection angle, which is suitable for addressing ions in a 1D line. To translate this deflection angle into some physical displacement at the ion plane, a telecentric scanning lens⁴⁹ is used. The telecentricity ensures that the beam k-vector direction is fixed across the scanned area, which is important for standing wave gate schemes to avoid unwanted travelling wave components. At the intermediate focus after the scanning lens, the unwanted orders are blocked by an adjustable slit aperture and the beam polarisation is cleaned and selected using a polarisation filter and half-wave plate. A plate beam splitter⁵⁰ is used within the periscope to pick off $\sim 5\%$ of the light for monitoring on a photodiode⁵¹, which is used in a feedback loop to stabilise the intensity at the ion plane (as discussed in section 3.6.2). The scan lens, in combination with a relay

⁴⁷**P/N**

⁴⁸**P/N**

⁴⁹**P/N**

⁵⁰**P/N**

⁵¹**P/N**

lens, forms a beam expander to increase the beam NA for the final objective lens to focus to a beam waist of $\approx 1.5 \mu\text{m}$ at the ion plane. Characterisation of the system (details given in section 4.2.4) showed that the addressing beam could be focused to a Gaussian waist of $1.50(1) \mu\text{m}$, measured by scanning a single trapped ion with Rabi experiments. The same beam was then tested in a five-ion chain, where the nearest-neighbour intensity crosstalk was measured to be $4.16(2) \times 10^{-4}$ **unit?**.

3.6.6 Global Addressing System

The single addressing provides the utility of spatially selecting a few ions to drive with the 729-nm light, however, the power efficiency scales as $O(n^2)$, with n being the number of addressed spots. As such, we limit the single addressing to simultaneously illuminate only one or two ions at a time. It is therefore convenient to also have a global beam that has dimensions larger than the ion chain to address all the ions in parallel. This global branch is currently counterpropagating to the single addressing path, and shares the same optical path as the imaging system (see figure 3.7). To increase the beam intensity at the ions, the global branch uses an anamorphic prism pair⁵² to elliptically shape the beam, with a 4:1 aspect ratio, with major axis along the ion chain direction. The beam is focussed onto the ion plane using an $f = xx$ lens⁵³, resulting in a major axis beam waist radius of $52(3) \mu\text{m}$ ⁵⁴. As with the other beams directed to the ion, we use a final piezo controlled mirror⁵⁵ for fine alignment and control when the MuMetal enclosure is closed.

3.7 Control System

Could tie up loose end here of control FPGA blocks.

⁵²*ThorLabs* PS883-B

⁵³**P/N**

⁵⁴The beam profile was measured by extracting per-ion Rabi frequencies while driving on resonance quadrupole transitions (see section 4.2.4) and assuming a Gaussian cross section.

⁵⁵*ThorLabs* POLARIS-K2S2P

3.8 Ion Loading

Atoms are provided to the trapping region by heating Calcium flakes housed in a resistive tube oven with the oven mount shown in figure 3.5. A thermocouple is tag welded onto the tube oven to provide a temperature reading for current feedback for fast stabilisation to the target temperature of $\simeq 200^\circ\text{C}$. A steady state of $\simeq 2\text{ A}$ across the $0.5\ \Omega$ load of the oven tube is required to maintain 200°C . The tube oven has a pinhole that is aligned to the ion flux aperture, visible in figure 3.5 with atom flux shown in green, and the atomic shield pinhole shown in figure 3.2. These apertures were included in the design to improve the operational lifetime of the trap by ensuring neutral Calcium metal cannot accumulate on the trap surface and cause shorts between RF, ground, and DC electrodes. The aperture design *luckily* failed. It failed as we observe atomic flux at all electrode locations suggesting the aperture is too large. It was *lucky* as the planned trapping region of *dc10* became non-viable after electrode *Fr dc11* developed a continuity fault. As such, ions are loaded at *dc5*.

The neutral atom flux is ionised using a two-step photoionisation scheme [41] that can selectively load the desired isotope of Calcium. Natural abundance Calcium, with typical isotope proportions of 96.9% : ^{40}Ca , 2.09% : ^{44}Ca , 0.647% : ^{43}Ca [41], was appropriate as we aim to exclusively load $^{40}\text{Ca}^+$. The first stage involves a frequency stabilised 423-nm beam that couples the $4s^2 \rightarrow 4s4p$ transition in neutral calcium. The natural linewidth of this transition ($\simeq 35\text{ MHz}$) is much smaller than the isotope shifts between abundant isotopes of Calcium (typically $> 100\text{ MHz}$ shifts), as such, through frequency selectivity, only the desired isotope can be excited. The second ionisation stage couples $4s4p \rightarrow \text{continuum}$ through a 373-nm UV laser source (although the exact wavelength does not matter as long as $< 389.8\text{-nm}$). We practically see no loading of other Calcium isotopes, and only see rare non-fluorescent ions (once every few weeks) after collisions with residual gas (we suspect this is CaH^+).

The 373-nm laser causes charge build-up on the trap electrodes, which can affect the trapping potential, and so a physical shutter, controlled by a TTL signal, is

used to block the beam when not actively loading ions.

4

Experimental Characterisation

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Before we can proceed with our planned experiments involving the motion and spin of the atoms, the apparatus must be characterised. This characterisation serves two purposes: it allows us to benchmark the performance of our system against state-of-the-art results, and it reveals any current limitations that must be addressed before we can successfully carry out the experiments.

As discussed above, we are interested in two paradigms: discrete variable quantum

computing (DVQC), that concerns the manipulation of qubits (or spins), and continuous variable quantum computing (CVQC), that concerns the manipulation of qumodes (or harmonic motion).

For DVQC, we must have access to the following elementary operations:

- Qubit state preparation, here via optically pumping the electronic state of the ion into one sublevel of the $4S_{1/2}$ manifold — the lower level of our qubit, section 4.2.1.
- Measurement of the qubit state, here via state-dependent fluorescence, section 4.2.2.
- Single qubit rotations, here via interaction with the 729-nm laser, section 4.2.4
- Two qubit entangling-gates, here via spin-dependent forces (SDFs), using the 729-nm laser and the motional modes of the ion, section 4.4.2.

For CVQC, we need:

- Motional state preparation, here via Doppler cooling and sideband cooling, section 4.3.1.
- Measurement of the motional state, which is not yet implemented on our system but will follow [fluhmann2020direct].
- Single and two qumode gates, which are also not yet implemented, but will utilise SDFs following [sutherland2021universal], section 4.4.1.

We split this chapter into interactions involving only the spin, then those focussed only on the motion, and finally we describe spin-motion interactions. First, however, we describe the ion, its level structure, and the relevant lasers to manipulate its state.

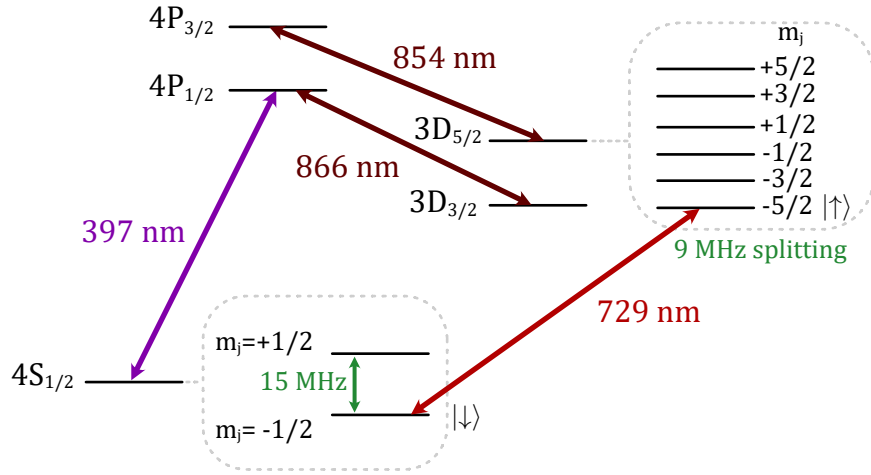


Figure 4.1: Electronic energy levels of $^{40}\text{Ca}^+$, which will be used in this thesis. The levels are split by the Zeeman effect due to a 5.4 G external magnetic field (they are shown explicitly only for the qubit manifolds). The transitions marked are required for cooling and control over the ion. The chosen qubit levels are labelled with $|\downarrow\rangle$, $|\uparrow\rangle$.

4.1 The Ion

The $^{40}\text{Ca}^+$ ion has a rich level structure, shown in figure 4.1, with many accessible quadrupole transitions $4S_{1/2} \leftrightarrow 3D_{5/2}$. The quadrupole transition is ideal for use as a qubit due to the long-lived metastable level ($\tau = 1.1$ s) [barton2000measurement], and the ability to optically couple the transition via a laser. As mentioned in section ??, the 729-nm beam is used for this purpose. By selecting the appropriate frequency of the 729-nm beam, qubit operations such as state preparation (section 4.2.1) and single qubit gates (section 4.2.4) may be driven, or by setting the detuning such that a motional sideband is near resonance, to couple the spin and motional degrees of freedom (section 4.4.1). Here, we discuss which quadrupole transitions are used, and the rationale behind their choice.

Zeeman splitting of the $4S_{1/2}$ and $3D_{5/2}$ states leads to 2 and 6 non-degenerate sublevels, respectively. Due to the magnetic field strength of 5.4 G (section 3.3), the S -levels are split by ~ 15 MHz, and the D -levels by ~ 9 MHz. This splitting is considerably larger than both the natural linewidth of the 729-nm transition ($\sim \text{Hz}$), and the Ti:Sapph laser linewidth (< 1 kHz), meaning the transitions can be selectively addressed by frequency tuning of the laser. There is some freedom

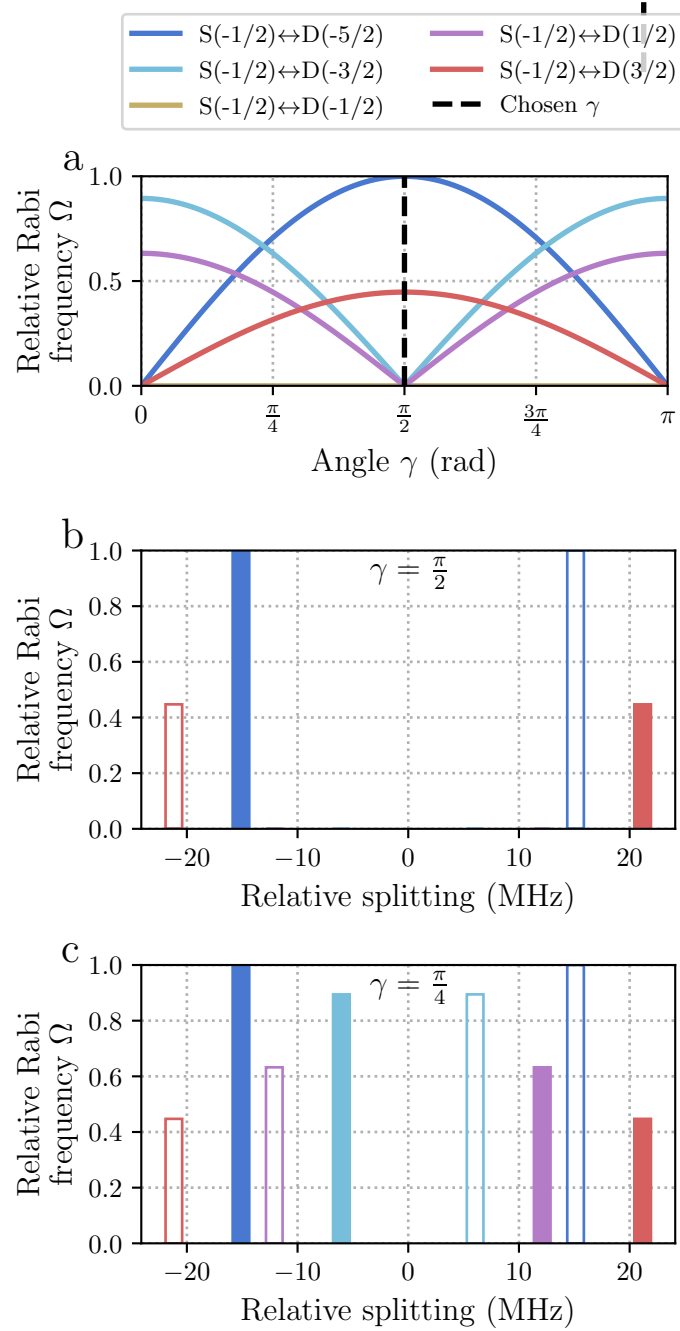


Figure 4.2: Normalised quadrupole Rabi frequencies, Ω , and frequency splittings for the $4S_{1/2} \leftrightarrow 3D_{5/2}$ Zeeman sublevels in $^{40}\text{Ca}^+$. Solid bars correspond to transitions with lower state $|4S_{1/2}, m_j = -1/2\rangle$, while open bars correspond to transitions with lower state $|4S_{1/2}, m_j = +1/2\rangle$. **a)** Relative Rabi frequencies as angle γ between the polarisation vector and the magnetic field is varied. The k-vector of the light is assumed to be perpendicular to the B-field. **b)** Relative frequency splittings and strengths when $\gamma = \pi/2$. This is the polarisation chosen for the current and future experiments to reduce the number of transitions which can cause off-resonant interactions when the qubit defined by $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ is used. **c)** Relative frequency splittings and strengths when $\gamma = \pi/4$. At this angle, all transitions are accessible apart from those with $\Delta m_j = 0$. This is the polarisation chosen for our initial experiments to allow access to all transitions for characterisation.

in choosing which of these 10 possible (with $\Delta m_j \leq 2$) quadrupole transitions we define as our qubit. Each transition has a varying coupling strength, here denoted by the relative Rabi frequency, Ω , depending on the polarisation and k -vector of the interacting light with respect to the B-field axis. The desired criteria for qubit and polarisation choice are as follows:

- Access to a closed-cycle transition for sideband cooling, (for details on this, see section 4.3.1). Closed-cycle here means $|4S_{1/2}, m_j = -1/2\rangle \rightarrow |3D_{5/2}\rangle \rightarrow |4P_{3/2}, m_j = -3/2\rangle \rightarrow |4S_{1/2}, m_j = -1/2\rangle$.
- The qubit transition needs a large quadrupole matrix element, leading to a high Rabi frequency, Ω , to enable laser-power-efficient, fast, gates. Polarisation, γ , may be optimised to increase this coupling.
- Ideally the qubit transition should have a low magnetic field sensitivity, to reduce the effect of magnetic field noise. Using a MuMetal enclosure around the experiment does relax this constraint, see section 3.3.
- To reduce the effects of off-resonant interactions due to nearby transitions, it is desirable to limit the number of accessible transitions near to the qubit transition. This is done by selecting a polarisation, γ , such that unwanted transitions are suppressed.

Given these criteria, we use the transition $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ with magnetic field sensitivity $-0.446 \text{ MHz G}^{-1}$. Closed-cycle sideband cooling is possible with this same transition, and state preparation via optical pumping is done via $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ with magnetic field sensitivity $-0.624 \text{ MHz G}^{-1}$.

Figure 4.2 *a*, shows the normalised Rabi frequencies of each transition as the angle γ between polarisation vector and B-field is varied. Here only linearly-polarised light with k -vector perpendicular to the B-field direction is considered. The polarisation of the 729-nm laser is set to $\gamma = \pi/2$ (black dashed line), which maximises the Rabi frequency of the qubit transition, while suppressing nearby

Parameter	Value
729-nm laser power	1000 μW
729-nm laser detuning	0 MHz*
729-nm pulse duration	5 μs
854-nm laser power	60 μW , $250 \times I_{\text{SAT}}$
854-nm laser detuning	0 MHz
854-nm pulse duration	40 μs
866-nm laser power	15 μW , $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
866-nm pulse duration	30 μs

Table 4.1: * With respect to qubit transition frequency.

Experimental parameters used for state preparation of the ion into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state.

transitions apart from that needed for state preparation. Figure 4.2 *b*, shows the relevant frequency splitting of the transitions at $\gamma = \pi/2$, which is the set polarisation for all experiments apart from initial quadrupole characterisation. For initial characterisation of the quadrupole transitions, the polarisation was set to $\gamma = \pi/4$, which allowed access to all transitions apart from those forbidden by the beam geometry, $\Delta m_j = 0$, see figure 4.2 *c*.

Other required lasers for manipulating the ion are: the 397-nm and 866-nm lasers used to form a closed-cycle for fluorescence, the 854-nm laser used to deshelve the ion from the $3D_{5/2}$ level for qubit state preparation. Details on these lasers can be found in section 3.6.

4.2 Spin

Discrete variable quantum computing consists of manipulating many two-level systems, which we refer to either as spins, or qubits. In this section the methods used to manipulate these spins are described, and their performances are benchmarked.

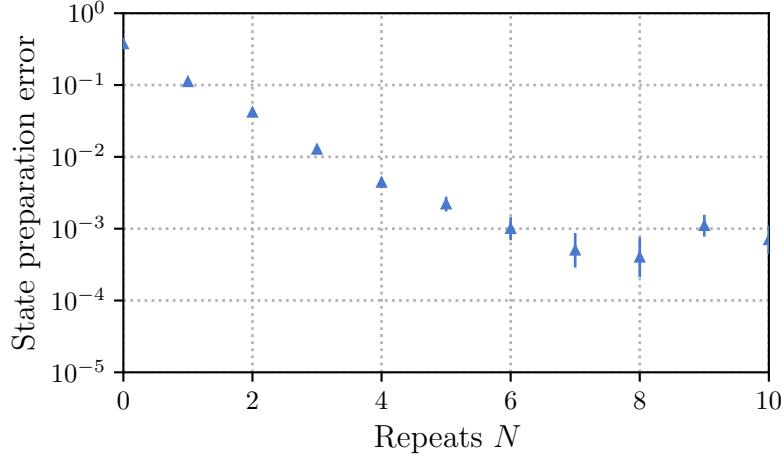


Figure 4.3: State preparation error as a function of the number of state preparation pulses, N . The error is defined as population not in the $|\downarrow\rangle$ state after the preparation sequence has been applied. The error is reduced to below 10^{-3} after $N = 7$ pulses, which is used in our experiments.

4.2.1 State Preparation

To utilise two levels of the ion as a qubit, the electronic state must first be prepared into the qubit manifold. Here, the ion is prepared into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state. This is done via effective optical pumping by pulsing the quadrupole transition $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$, followed by deshelling pulses using the 854-nm, 866-nm lasers on resonance. The 729-nm laser pulse duration and intensity is set to invert the populations, i.e. perform a π -pulse. Details for laser parameters are summarised in table 5.1. These three pulses are repeated N times sequentially. Figure 4.3 shows the state preparation error with increasing N . The error is found by applying a π -pulse on the state preparation transition directly after the state preparation sequence and looking at the population in the $|\downarrow\rangle$ state. Any reduction from $|\downarrow\rangle = 1$ indicates residual population in the $|4S_{1/2}, m_j = +1/2\rangle$ state, or the D levels. After $N = 7$ pulses, which is used in our experiments, there is a state preparation error of $< 10^{-3}$. This is further verified by randomised benchmarking in section 4.2.4. The state preparation error is likely limited by accidental excitation of the $|4S_{1/2}, m_j = -1/2\rangle$ state to the $D_{5/2}$ level. This may

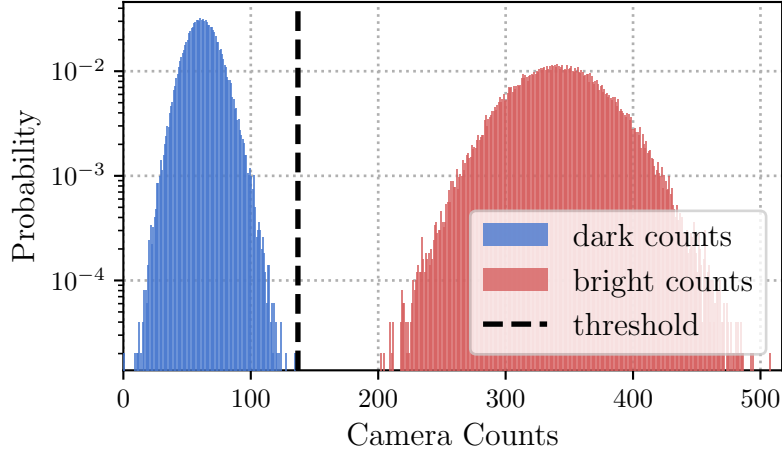


Figure 4.4: Readout histogram with 100,000 individual measurements for bright (red) and dark (blue) populations. The camera counts are presented as a histogram and fitted to Poissonian distributions to calibrate a threshold for discerning if the ion is bright or dark with the lowest probability for false measurement. Dark counts are mainly due to 397-nm light scattered from the nearest trap electrodes. With 100 μ s readout duration, and laser parameters set to Doppler cooling parameters, a threshold count of 137 photons is found, giving an expected statistical error of 6×10^{-9} given the fitted Poissonians.

be due to off-resonant excitation of another transition, or due to spectral purity of the 729-nm laser.

Another protocol for state preparation is optical pumping on the dipole transition (397-nm), which often is much faster. However, in this setup it is impractical as polarisation selectivity of the 397-nm transition is not possible due to the constraint of beam geometry from the in vacuum optics, see section 3.4, and frequency selectivity is not possible due to the low magnitude fixed B-field providing Zeeman shifts less than the natural linewidth of the 397-nm transition, see section 3.3.

4.2.2 Measurement

To measure the qubit state, the 397-nm and 866-nm lasers are applied, and the scattered 397-nm photons are imaged by the objective onto a camera (see section 3.5). 397-nm light will only be scattered if the qubit is in $|\downarrow\rangle$. Use of a high numerical aperture ($\text{NA} = 0.6$) objective greatly increases the collection efficiency. This efficiency allows the 397-nm beam to be red-detuned to Doppler cooling parameters

Parameter	Value
397-nm laser power	3 μ W, $1 \times I_{\text{SAT}}$
397-nm laser detuning	-17.5 MHz
866-nm laser power	15 μ W, $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
Doppler cooling duration	1000 μ s
Readout duration	100 μ s
Camera threshold	137 counts

Table 4.2: Experimental parameters used for Doppler cooling and state-selective fluorescence of the ion.

(see section 4.3.1), while still maintaining low readout duration (100 μ s) and low readout error. Cooling during readout increases the experiment’s repetition rate by reducing the amount of dedicated cooling time required.

The parameters used for readout are summarised in table 4.2, and a typical histogram of readout counts for one ion can be seen in figure 4.4. The readout threshold is calibrated by taking bright counts with the 397-nm and 866-nm lasers on, and dark counts with the 397-nm laser on and the 866-nm laser off. This assumes dark counts are due to 397-nm scattering from the trap electrodes. Fitting Poissonian distributions to the bright and dark measurements, an optimal count threshold of 137 is found, shown by black dashed line. This threshold gives an expected statistical readout error of $\epsilon_{\downarrow} = \epsilon_{\uparrow} = 6 \times 10^{-9}$, where $\epsilon_{\downarrow, \uparrow}$ are the expected error associated with measuring bright or dark. This measurement ignores errors due to the finite lifetime of the metastable $3D_{5/2}$ level, which dominate $\epsilon_{\uparrow}$. The expected error due to the finite lifetime, τ , is $\epsilon_{\text{decay}} = 1 - e^{-t/\tau}$, where t is the readout duration. For $^{40}\text{Ca}^+$, $\tau \sim 1.1$ s, $\epsilon_{\uparrow} \approx \epsilon_{\text{decay}} = 10^{-4}$. Assuming the metastable lifetime is limited only by spontaneous decay overlooks leakage from the 854-nm deshelving beam, which could significantly shorten τ [sotirova2024high-fidelity].

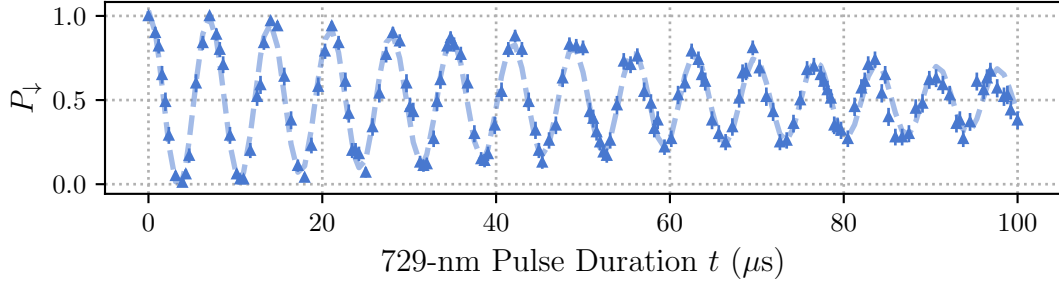


Figure 4.5: Long duration, t , Rabi flop on the qubit transition to extract Rabi frequency. Estimated probability of being in $|\downarrow\rangle$, $P_{\downarrow}$, at each time step using 200 individual experimental shots. Fitted dotted line using decaying oscillation model. Extracted Rabi frequency $\Omega/2\pi = 0.2864(2)$ MHz, and decay rate $\lambda = 0.0107(7) \mu\text{s}^{-1}$.

4.2.3 Calibration Routines

Micromotion Compensation

Magnetic Field and Laser Offset

Beam Pointing Calibration

4.2.4 Single Qubit Rotations

A qubit is fully described by $|\psi\rangle = \alpha|\downarrow\rangle + \beta|\uparrow\rangle$, where α and β are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$. The qubit state can be manipulated by applying single qubit rotations, $R_{x,y}(\theta)$, commonly referred to as gates. In the Bloch sphere, x - and y -rotations are performed by perturbations resonant with the qubit transition, here, using the 729-nm laser. The rotation angle, θ , is given by $\theta = \Omega t/2$, where Ω , is the Rabi frequency, and t is the pulse duration. The rotation axis, is determined by the relative phase of the light field and the qubit. We define the initial relative phase to correspond to a y -rotation.

Gate Time Calibration

The pulse duration, for a $R(\pi/2)$ ($\pi/2$ -pulse), is calibrated by Rabi oscillations. Figure 4.5 shows the population in $|\downarrow\rangle$, $P_{\downarrow}$, calculated from 200 experimental shots. To find the Rabi frequency, Ω , the data is fitted to a decaying oscillation model,

$$P_{\downarrow} = \frac{1 + e^{-\lambda t} \cos(\Omega t/2)}{2}, \quad (4.1)$$

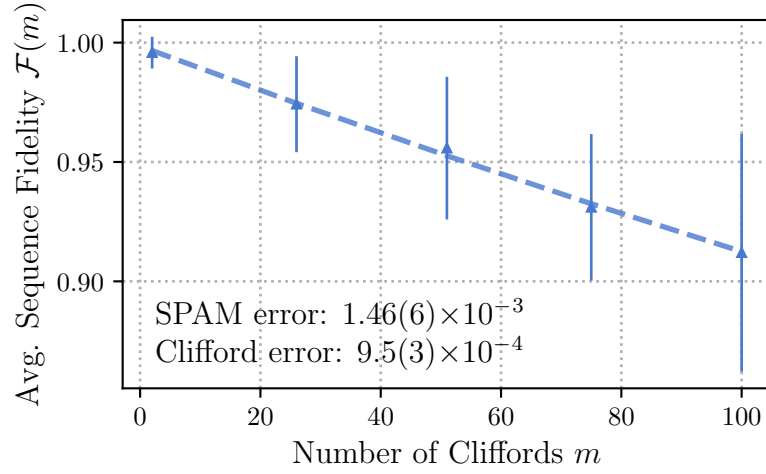


Figure 4.6: Estimating single qubit Clifford fidelities, $\mathcal{F}$, and errors due to state preparation and measurement, using randomised benchmarking. The data points are the average survival population after applying a sequence of m random Clifford gates, and a final inverting Clifford gate. The dashed line is the fit to the decay model given by equation 4.2. The error bars are given by the standard deviation of the survival populations.

where λ is the decay rate due to decoherence effects [wineland1998experimental]. The Rabi frequency is found to be $\Omega/2\pi = 0.2864(2)$ MHz, and the decay rate $\lambda = 0.0107(7)$ $1/\mu\text{s}$. From this $R(\pi/2) = 1.7$ μs can be estimated. These measurements were taken with 1 mW of 729-nm power, with beam radius $\sim 15\mu\text{m}$ at the ion.

Randomised Benchmarking

To evaluate the quality of both our state preparation and single qubit rotations, randomised benchmarking (RBM) [knill2008randomized] is employed. RBM consists of applying random combinations of gates from the single-qubit Clifford group to estimate an average error per gate. The single-qubit Clifford group is the set of unitaries which map the Pauli matrices to one another through conjugation. The RBM protocol used here is described in [hughes2021benchmarking]. The sequence of operations is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 4.2.1.

2. Apply a random sequence of m single qubit Clifford gates, where m is the length of the sequence.
3. Apply a final “inverting” Clifford gate, which is chosen such that the full sequence performs the identity operation.
4. Measure the population in $|\downarrow\rangle$, $P_\downarrow$, via state-selective fluorescence, section 4.2.2.
5. Repeat steps 1-4 multiple times to find the average survival population, $\langle P_\downarrow \rangle$.
6. Repeat steps 1-5 for many different random sequences with a range of sequence lengths, m .
7. Fit the decay of the average survival population, $\mathcal{F}(m) = \langle P_\downarrow(m) \rangle$, to a decay model to find the average error per Clifford gate.

The decay model used to find the fidelity versus number of Clifford gates is given by **[hughes2021benchmarking]**,

$$\mathcal{F}(m) = \frac{1}{2} (1 + (1 - 2\epsilon_{\text{SPAM}})(1 - 2\epsilon_c)^m), \quad (4.2)$$

where $\mathcal{F}(m)$ is the fidelity of the sequence of length m , ϵ_{SPAM} is the state preparation and measurement error, and ϵ_c is the average error per Clifford gate. The Clifford gates are decomposed into sequences of $\pi/2$ - and π -pulses about either the x - or y -axes. Up to $m = 100$ Clifford gates are probed, and the decay of the fidelity, $\mathcal{F}(m)$, is fit using the above model. The error per Clifford is found to be $\epsilon_c = 9.5(3) \times 10^{-4}$, while the SPAM error is $\epsilon_{\text{SPAM}} = 1.46(6) \times 10^{-3}$. The decay plot for this RBM sequence can be seen in figure 4.6. The independently characterised state preparation (section 4.2.1) and measurement (section 4.2.2) errors of $\sim 10^{-3}$, and $\sim 10^{-4}$ respectively, are consistent with the measured ϵ_{SPAM} .

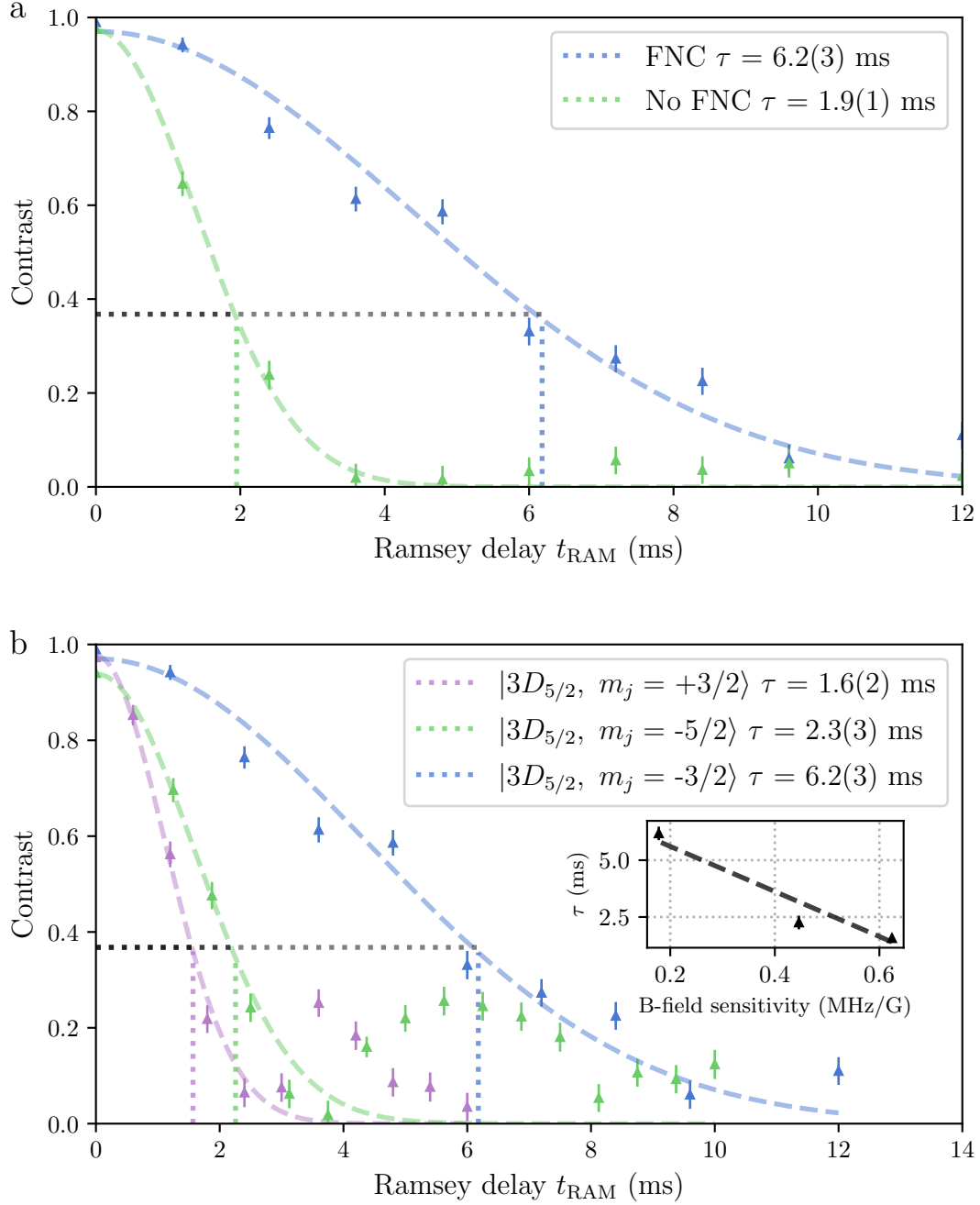


Figure 4.7: Coherence time measurements. **a)** Coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without fibre noise cancellation (FNC). **b)** Coherence time of three quadrupole transitions with varying magnetic field sensitivities. The inset plot shows a linear fit between the coherence time and the magnetic field sensitivity of the transition.

sec:Single Addressing char

4.2.5 Spin Coherence Times

Add section of spin-lock scans of laser. Decoherence, as can be seen in figure 4.5, can be caused by laser frequency and power fluctuations, magnetic field noise, and spontaneous emission from the $3D_{5/2}$ level. The coherence time of the qubit is a useful diagnostic for finding unwanted noise sources, and for benchmarking the performance of the qubit as a store of quantum information.

To measure the coherence times, we perform Ramsey sequences with varying delay durations. The Ramsey sequence is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 4.2.1.
2. Apply a $R_y(\pi/2)$ pulse to prepare the qubit in $|+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle)$ superposition state.
3. Wait for a time t_{RAM} .
4. Apply a second $R_\phi(\pi/2)$ -pulse, with a variable axis of rotation, ϕ .
5. Measure the population in $|\downarrow\rangle$, $P_\downarrow$, via state-selective fluorescence, section 4.2.2.
6. Repeat steps 1-5 for different values of ϕ and fit the resulting fringe to find the contrast.
7. Repeat steps 1-6 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

With this tool, the efficacy of the fibre noise cancellation (FNC, see section ??) can be measured. Figure 4.7 *a* shows the coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without FNC. The coherence time is increased from 1.9(1) ms to 6.2(3) ms with FNC, showing that without FNC, the coherence times are limited by laser phase noise.

With FNC enabled, to see if the coherence time is now limited by magnetic field

noise, transitions with varying magnetic field sensitivities are compared. Figure 4.7 *b* shows the coherence time of three quadrupole transitions with varying $D_{5/2}$, m_j numbers: $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = \{-5/2, -3/2, +3/2\}\rangle$. These transitions have magnetic field sensitivities -0.446 , -0.178 , $+0.624$ MHz G $^{-1}$ respectively. The contrast decays for all three qubits are shown in the figure. The inset plot shows a well correlated linear fit between the coherence time and the magnetic field sensitivity, suggesting that the coherence time is limited by magnetic field noise. For both $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ and $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = +3/2\rangle$, revivals in contrast can be seen which suggest that the dominant noise source is oscillatory and coherent over the Ramsey scan duration.

These measurements were taken without the MuMetal enclosure (section 3.3) fully assembled around the vacuum system, which we expect will greatly reduce external magnetic field noise, and therefore increase the coherence times.

4.3 Motion

Motion of ions trapped in the same potential can be described by a set of normal modes [james1998quantum]. These modes are labelled with respect to the ion chain geometry: axial modes oscillate parallel with the chain, and radial modes oscillate perpendicular to the chain. Due to the geometry of the 729-nm beam access with respect to the ion chain (see section 3.4), only the radial modes can be addressed in our system. For the *NPL* trap (see section 3.2), the radial modes frequencies are non-degenerate. In our system, for a single ion, the radial mode frequencies are $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and $\omega_{\text{lower}}/2\pi = 2.5$ MHz, and the axial mode frequency is $\omega_{\text{axial}}/2\pi = 1.6$ MHz. All work here is using the upper radial mode.

The motional mode, i , leads to sidebands in the 729-nm spectrum at $\omega_0 \pm \omega_i$ [ozeri2011tutorial], where ω_0 , the carrier, is the frequency of the 729-nm laser when on resonance with the qubit transition. The $+\omega_i$ sideband is referred to as the blue sideband (BSB), and the $-\omega_i$ is the red sideband (RSB). The RSB, BSB, and RF fields near to resonance with the motional modes, are the tools used in this section to manipulate

the motion of the ion.

The motional state preparation, heating rate, frequency stabilities, and coherence times are characterised in this section.

4.3.1 Cooling

For any interaction that involves the motion of the ion, it is essential to both prepare and measure the motional state with high fidelity. For all motional interactions considered in this thesis, the ion is initially prepared as close as possible to the motional ground state, with a mean phonon number $\bar{n} \approx 0$. This is done by an initial stage of Doppler cooling, followed by pulsed sideband cooling on the 729-nm transition. The final motional state is characterised by measuring the population in Fock states using RSB and effective BSB pulses.

Doppler Cooling

Doppler cooling relies on the fact that light incident on a moving ion appears frequency-shifted in the ion's rest frame. For Doppler cooling of $^{40}\text{Ca}^+$, both the 397-nm and 866-nm lasers are applied. The 397-nm laser is red-detuned from the dipole $|4S_{1/2}\rangle \leftrightarrow |4P_{1/2}\rangle$ transition, leading to preferential scattering of 397-nm photons that have a net cooling effect [wineland1979laser]. The excited $4P_{1/2}$ level decays via two channels: emission of a 397-nm photon back to $|4S_{1/2}\rangle$, or an 866-nm photon to $|3D_{3/2}\rangle$, with a branching ratio of 14.5 [ramm2013precision]. Since many photon scattering events are required, the cooling cycle must be closed. To prevent population trapping in the metastable $|3D_{3/2}\rangle$ level, the 866-nm laser is applied on resonance during cooling.

Equilibrium is reached when the cooling rate balances photon recoil heating. Assuming the 397-nm transition is broadened only by its natural linewidth γ , the minimum Doppler temperature is,

$$T_{\text{Doppler}} \approx \frac{\hbar\gamma}{2k_B}, \quad (4.3)$$

Parameter	Value
729-nm laser power	3.0 mW
729-nm laser detuning	-4.0 MHz*
729-nm pulse duration $N = 1$	2.5 μ s
729-nm pulse duration $N = N_{\max}$	33 μ s
Repumping 866-nm duration	14 μ s
Deshelving 854-nm duration	5 μ s
Number of cooling cycles $N_{\max}$	50

Table 4.3: *729-nm detuning is relative to the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ transition frequency.

Experimental parameters used for sideband cooling. Other parameters for the repump and deshelve pulses are the same as for state preparation, see table 5.1. The 729-nm laser is detuned to the RSB resonance for the upper radial mode, $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and the pulse duration for $N = N_{\max}$ is chosen to be the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

where $\hbar$ is the reduced Planck constant, and k_B is Boltzmann's constant [wineland1979laser]. For $^{40}\text{Ca}^+$, the natural linewidth of the 397-nm transition is $\frac{\gamma}{2\pi} = 21$ MHz[morton2003atomic], giving $T_{\text{Doppler}} = 0.5$ mK. For a radial mode frequency of $\frac{\omega}{2\pi} = 4$ MHz, the expected mean phonon number is,

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_B T} - 1}, \quad (4.4)$$

yielding $\bar{n} = 2.3$ at the Doppler limit. The parameters used here for Doppler cooling are summarised in table 4.2.

Sideband Cooling

To further cool the ions toward their motional ground state, resolved sideband cooling is used. As mentioned above, the motion of the ion modulates the absorption spectrum of the ion, leading to red and blue sidebands (RSB and BSB). For the $|4S_{1/2}\rangle \leftrightarrow |3D_{5/2}\rangle$ transition, at appropriate motional mode frequencies, interaction strengths, and using a narrow linewidth laser (section ??), these sidebands can be resolved spectroscopically. The pulsed sideband technique employed consists of RSB pulses on the 729-nm transition, followed by deshelving, and repumping pulses on the 854-nm and 866-nm transitions respectively. For efficient cooling, initial short

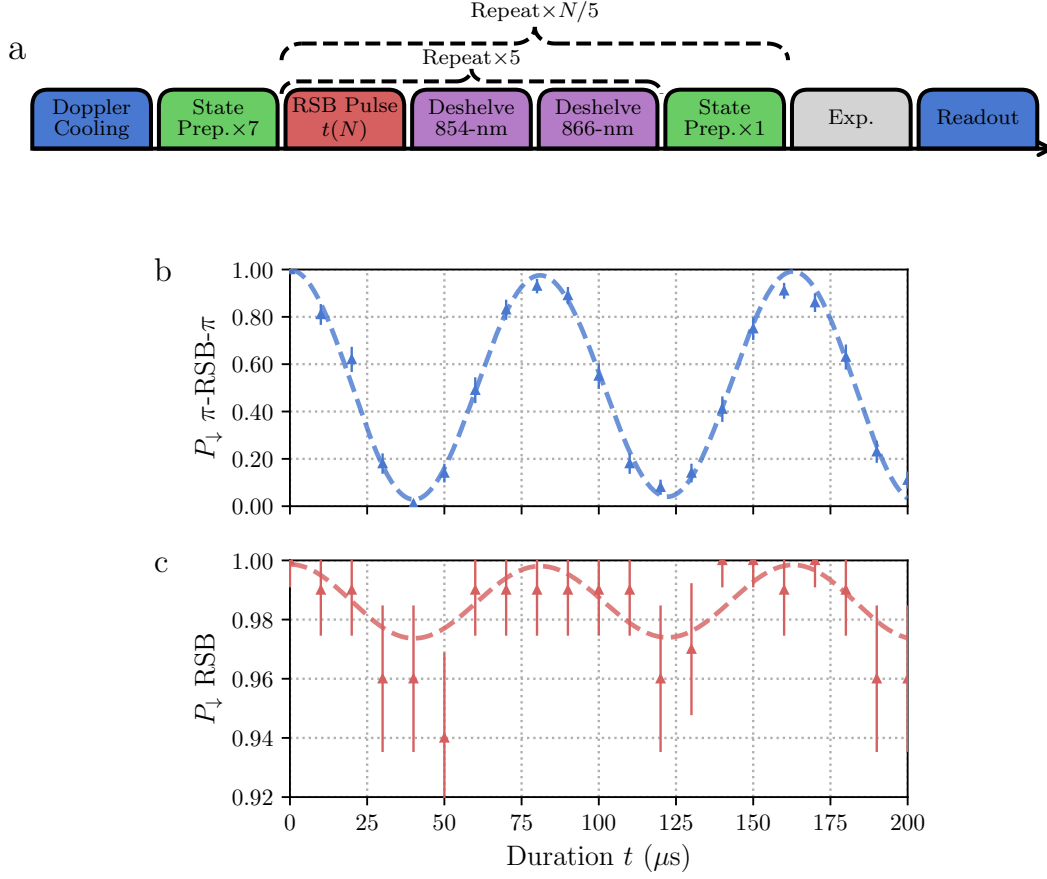


Figure 4.8: **a)** Example pulse sequence for one experimental shot including pulsed sideband cooling. The *Exp.* block represents an arbitrary sequence of operations, for the experiment under study, such as a thermometry sequence. The blue blocks represent Doppler cooling and readout (section 4.2.2), the green blocks represent qubit state preparation (section 4.2.1), the red blocks are RSB pulses with varying durations $t(N)$, and the purple blocks are deshelling pulses. After every 5 sideband cooling pulses, there is an interleaved state preparation sequence to ensure there is no population trapped outside of the closed cooling cycle. The RSB length is linearly increased with pulse number N , upto the expected RSB $R(\pi)$ -pulse duration for $n = 1$. **b, c)** Thermometry scans after Doppler and sideband cooling. The red points are the populations measured via on-resonance RSB pulses, and the blue points are the populations after a sequence of $R(\pi)$ -RSB- $R(\pi)$ with varying RSB probe durations. The dashed lines are fits to a thermal Fock state distribution, with truncation at $n = 100$. The extracted mean occupation number is $\bar{n} = 0.03(1)$, and $\eta\Omega/(2\pi) = 11.6(1)$ kHz.

729-nm pulses are applied to preferentially excite the thermal population with high phonon number, as the pulse duration for RSB $R(\pi)$ is proportional to $\sqrt{n}$, where n is the phonon number. The RSB pulse length is linearly increased to the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

An example pulse sequence for a single experimental shot using sideband cooling, can be seen in figure 4.8 *a*. Experimental parameters used are summarised in table 4.3.

To verify the efficacy of our sideband cooling, thermometry experiments are performed by probing Fock state populations. This is done using on resonance RSB pulses and effective BSB pulses composed of a π -RSB- π pulse sequences. The time dynamics of population flopping is measured as RSB pulse length, t , is varied. In the case of Fock state $|0\rangle$, there should be full contrast “BSB” oscillations, and no visible oscillations when probed with RSB pulses. A thermal Fock state distribution [meekhof1996generation] (with truncation at Fock state = 100) is fitted to these signals to extract the mean occupation number, and $\eta\Omega$, the carrier Rabi frequency multiplied by the Lamb-Dicke parameter. A typical thermometry scan after Doppler and sideband cooling can be seen in figure 4.8 *b, c*, where *b* shows the population after effective “BSB” pulses, and *c* shows the population after RSB(t) pulses with duration t . These oscillations are fit to thermal Fock state distributions, shown by the dashed lines. The mean occupation number after sideband cooling is found to be $\bar{n} = 0.03(1)$, and $\eta\Omega = 11.6(1)$ kHz.

4.3.2 Heating Rates

After cooling the motion, the ion will return to a temperature in equilibrium with the environment. However, this process can be slowed by adequate isolation of the ion. In UHV ion trap systems (see section 3.4), motional heating is predominantly caused by electric field noise from the ion trap [wineland1998experimental]. Noise due to thermal charge fluctuations at the trap surface can be mitigated by increasing ion-electrode distances. The *NPL* trap (section 3.2), used here, has an

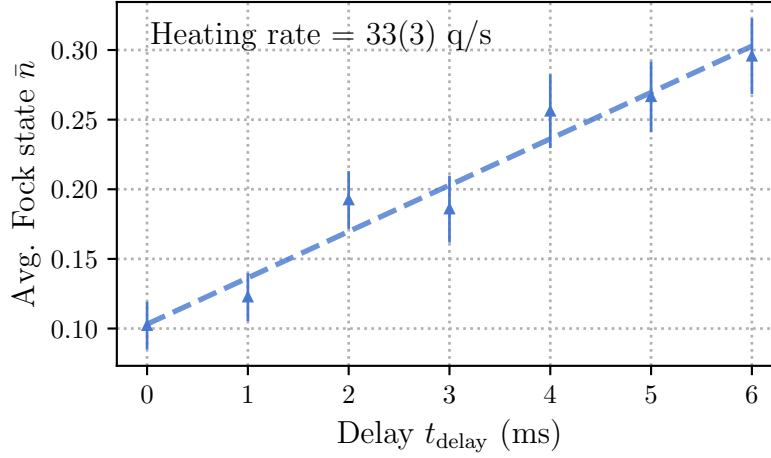


Figure 4.9: Heating rates of upper radial mode, measured by varying the delay between cooling and thermometry pulses. The dashed line is a linear fit to the data, with slope 33(3) quanta/s. The error bars are given by the standard deviation of the fitted $\bar{n}$.

ion-electrode distance of 250 μm , larger than most surface traps, but less than that of a macroscopic blade or rod style trap [milne2021construction]. Heating rates were probed by a series of thermometry experiments, explained above, with varying delay times, t_{delay} , between cooling and thermometry pulses. A typical plot can be seen in figure 4.9. The heating rate of the system was found by a linear fit, shown by the dashed line, to be 33(3) quanta per second on the upper radial 4 MHz mode on one ion. As will be discussed in section 4.3.4, the heating rate sets a limit for the motional coherence time.

4.3.3 Motional Mode Stability

Due to thermal drifts and microphonics introducing amplitude and frequency noise on the RF chain, drifts of the radial motional mode frequencies are seen over time. As will be discussed in the following section 4.3.4, this can lead to dephasing of the motional state. Here, the current motional mode stability is characterised to estimate its contribution to dephasing, and to set a benchmark for future stability improvements.

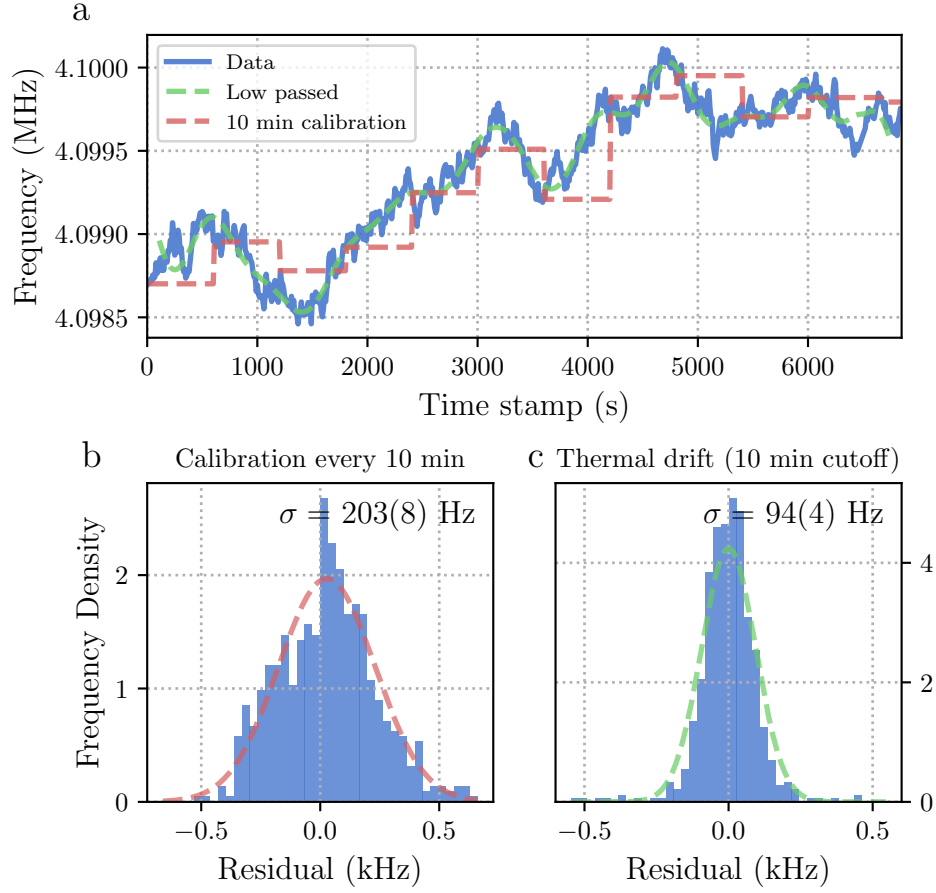


Figure 4.10: **a)** Motional mode frequency drift over time. The radial motional mode frequency is measured by exciting motion using an RF field in resonance with the motional frequency, and looking at increased fluorescence. The resulting frequency trace is shown in blue. The green line shows a low pass filter of the data, with cut-off frequency of 1.67 mHz, corresponding to an assumed 10 minutes slow thermal drift. The motional mode frequency is typically calibrated every 10 minutes, which is indicated by the red line. **b)** Residuals from the measured data and expected calibrated mode frequencies. The standard deviation of these residuals suggests the typical mis-set detuning of the mode frequency in current experiments with automated mode frequency calibrations every 10 minutes. **c)** Residuals from the measured data and the low passed trace. The standard deviation of these residuals are due to noise with period less than 10 minutes.

The motional mode frequency is measured via an RF “tickle” experiment. This consists of short ($\sim 5 \mu\text{s}$) RF field pulses applied to one of the trap DC electrodes which has a non-zero electric field projection to the mode we wish to probe. The frequency of the RF field is scanned around the expected motional mode frequency, and when resonant, drives the motion of the ion. This resonance is measured by observing an increase in fluorescence counts using a 397-nm laser red-detuned by 100 MHz. To measure mode frequency drift, this sequence is repeated every 10 seconds over multiple hours. A typical plot can be seen in figure 4.10 *a*. Over this two hour period, the frequency had an absolute drift of 1.5 kHz. Typically, we calibrate the mode frequency every 10 minutes, which is indicated with the red dashed line. Residuals between this calibrated line and the data, shown in *b*, have a standard deviation of $\sigma/(2\pi) = 203(8)$ Hz, giving the expected uncertainty of the mode frequency detuning given our current periodic calibration routine. A slow drift with frequency cut-off of 1.67 mHz is also fitted to the data, shown in green. The residuals from this fit, shown in *c*, have a standard deviation of $\sigma/(2\pi) = 94(4)$ Hz. This suggests that the motional mode frequency is relatively stable over time scales shorter than 10 minutes, but has a slow drift on longer time scales, which is likely due to thermal fluctuations.

4.3.4 Motional Coherence Times

We will utilise the motional modes of the ion to store quantum information. As with the section on spin coherence times 4.2.5, the fidelity of operations, and success probability of algorithms will be limited by the coherence time of the motional state. To measure motional coherence, a Ramsey sequence is performed between states $|\downarrow, 0\rangle$ and $|\downarrow, 1\rangle$, where the first element is the qubit state, and the second element is the motional Fock state. To prepare these states the following pulse sequence is applied:

1. State prepare system to $|\downarrow, 0\rangle$ via optical pumping and sideband cooling.

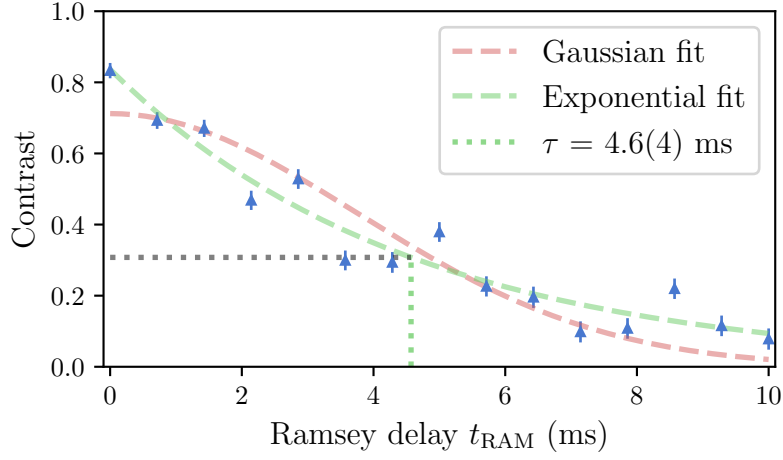


Figure 4.11: Motional coherence time measured by Ramsey sequence on the motional state. The blue points are the population in $|\downarrow\rangle$ after a Ramsey sequence with varying delay time, t_{RAM} . The dashed line is an exponential decay fit, with a coherence time $\tau = 4.6(4)$ ms.

2. Apply $R_y(\pi/2)$ -pulse on carrier transition to prepare the qubit in $(|\downarrow, 0\rangle + |\uparrow, 0\rangle)/\sqrt{2}$.
3. Apply $R(\pi)$ -pulse on the RSB to prepare the motional state in $(|\downarrow, 0\rangle + |\downarrow, 1\rangle)/\sqrt{2}$.
4. Delay for some time t_{RAM} .
5. Apply $R(\pi)$ -pulse on the RSB to map accumulated phase onto the qubit state.
6. Apply $R_\phi(\pi/2)$ -pulse on the carrier, with variable axis of rotation, ϕ .
7. Measure the population in $|\downarrow\rangle$ via state-selective fluorescence, section 4.2.2.
8. Repeat steps 1-7 for different values of ϕ to find the contrast of the Ramsey fringe.
9. Repeat steps 1-8 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

There are two main mechanisms for motional decoherence[turchette2000decoherence]: motional heating, as characterised in section 4.3.2, and motional dephasing due to mode frequency instability, as discussed in section 4.3.3.

A heating rate dominated coherence time is given by, $\tau_{\text{HEAT}} = (\sqrt{e}-1)/\dot{n}$ [turchette2000decoherence] where $\dot{n}$ is the heating rate. A dephasing dominated coherence time, if noise is correlated on timescales long compared to the Ramsey delay [omalley2015qubit], is characterised by a Gaussian decay profile with a coherence time of $\tau_{\text{DEPH}} = \sqrt{2}/\sigma_\omega$. Evaluating both of these models with $\dot{n} = 33(3)$ q/s, and $\sigma_\omega/(2\pi) = 94(4)$ Hz, the expected motional coherence times of $\tau_{\text{HEAT}} \approx 20$ ms and $\tau_{\text{DEPH}} \approx 2.3$ ms respectively, are found. From these auxiliary characterisations of the motional modes, it is expected that the coherence time will be limited by motional dephasing. Figure 4.11 shows the decaying coherence when probed by the above sequence. It can be seen that the contrast has a good fit assuming an exponential decay (green dashed line), defined by $C(t_{\text{RAM}}) = C(0)e^{-t_{\text{RAM}}/\tau}$, where $C(t_{\text{RAM}})$ is the contrast at time t_{RAM} , and τ is the coherence time. The fit yields a motional coherence time of $\tau = 4.6(4)$ ms, defined as the time at which the contrast decays to $1/e$ of its initial value. An exponential decay suggests that the noise source causing the dephasing is correlated only for durations much shorter than the Ramsey delay times [omalley2015qubit]. The Gaussian decay, (red dashed line), which was appropriate for the previous spin coherence time measurements in section 4.2.5, is a poor fit here.

4.4 Spin-Motion

For full control of the spin-motion hybrid system, we require interactions that couple the two. Perhaps the simplest of this class of interactions are the red- and blue-sidebands (RSB and BSB respectively). The RSB interaction was used previously in the thermometry and sideband cooling sections 4.3.1. The RSB (BSB) consists of a single frequency laser tuned at the carrier frequency minus (plus) the motional mode frequency, ω_m . The interaction is well described by the (Anti-)

Jaynes-Cummings Hamiltonian, and effectively couples spin flips with the addition or subtraction of a motional quanta depending on the initial spin state. Here, another such interaction coupling spin and motion, known as the spin-dependent force, SDF, is introduced. While the RSB and BSB interactions couple the spin and motion in the motional Fock basis, the SDF couples them through spin-dependent displacements, thus in a coherent state basis. The direction of the displacement is determined by the spin state and an effective detuning parameter.

4.4.1 Spin-Dependent Forces

The optical Mølmer-Sørensen (MS) scheme [sorensen2000entanglement] is used to generate the SDF via a bichromatic laser field. Bichromatic refers to the simultaneous application of two tones symmetrically detuned around the qubit carrier frequency, with absolute detuning approximately equal to the motional mode frequency, $\delta \approx \omega_m$. The resulting interaction, when ignoring off-resonant and higher order couplings, is given by,

$$\hat{H}_{MS} = i\hbar\eta\Omega\hat{\sigma}_\phi \cos(\delta t) \left(\hat{a}e^{-i\omega_m t} + \hat{a}^\dagger e^{i\omega_m t} \right), \quad (4.5)$$

where η is the Lamb-Dicke parameter, Ω is the carrier Rabi frequency, $\hat{a}(\hat{a}^\dagger)$ is the lowering (raising) operator, and $\hat{\sigma}_\phi$ is the Pauli operator with $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$. Applying the rotating wave approximation, and defining $\delta_g = \delta - \omega_m$, the interaction Hamiltonian can be approximated to,

$$\hat{H}_{MS} \approx \frac{i\hbar\eta\Omega}{2} \hat{\sigma}_\phi \left(\hat{a}e^{-i\delta_g t} + \hat{a}^\dagger e^{i\delta_g t} \right). \quad (4.6)$$

The trajectory of this displacement can be controlled by varying δ_g : on resonance, $\delta_g = 0$, corresponds to linear trajectories, whilst off resonance, $\delta_g \neq 0$, corresponds to cyclic trajectories where after some duration $t = 2\pi/\delta_g$, the motion returns to the initial state (with perhaps some phase shift [haljan2005spin-dependent]). The dynamics of the SDF may be measured by projective spin measurements, the population in $|\downarrow\rangle$, $P_\downarrow$, after applying the SDF for duration t is described by [burd2020squeezing],

$$P_{\downarrow, \text{th}} = \frac{1}{2} \left[1 + e^{-4(\bar{n} + \frac{1}{2})|\alpha(t, \delta_g)|^2} \right], \quad (4.7)$$

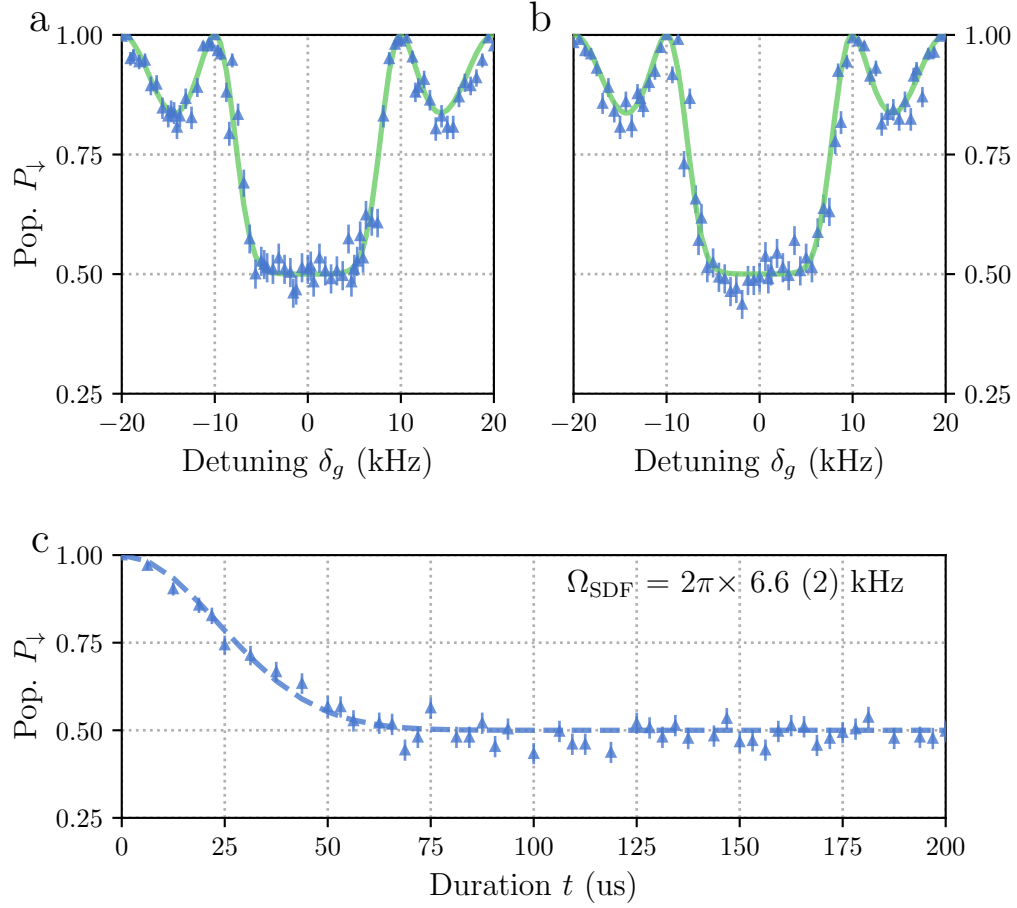


Figure 4.12: Detuning and duration scans of the spin-dependent force applied to an initial thermal state with $\bar{n} = 0.03(1)$. **a)** Detuning scan, where the detuning, δ_g , is varied, and the duration is fixed at $t = 100 \mu\text{s}$. The green line is a fit to the expected behaviour given by equation 4.7. **b)** Detuning scan with a mis-set qubit frequency of 0.5 kHz. The green line is the same fit as in *a* to guide the eye. **c)** Duration scan, where the duration, t , is varied, and the detuning is fixed at $\delta_g = 0$. The dashed line is a fit to the expected behaviour given by equation 4.7 with only Ω_{SDF} as a free parameter.

where $\bar{n}$ is the initial thermal phonon occupation, and $|\alpha(t, \delta_g)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, is the absolute displacement of the initial state. This control is utilised in both two-qubit entangling gate experiments, as well as in the creation of squeezed states.

Calibrating the SDF

Geometric phase gates, such as the MS interaction, are widely employed in ion trap experiments due to their robustness against variations in initial motional state. However, the SDF is sensitive to various frequency and power miscalibrations and drifts of these between calibrations. Here, we describe the workflow for calibrating and optimising the SDF behaviour.

In practice, the SDF interaction involves relatively few degrees of freedom: the central qubit frequency, the motional mode frequency, the power in each tone, and the durations of the pulses must all be calibrated. The complexity comes from the fact that the ion is a multi-level system, possesses multiple motional modes, and that each experimental parameter can only be controlled with finite precision.

The effect of nearby motional modes is negated by either selecting the interaction mode to be well separated from the others either in frequency, by beam geometry, by operating the SDF near to resonance of the desired mode, or by using amplitude shaping of the pulse. Smoothing a square pulse through amplitude shaping narrows the pulse frequency bandwidth, and thus reduces unwanted off-resonant excitation of other modes.

The power balance of the SDF tones is complicated due to our use of AOMs. AOMs generate frequency shifts in the laser beam at the expense of small frequency-dependent angular shifts. After the AOM, the beam is coupled into a single mode fibre (see figure 3.10), and so the AOMs effectively introduce a frequency-dependent loss. To calibrate the power balance, a pick off of the bichromatic beam after the ion is monitored on a high bandwidth photodiode. The beatnote contrast is measured for the two tones and the power balance is optimised by maximising this contrast. The far off-resonant levels of our ion lead to light-shifts of our qubit frequency. To account for the light-shift the central qubit frequency must be calibrated at the optical power used for the interaction. This is done with the SDF interaction itself. When the qubit frequency is set incorrectly, the two tones will no longer have the same absolute detuning from their respective RSB or BSB. This manifests as a “skewness” of the SDF detuning trace. By varying the qubit frequency and

inspecting these detuning scans, the desired SDF behaviour can be found.

To verify the behaviour of the calibrated SDF, the experimental data is compared with theory. Both “detuning” and “duration” scans are used. The “detuning” scan is performed by varying the detuning, δ_g , of the interaction, whilst keeping the SDF duration, t , constant, and vice-versa for the “duration” scan.

Figure 4.12 *c* shows the measured duration scan described by equation 4.7, with $\delta_g = 0$ and the motional state assumed to be thermal with average Fock state $\bar{n} = 0.03(1)$ which was found previously from thermometry measurements. Here, the displacement, $|\alpha(t)| = \Omega_{\text{SDF}}t/2$, where Ω_{SDF} is the SDF amplitude. The dashed line fit is given by floating Ω_{SDF} only, and finds $\Omega_{\text{SDF}}/(2\pi) = 6.6(2)$ kHz. These measurements were performed with total 729-nm power of 2 mW and $\delta_{\text{LS}}/(2\pi) = 3.5$ kHz, the light shift due to the 729-nm laser relative to the unperturbed qubit frequency.

The detuning scan, shown in figure 4.12 *a*, has the same experimental parameters, but with fixed duration $t = 100$ us. The fit in green is taking the displacement to be $|\alpha(t)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, given the value for Ω_{SDF} measured from the duration scan. Here, a qualitatively good agreement between the measured data and the expected behaviour is seen, with the main features being the “closure” at $\delta_g = 2\pi/t$ where the motional state returns to the initial state, and the central flat region around $P_{\downarrow, \text{th}} = 0.5$ where the two motional wave packets are non-overlapping.

Figure *b* shows an example detuning scan when the qubit frequency is mis-set by 0.5 kHz. The central region now displays a clear asymmetry around $\delta_g = 0$. The green line here is the fit from *a* to guide the eye.

4.4.2 Two-Qubit Entangling Gates

We perform two-qubit entangling gates using the Mølmer-Sørensen (MS) interaction [sorensen2000entanglement]. This interaction is the same described SDF from the previous section, but applied globally to two ions. The MS interaction relies on the spin dependent geometric phase accumulated during the motional displacement [ozeri2011tutorial]. To create a two-qubit entangled state, a

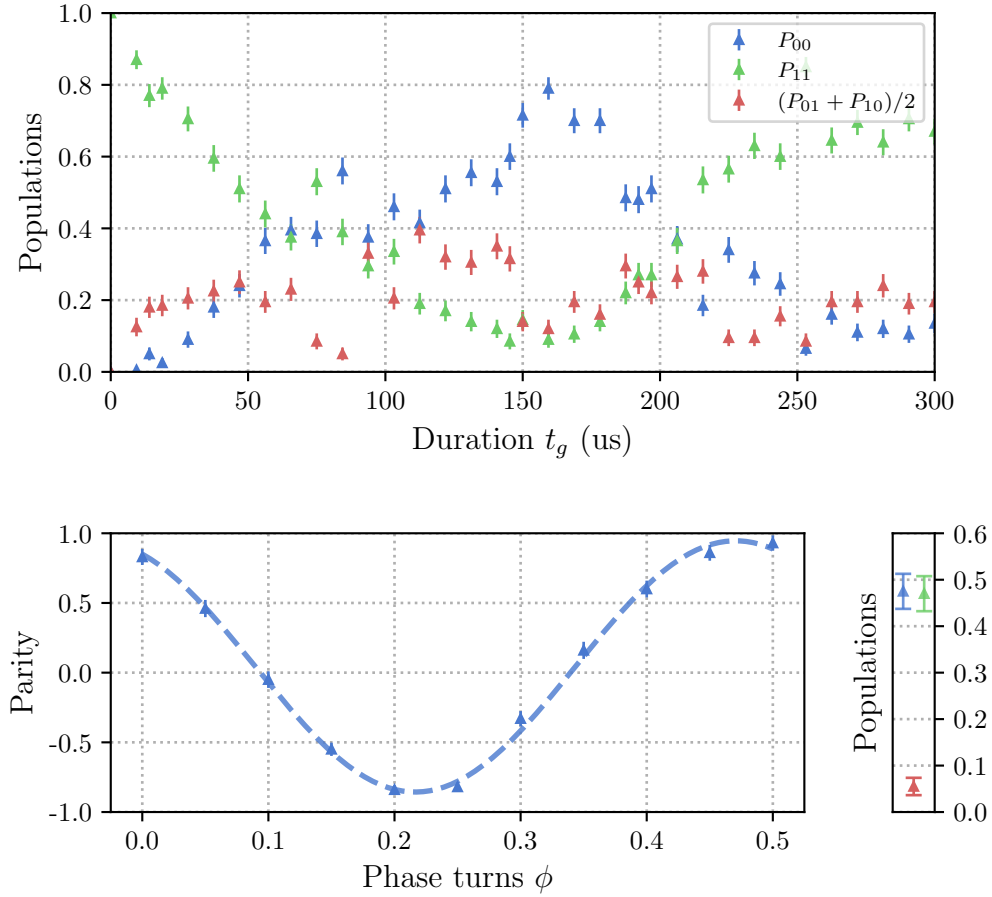


Figure 4.13: Experimental data for $\mathcal{F} = 92(2)\%$ Mølmer-Sørensen (MS) two-qubit entangling gate. **a)** Populations versus duration scan of the MS interaction. Each point corresponds to the average of 200 shots, and the error bars are given by the standard deviation of these averages. The blue points are the populations of both ions being in the $|0\rangle$ state, P_{00} , the green points are the populations of both ions being in the $|1\rangle$ state, P_{11} , and the red points are the ions being the the opposite states, $(P_{01} + P_{10})/2$. The desired entangling gate corresponds here to $t_g = 75\mu\text{s}$. **b)** Parity oscillations after the MS interaction. The parity is defined as $P_{00} + P_{11} - P_{01} - P_{10}$. The fitted contrast here is $0.90(2)$. **c)** Populations at the desired entangling gate duration, $t_g = 75$ us.

differential geometric phase of $\pi/2$ must be accumulated between the two-qubit basis states [ozeri2011tutorial]. To ensure there is no residual motional entanglement, the final motional state must return to the initial state. In practise, using an SDF detuned by δ_g , this is achieved by applying the MS interaction for a duration $t_g = 2n\pi/\delta_g$, where n is the number of “loops”. The MS gate is a universal two-qubit gate, and along with only single qubit gates, constitutes a universal gate set for discrete quantum algorithms.

Here we quote the current fidelity of experimentally demonstrated two-qubit gates on our system. The fidelity serves to quantify the similarity of two density matrices. Here for a two-qubit entangling gate, we target the creation of the Bell state $|\Phi_{\phi_0}^+\rangle = 1/\sqrt{2}(|00\rangle + e^{i\phi_0}|11\rangle)$, from an initial state of $|00\rangle$, where ϕ_0 is some fixed phase. The fidelity between our mixed state ρ , and the pure Bell state may be given by,

$$\mathcal{F} = \langle \Phi_{\phi_0}^+ | \rho | \Phi_{\phi_0}^+ \rangle = \frac{1}{2} (\rho_{00,00} + \rho_{11,11}) + \frac{1}{2} (e^{i\phi_0} \rho_{11,00} + e^{-i\phi_0} \rho_{00,11}), \quad (4.8)$$

where $\rho_{ij,kl} = \langle ij | \rho | kl \rangle$. To extract the fidelity experimentally, a partial state tomography protocol is used [sackett2000experimental], where the first bracketed term of equation 4.8 is measured by performing projective measurements of the two ions after the gate sequence to extract populations, and the second bracketed term, known as the coherence terms, are measured by applying a global analysis $R_\phi(\pi/2)$ pulse to the two ions and applying parity measurements. Parity measurements are defined as $P = P_{00} + P_{11} - P_{01} - P_{10}$, where P_{ij} is the population of the two ions in state $|i\rangle|j\rangle$ after the gate sequence and variable $\pi/2$ -pulse. Creating a Bell state is essential for two-qubit entanglement, but its basis can vary as long as it is consistent across experiments. This allows the Bell state phase, ϕ_0 , to float, which in practice means using only the amplitude of the fitted parity oscillations to compute fidelity, ignoring any phase offset.

The best two-qubit entangling gate fidelity currently achieved on our system is $\mathcal{F} = 92(2)\%$. As shown in figure 4.13, the magnitude of the parity scan was measured to be $0.90(2)$, while the populations $(\rho_{00,00} + \rho_{11,11}) = 0.95(2)$. Each

population point in these figures are found by taking the average of 200 shots of the gate sequence.

These results serve as both a proof of principle for full spin control on our system, but also as a benchmark to which we can compare future system improvements. For future work, we will require the use of this entangling gate either as Bell state preparation for input to analogue simulation experiments [**bazavan2023synthetic**], or as a primitive gate for the spin component of hybrid algorithms [**varona2024towards**, **brenner2024factoring**]. In both these cases, especially any use that requires multiple concatenated entangling gates, we will likely require improved gate fidelities. Current gate fidelities are limited by the motional dephasing. The gate error due to motional dephasing follows (see section 4.4.6 in [**ballance2017high-fidelity**])

$$\epsilon_{\text{DEPH}} = 0.686 \frac{t_g}{\tau_{\text{DEPH}}}, \quad (4.9)$$

where for these experiments we estimate a motional coherence time of approximately $\tau_{\text{DEPH}} = 1$ ms. Given the gate duration of $t_g = 75$ μs , this gives a dephasing error of $\epsilon_{\text{DEPH}} = 0.05$, or approximately 5%. This is the dominant error source in our current gate fidelity, and can be improved by decreasing the gate duration, or by improving the mode frequency stability. We have not yet attempted to optimise these gates, nor have we attempted to fully account for the observed errors. However, this work serves as a benchmark for future improvements, and a demonstration of the full spin control capabilities of our system.

5

k-copy Randomised Benchmarking

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Building on the general twirling formalism and noise models of Chapter ??, this chapter introduces *k-copy randomised benchmarking*: an RB protocol in which identical single-qubit Clifford sequences are applied in parallel across multiple qubits, and pairwise measurement correlations are retained rather than averaged away. We derive the resulting population dynamics for the common-noise model of Section 2.8.2, and present experimental demonstrations on a trapped-ion processor, including isolation of correlated versus local error, a parity-based confirmation of

Parameter	Value
729-nm laser power	1000 μW

Table 5.1: Summary table of other methods

the underlying symmetric-subspace occupation, a controlled noise-injection scan, and scaling to an eight-ion register.

5.1 Existing approaches to correlated-noise characterisation

Define spatial correlations (do in theory section?)

Existing approaches to correlated-noise characterisation span a trade-off between expressiveness and scalability. Tomographic and noise-learning techniques can reconstruct detailed structure in the error channel, but their cost grows rapidly with system size [54–57]. Random circuit methods, including randomised benchmarking, are substantially more scalable, but typically compress the action of noise into a small number of averaged parameters [4, 11, 58–60].

A related RB protocol, termed “Correlated RB” [58] (itself built on “Simultaneous RB” [60]), uses *independent* local twirling of multiple system copies to estimate the distance of the measured channel from a product (uncorrelated) channel. This differs fundamentally from the *identical*-sequence twirling strategy developed in Chapter ??: independent local twirling averages away exactly the symmetry information that identical twirling preserves.

A complementary source of inspiration comes from correlation spectroscopy and decoherence-free-subspace techniques, where common-mode noise is revealed through its action on collective and differential degrees of freedom [61]. In such settings, symmetry is not merely a simplifying assumption: it is the very mechanism that separates common fluctuations from local noise. This observation — that enforcing a symmetry can make correlated dynamics directly visible without reconstructing the full many-qubit channel — is the guiding idea carried into Chapter ??, where it is realised concretely as *k-copy randomised benchmarking*.

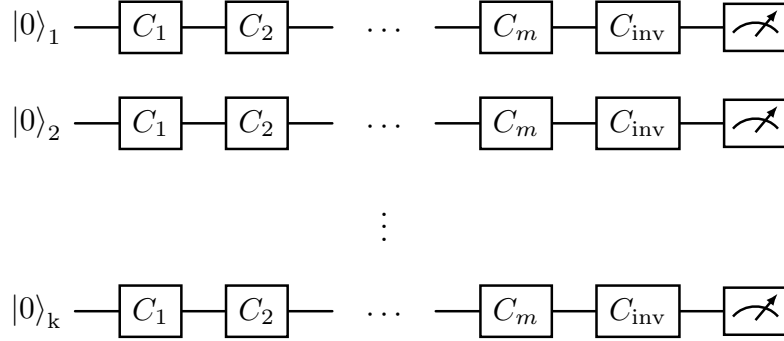


Figure 5.1: Circuit diagram for *k*-copy RB.

Add a comparative summary table at the end of this section (methods $\times$ {scalability, expressiveness, assumptions, experimental requirements}) contrasting tomography, Simultaneous RB, Correlated RB, cycle benchmarking, and *k*-copy RB. This would give the reader a single reference point before moving into Chapter ??.

5.2 The *k*-copy RB protocol

We introduce a symmetry-based viewpoint into randomised benchmarking through *k*-copy RB. In *k*-copy RB, the same single-qubit Clifford sequence is applied in parallel to multiple qubits. This diagonal Clifford twirl preserves symmetry sectors that are invisible to standard single-qubit RB (Section 2.6) but become accessible through pairwise measurement outcomes. For two qubits, this symmetry separates local single-qubit decay from collective two-copy dynamics.

Concretely, the protocol applies a standard single-qubit Clifford RB sequence [10, 11] simultaneously to multiple qubits, with the *same* randomly sampled Clifford applied to every qubit at each step. For each sequence length M , r random Clifford sequences are generated and each appended with its inverting Clifford, as in conventional RB, with each sequence repeated s times. As in standard RB,

the target computational state is randomised between $|0\rangle$ and $|1\rangle$ per sequence to reduce sensitivity to SPAM asymmetries (Section 2.6.3).

(a) Circuit diagram for k -qubit parallel Clifford RB. (b) Schematic single-qubit final states before measurement, sequence by sequence. (c) Bloch-sphere ensembles for correlated vs. uncorrelated qubit pairs. The k -copy protocol: identical parallel single-qubit Clifford RB sequences applied across k qubits (adapted from manuscript Fig. 1). **Redraw for thesis; consider splitting panel (c) out into its own figure with expanded discussion of the joint-angle-distribution picture (see note at end of Section 5.2.1).**

The distinguishing feature of k -copy RB relative to standard or Simultaneous RB (Section 5.1) is the combination of identical Clifford sequences with shot-resolved post-processing of *correlated* outcomes. For each qubit pair (i, j) , every shot is assigned to one of four outcomes, $P_{SS}^{(i,j)}$, $P_{SF}^{(i,j)}$, $P_{FS}^{(i,j)}$, or $P_{FF}^{(i,j)}$, where S and F denote success and failure relative to the sequence-dependent target outcome. Standard single-qubit RB averages this data into independent survival probabilities (Eq. (2.2)), collapsing the dynamics to a single exponential decay. By retaining pairwise correlations between successes and failures instead, k -copy RB distinguishes shared, sequence-dependent deviations from independent local decay. For a register of k qubits, a single shot record simultaneously contains outcomes for all $k(k-1)/2$ pairs, so extracting pairwise correlations requires only additional classical post-processing and no extra trials on the quantum register (this scaling property is used explicitly in Section 5.5.3).

Formally, applying identical Cliffords across k qubits implements the twirled representation of the single-qubit Clifford group, $Cl^{\otimes k}$. For two qubits, this is exactly the representation analysed generally in Section 2.7.2: the twirl decomposes the two-qubit Liouville space into the symmetry-informed subspaces of Table 2.1. The remainder of this chapter uses that decomposition, together with the common-noise model of Section 2.8.2, to derive concrete, fittable predictions for the correlated populations measured in a k -copy RB experiment.

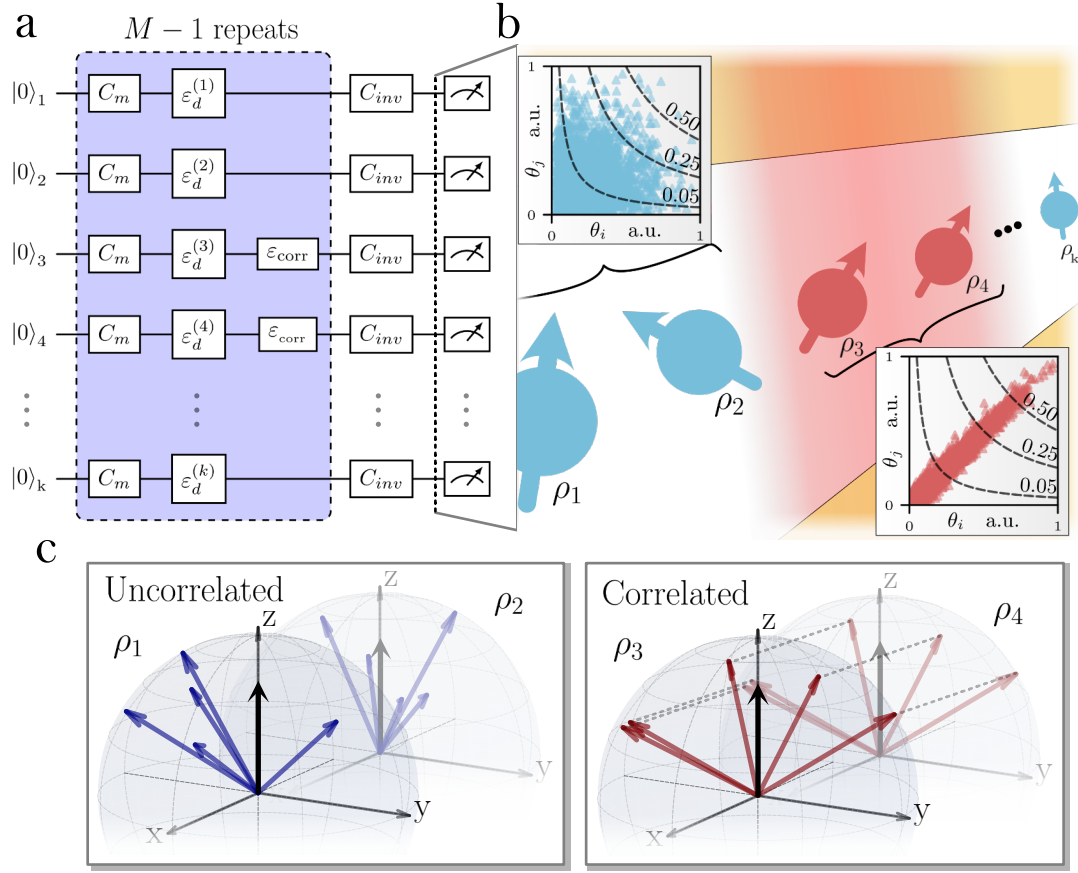


Figure 5.2: Bloch sphere schematic of corr vs uncorr noise. Include schematic of ion chain prev fig 1c.

Add a short worked numerical example here (e.g. a simulated two-qubit trace showing $P_{SS}, P_{SF}, P_{FS}, P_{FF}$ for a chosen $\{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$) to make the abstract shot-bookkeeping description concrete before moving to the analytic derivation.

5.2.1 Illustrating correlated vs. uncorrelated signatures

The distinction between correlated and uncorrelated noise is illustrated schematically by considering a register in which qubits $i = 3$ and $i = 4$ are dominated by a common rotation error while other qubits are dominated by independent noise. All qubits can exhibit identical single-qubit RB error rates (i.e. an identical depolarised Bloch vector under standard RB), while possessing entirely different pairwise correlation structure: independent noise produces uncorrelated sequence-to-sequence deviations,

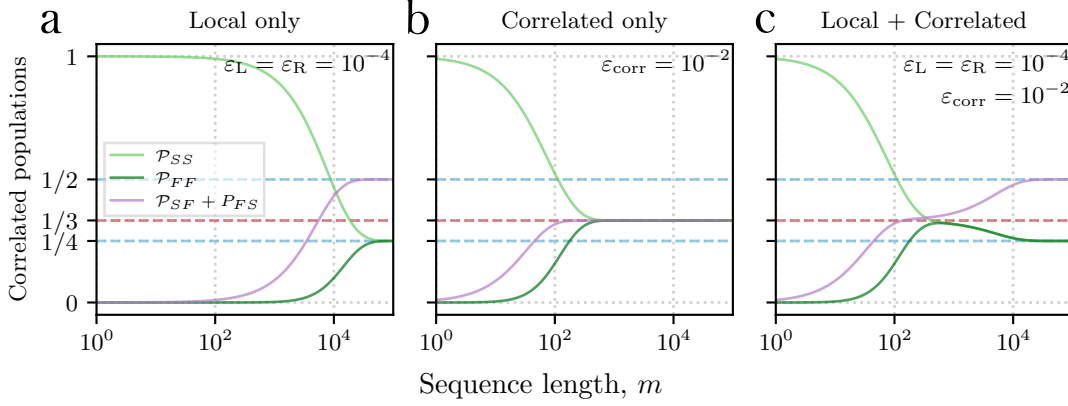


Figure 5.3: Numerical simulation of intermediate asymptotes measured in Z basis.

whereas common noise produces shared deviations between qubits acted on by the same random Clifford sequence. The identical-sequence condition is therefore essential to the protocol: it preserves exactly the symmetry information that would be averaged away under independently sampled local Clifford sequences (as in Simultaneous/Correlated RB, Section 5.1).

This is currently a qualitative, schematic argument (matching the level of detail in the PRL). A thesis chapter could usefully add the joint-angle-distribution analysis that was drafted but cut from the manuscript: uncorrelated noise exponentially suppresses higher-weight error terms $p_w = \prod_i p_i$ relative to single-qubit rates, whereas correlated noise violates this independence assumption and can scale linearly with weight w in the worst case, with consequences for QEC thresholds assuming independent errors ($p_t \propto \theta_i^2 \theta_j^2$). This connects naturally to the > 2 -qubit scaling results in Section 5.5.3.

5.3 Analytic predictions for common noise

5.3.1 Correlated population decay

Using the block-diagonal Liouville structure of Eq. (2.8) and the scalar decay rates it implies, the probability of both qubits successfully returning to their initial state

after m Cliffords, for a general error channel diagonal in the irrep basis of Table 2.1, is

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (5.1)$$

with corresponding expressions for $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$ following the same structure with sign changes on the α_L^m, α_R^m and $\alpha_{dt}^m, \alpha_{tr}^m$ terms respectively.

Combining the local depolarising channel (Eq. (2.10)) with the twirled common-noise channel (Section 2.8.2) gives, to first order, the sector decay rates

$$\alpha_L = 1 - \varepsilon_L - \varepsilon_{\text{corr}}, \quad \alpha_R = 1 - \varepsilon_R - \varepsilon_{\text{corr}}, \quad (5.2)$$

$$\alpha_{tr} = 1 - \varepsilon_L - \varepsilon_R, \quad \alpha_{dt} = 1 - \varepsilon_L - \varepsilon_R - 3\varepsilon_{\text{corr}}, \quad (5.3)$$

where $\varepsilon_{\text{corr}}$ is defined by Eq. (2.13). Because the symmetric trace sector is exactly invariant under common rotation noise ($\alpha_{tr} = 1$ when $\varepsilon_L = \varepsilon_R = 0$), Eq. (5.1) predicts an intermediate asymptote, $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/3$, distinct from the fully depolarised asymptote $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$ reached once any finite local error rate is present. This intermediate asymptote is the central experimental signature of correlated noise exploited throughout this chapter, and corresponds to the two-qubit state

$$\rho_{\text{sym}} = \frac{1}{3} \left(|00\rangle\langle 00| + |11\rangle\langle 11| + |\Psi^+\rangle\langle \Psi^+| \right), \quad (5.4)$$

where $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ is a Bell state. The presence of this Bell-state component reflects residual coherence induced between the two qubits by the shared rotation error, and motivates the independent, coherence-sensitive probe developed in Section 5.3.2.

Insert the full derivation connecting Eq. (5.1) to the vectorised-state inner product $\mathcal{P}_{L,R} = \langle\langle L, R | \Lambda_T^m | 00 \rangle\rangle$ and the explicit transfer matrices, rather than quoting the result (this is worked in the manuscript SM and can be reproduced here in full for thesis completeness, including the $\mathcal{P}_{0,1}, \mathcal{P}_{1,0}, \mathcal{P}_{1,1}$ companion expressions).

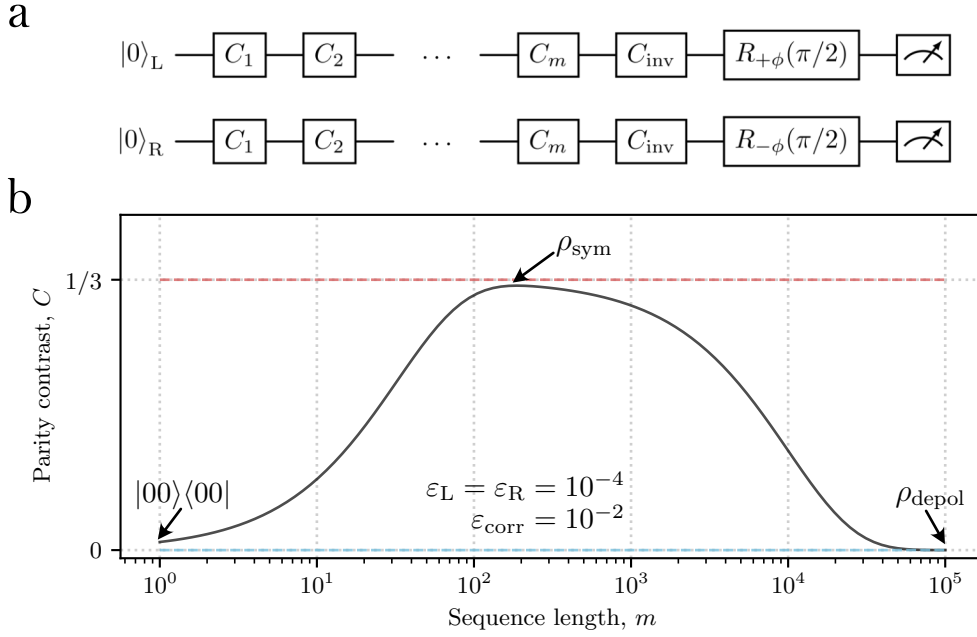


Figure 5.4: circuit diagram for parity measurement. Could there also be a nice schematic of the Bloch sphere picture of the differential rotation?

5.3.2 Parity contrast: an independent probe of coherence

Population decays alone indicate an intermediate asymptote; a parity-based measurement independently verifies that this asymptote corresponds specifically to occupation of the symmetric trace sector, by directly probing the residual coherence $|01\rangle\langle 10|$ built up between the two qubits. Differential rotations of the form $\hat{R}_{+\phi}(\pi/2) \otimes \hat{R}_{-\phi}(\pi/2)$, with rotation axis $\hat{\sigma}_\phi$ in the XY -plane, are applied prior to measurement, and the contrast of the resulting fringe (as ϕ is varied) is extracted. This contrast has the predicted dependence

$$\mathcal{C} = \frac{1}{3} (\alpha_{tr}^m - \alpha_{dt}^m), \quad (5.5)$$

with a maximum value $\mathcal{C}_{\text{max}} = 1/3$, reached in the same regime in which the population asymptote of Eq. (5.1) sits at $1/3$. Equations (5.1) and (5.5) may be combined to isolate the individual sector decay rates α_{tr} and α_{dt} as single exponentials, rather than fitting the multi-exponential population decay directly.

Reproduce the full parity-contrast derivation from the manuscript SM (App. “Parity analysis”): the transfer-matrix expression $\Pi(\phi) =$

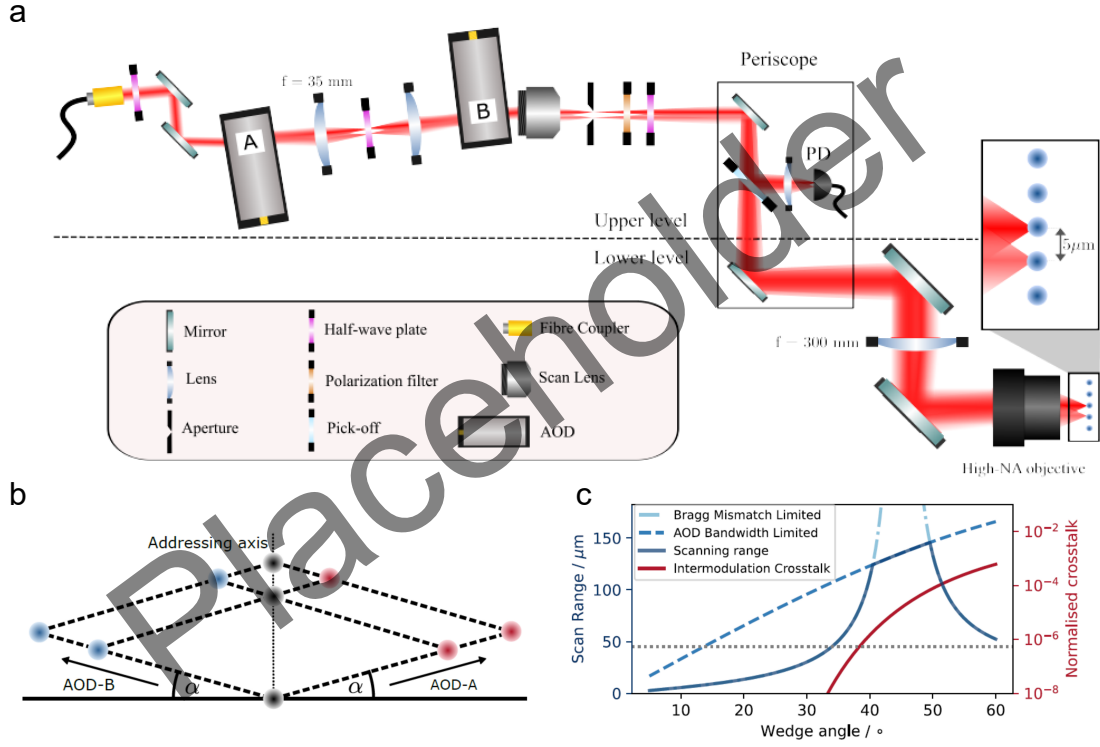


Figure 5.5: Block diagonal Liouville structure of the two-copy Clifford twirl, showing the sub-representations and their associated decay rates.

$\langle\langle\pi|\Lambda_{\Pi}(\phi)\Lambda_T^m|00\rangle\rangle = \mathcal{C} \cos(2\phi)$, the Bloch-sphere-averaging argument giving $\mathcal{C}_{\max} = 1/3$, and the isolation identities $\Pi_Z - C = \alpha_{dt}^m$, $\Pi_Z + 2C = \alpha_{tr}^m$ using the computational parity $\Pi_Z = \mathcal{P}_{0,0} + \mathcal{P}_{1,1} - \mathcal{P}_{0,1} - \mathcal{P}_{1,0}$. This is a good candidate for a fully worked thesis subsection given its role in confirming the central experimental claim.

5.3.3 Isolating sectors via initial-state choice

Beyond parity measurements, the sector decomposition of Table 2.1 suggests that preparing initial states with overlap on only a subset of sub-representations can isolate individual decay constants directly, without needing to fit multi-exponential curves. For example, preparing the singlet state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ gives a density matrix with overlap only on the identity and symmetric-trace sub-representations, so that the correlated populations decay as a single exponential in α_{tr} alone, insensitive to $\varepsilon_{\text{corr}}$.

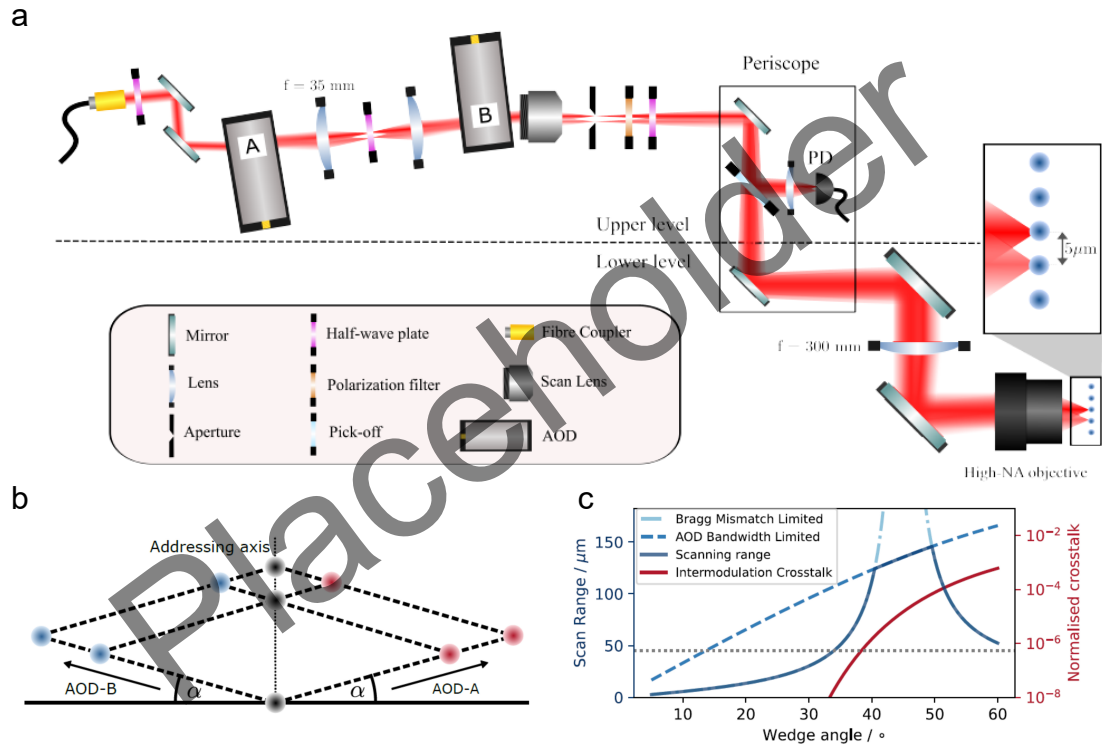


Figure 5.6: Asymptotic population decay for $k > 2$ qubits via numerical simulation.

This subsection is included as a brief pointer to a technique discussed only in outline in the manuscript SM (by analogy with *character RB* [62]). Expand into a full worked treatment if this state-engineering approach is pursued experimentally; otherwise keep as a short “future extensions” note and cross-reference from the Outlook (Section 5.6).

5.3.4 Scaling to $k > 2$ qubits

5.4 Analytic predictions for other noise types

Discuss ZZ-type errors, non-unitals, generalised depolarising channels, Stochastic vs coherent errors, etc.

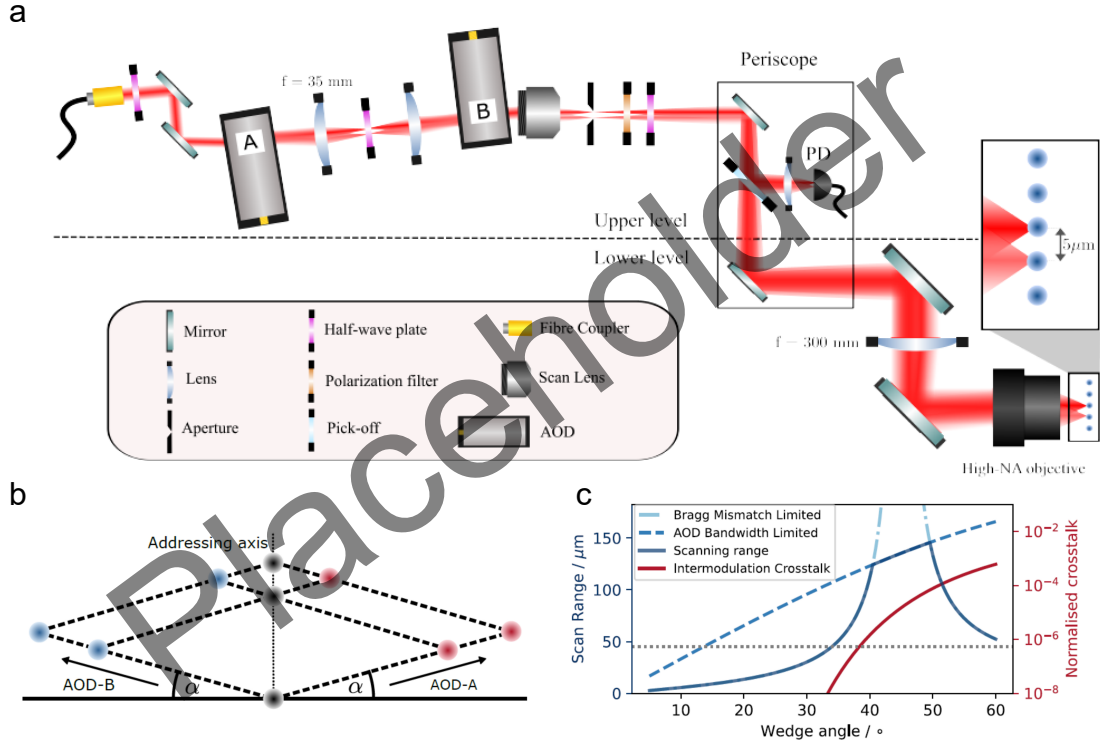


Figure 5.7: **Liouville-block structure for other noise types.**

5.5 Results

5.5.1 Discerning correlated and uncorrelated error

The 24 single-qubit Clifford gates are decomposed into native single-qubit rotations of the form $\hat{R}_x(\pi/2)$ and $\hat{R}_z(\pi/2)$: $\hat{\sigma}_x$ rotations are performed natively on resonance with the qubit transition, with relative phase set by modulating the 729 nm beam via an acousto-optic modulator (AOM), while $\hat{\sigma}_z$ rotations are performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations.

Z-basis measurements

k-copy RB is first demonstrated in a regime where common noise is the dominant error source, induced via a fixed 10 kHz frequency detuning of the 729 nm laser. Standard single-qubit RB decay curves (Section 2.6) are recorded alongside two-qubit correlated populations extracted via *k*-copy RB.

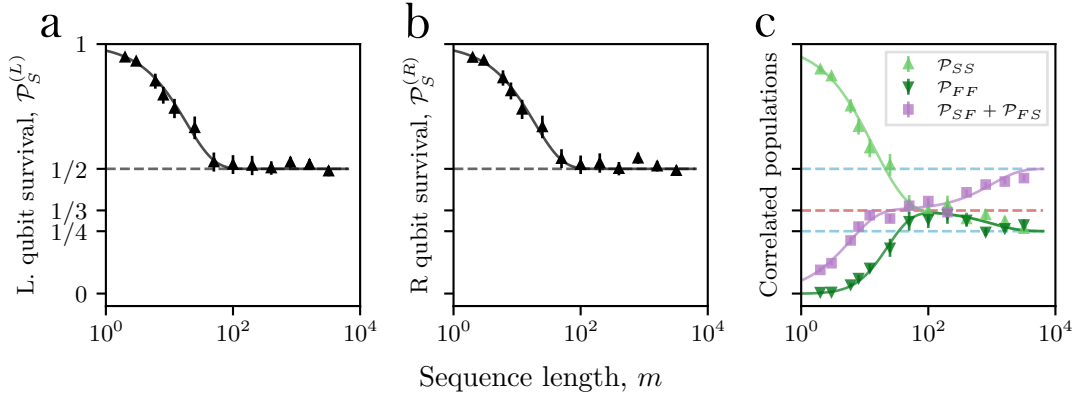


Figure 5.8: Exp data from fig 2 a,b,c

(a,b) single-qubit RB decays; (c) two-qubit correlated success population P_{SS} vs. sequence length; (d) parity contrast vs. sequence length, with example fringe insets. Correlated two-qubit probabilities discern correlated and uncorrelated error channels (adapted from manuscript Fig. 2). **Redraw for thesis; consider separating panels (a,b) [standard RB, Section 2.6 recap] from (c,d) [the k -copy-specific result] into two figures for pedagogical clarity.**

The correlated populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|00\rangle$ approaching ρ_{sym} (Eq. (5.4)), followed by a slower decay to the fully depolarised two-qubit state by $m \approx 1000$. These two decay rates, corresponding to correlated and uncorrelated error respectively, are indistinguishable in the single-qubit decays and are only revealed through the correlated measurements made during k -copy RB. Fitting Eq. (5.1) gives extracted parameters $\varepsilon_{\text{corr}} = 0.051(3)$, $\varepsilon_L = 0.001(1)$, and $\varepsilon_R = 0.000(1)$.

Coherence measurements

A partial-tomography parity measurement (Section 5.3.2) is used to independently confirm occupation of the symmetric sector, quantifying overlap of the final state with $|\Psi^+\rangle$. The expected contrast asymptote $\mathcal{C} \rightarrow 1/3$ (Eq. (5.5)) is observed for $m \in [10, 100]$, corresponding to residual two-qubit coherence. Fitting Eq. (5.5) gives $\varepsilon_{\text{corr}} = 0.043(3)$ and $\varepsilon_L = \varepsilon_R = 0.0016(1)$; the increased depolarising rate

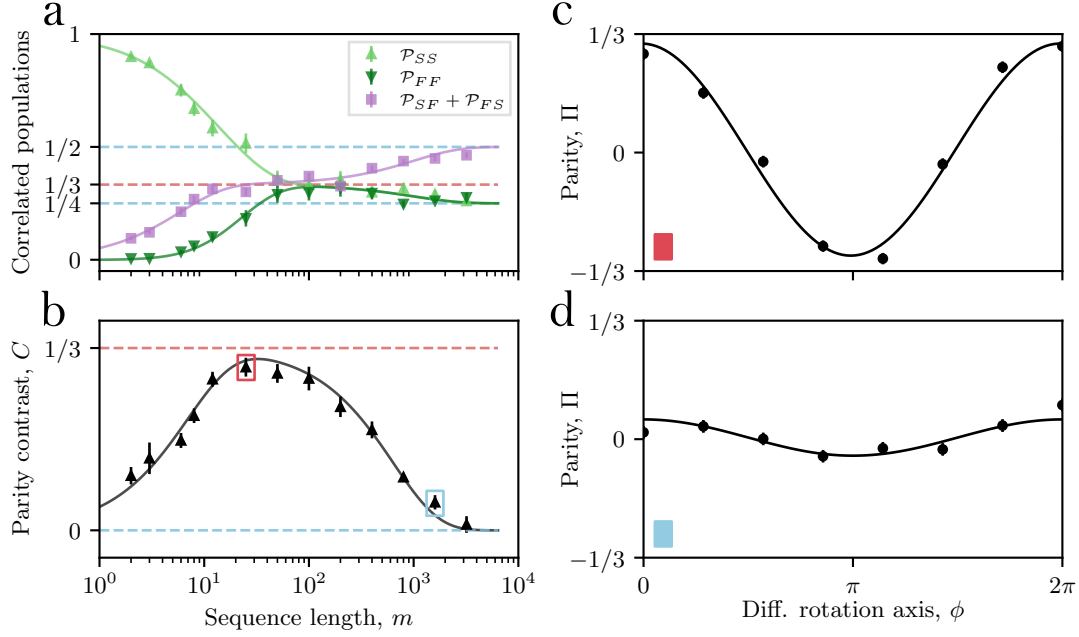


Figure 5.9: Exp data from fig 2 d + parity oscillations.

relative to the population fit is attributed to imperfect execution of the $R_{\pm\phi}(\pi/2)$ pulses used for the addressed differential rotation.

Add explicit discussion/reconciliation of the discrepancy between the $\varepsilon_L, \varepsilon_R$ values extracted from the population fit vs. the parity fit — the manuscript attributes this briefly to addressing-beam imperfections but a thesis chapter could quantify this more carefully (e.g. independently characterising the single-addressing system’s contribution).

5.5.2 Controlled noise injection

Numerical simulations of correlated vs. uncorrelated noise

laser phase jitter

laser detuning scan

k-copy RB can isolate errors arising from shared control fields, making it useful both for device characterisation and for optimising global control parameters. This is demonstrated by scanning the 729 nm laser frequency used for single-qubit rotations away from the bare qubit resonance.

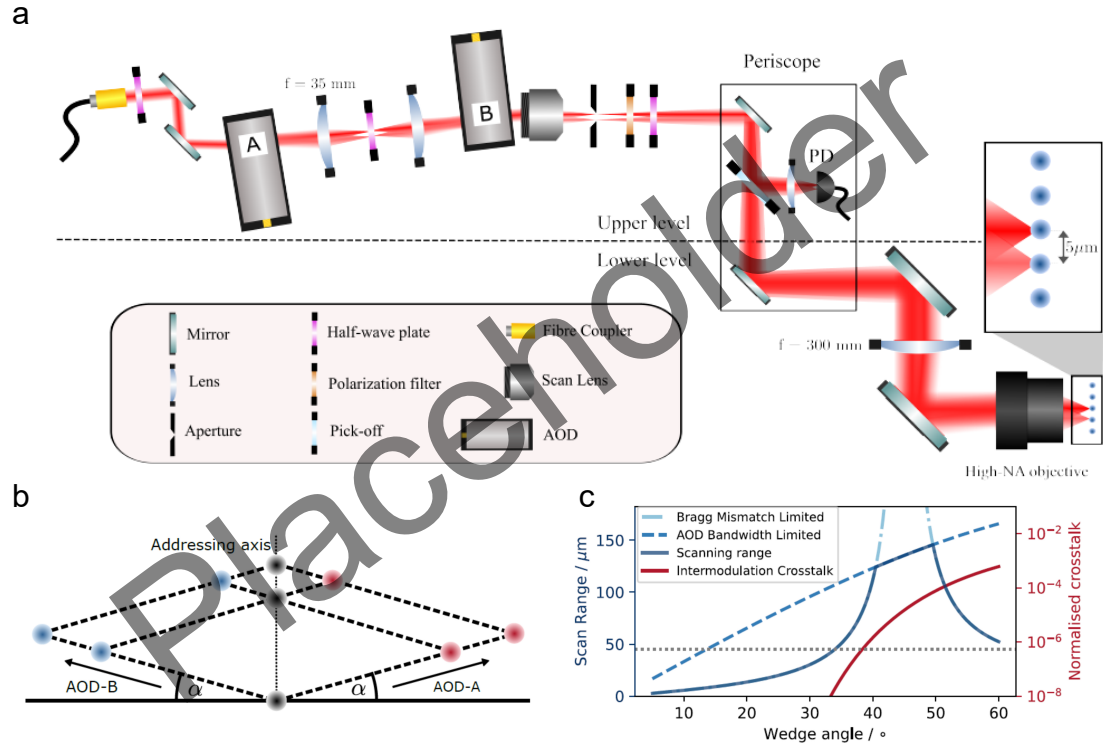


Figure 5.10: Numerical sim of param space and extracted fit uncertainties.

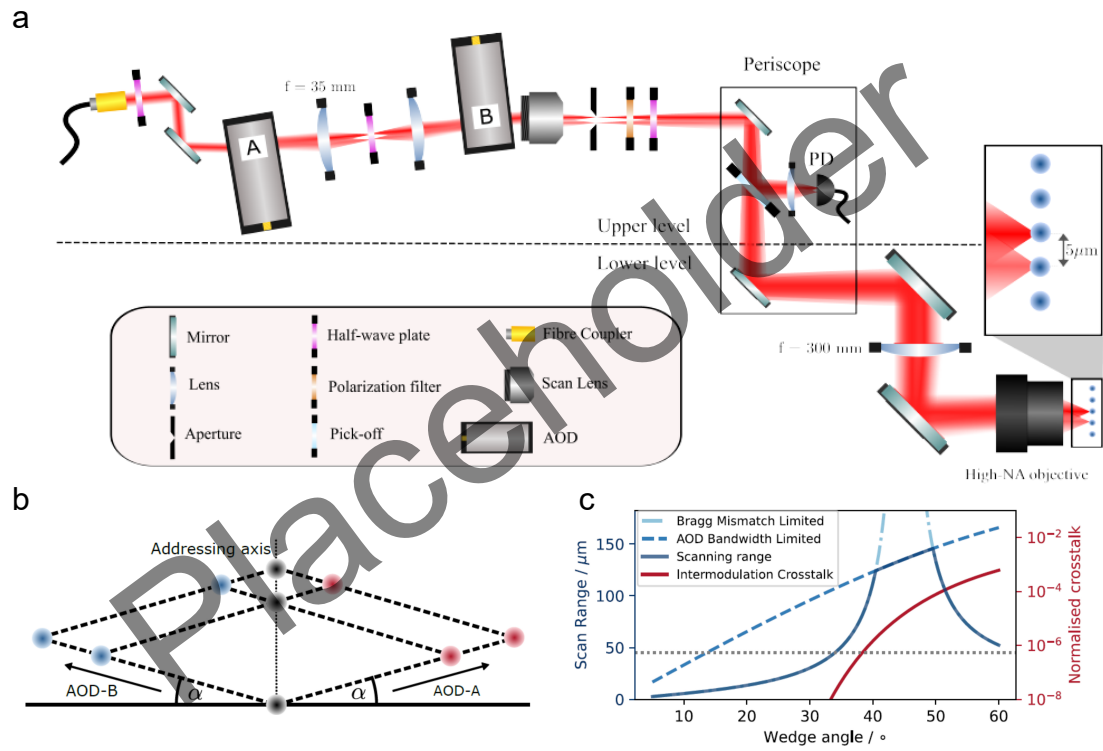


Figure 5.11: Injecting phase jitter calibration

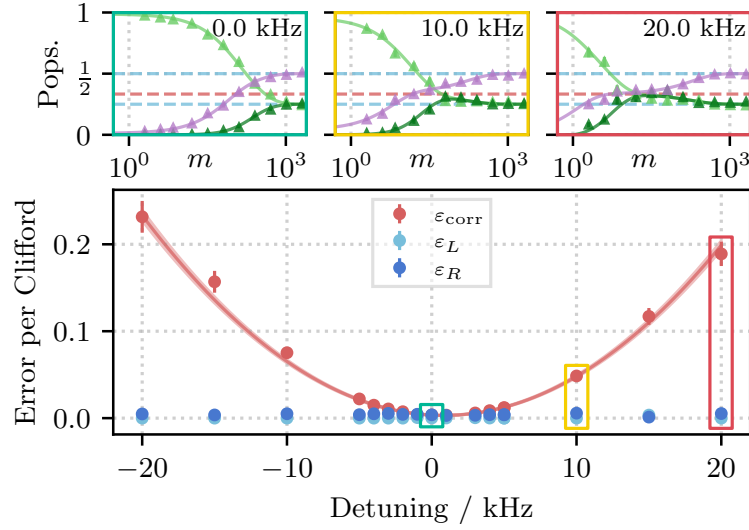


Figure 5.12: figure 3 from paper.

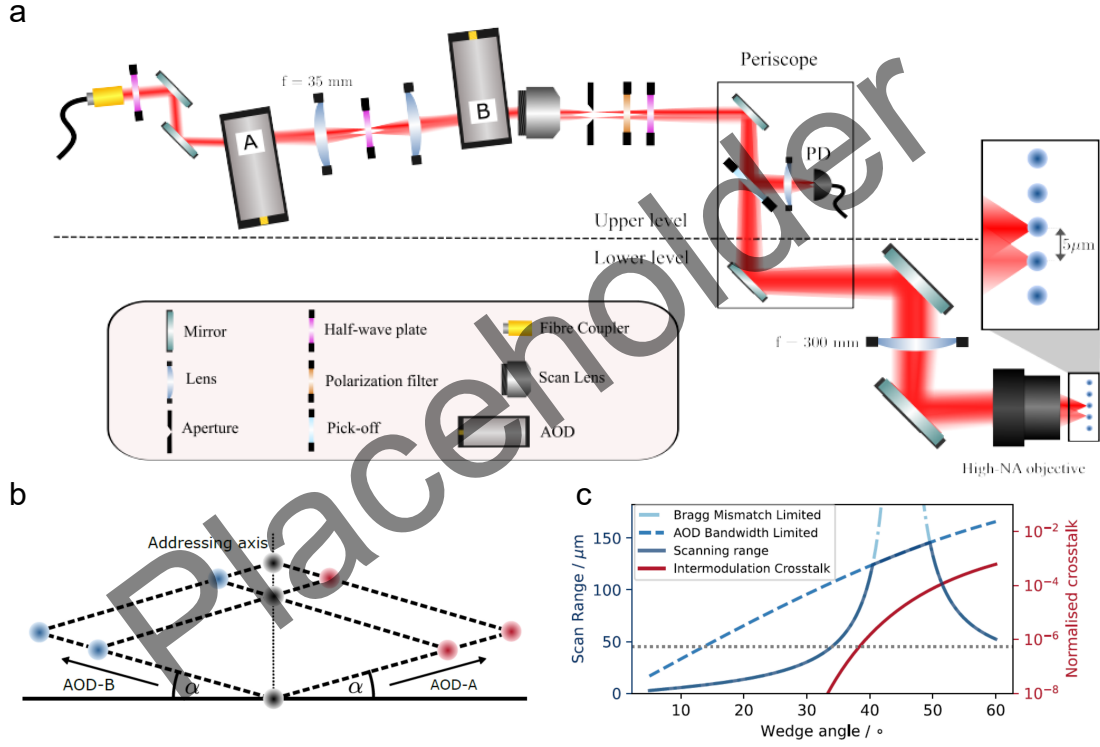
Upper: example correlated populations for several detuning values. Lower: extracted correlated ($\varepsilon_{\text{corr}}$, red) and local ($\varepsilon_L, \varepsilon_R$, blue) error per Clifford vs. detuning, with quadratic fit. Correlated errors injected via detuning of a shared drive laser (adapted from manuscript Fig. 3). **Redraw for thesis.**

Each data point is obtained from a complete k -copy RB experiment with 11 sequence lengths up to $m = 1024$, using $r = 50$ random sequences and $s = 50$ shots per sequence, with correlated populations fit to Eq. (5.1) to extract $\{\varepsilon_{\text{corr}}, \varepsilon_L, \varepsilon_R\}$ at each detuning. The correlated error is observed to increase quadratically with detuning, while the local depolarising errors remain approximately constant, consistent with detuning of a shared drive producing common noise as defined in Section 2.8.2. This dependence is modelled as

$$\varepsilon_{\text{corr}} = \varepsilon_0 + \frac{\kappa}{3} \left(\frac{\delta_L - \delta_0}{\Omega} \right)^2, \quad (5.6)$$

where ε_0 is the residual correlated error not originating from detuning, δ_0 accounts for the light-shifted resonance, $\Omega = 2\pi \times 100(1)$ kHz is the independently measured Rabi frequency, and κ depends on the Clifford decomposition to hardware-native gates. Fitting gives $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $\kappa = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)$ kHz.

Derive Eq. (5.6) explicitly from the detuned-carrier propagator and average-fidelity argument given in the manuscript SM (“Noise injection

Figure 5.13: **Calibrating stark shifts**

methods”), i.e. $\hat{U}_\delta(t) = \exp(-\frac{i}{2}t(\Omega\hat{\sigma}_x - \delta\hat{\sigma}_z))$, the Haar-averaged fidelity $\mathcal{F}_{\text{avg}} = (|\text{Tr}(U_{id}^\dagger U_\delta)|^2 + 2)/6$, and the small- $\eta = \delta/\Omega$ expansion $\mathcal{F}_{\text{avg}} = 1 - \eta^2/3 + O(\eta^4)$. This was only sketched in the SM and deserves a full worked derivation here.

Single addressed stark shifts

5.5.3 Scaling to a multi-ion register

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Pairwise *k*-copy RB extends simply to > 2 qubits: correlated populations are fit to Eq. (5.1) independently for each qubit pair. This provides a practical diagnostic of pairwise spatial correlations across a large register, though it does not uniquely determine higher-order (≥ 3 -body) correlated error terms (see Outlook, Section 5.6). For a register of *k* qubits, all $k(k-1)/2$ pairwise outcomes are extracted from the same shot record — no additional trials on the quantum register are required,

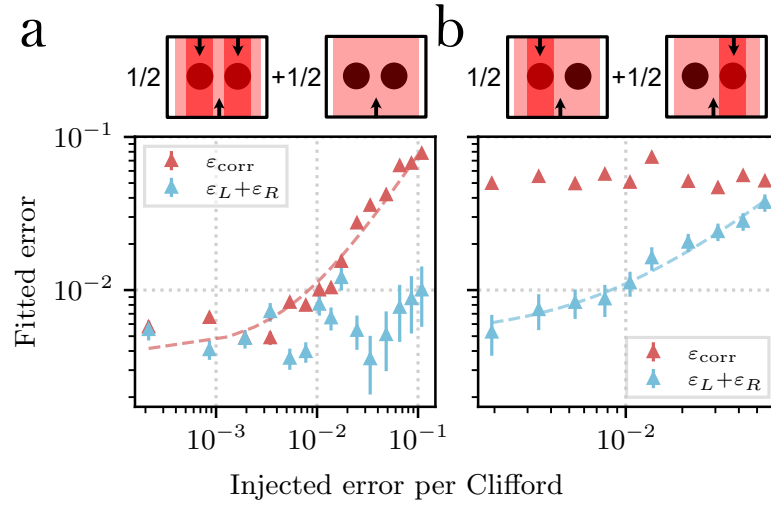


Figure 5.14: Stark shift supplemental figure.

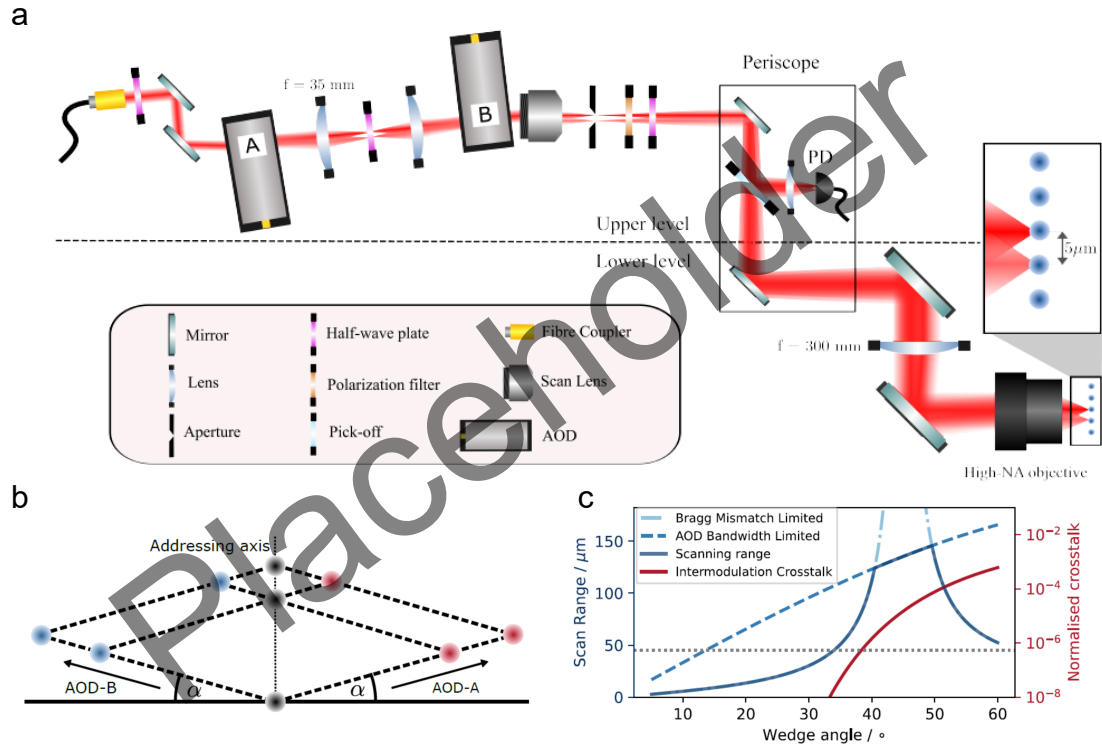


Figure 5.15: Calibrating global beam

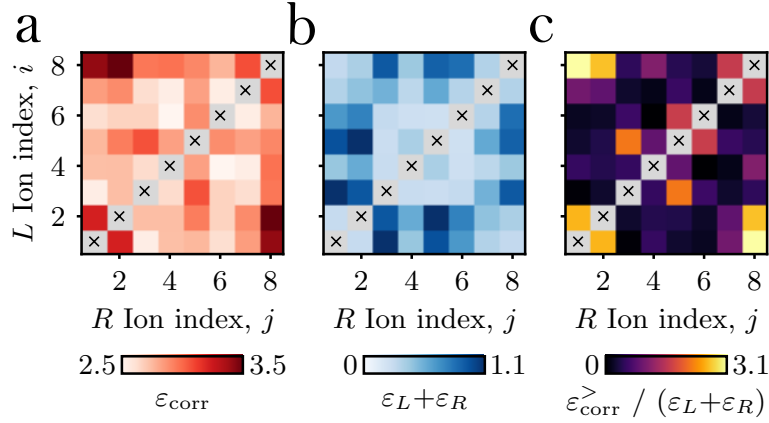


Figure 5.16: Figure 4 from paper.

though classical post-processing scales with the number of pair combinations. The protocol itself requires only synchronised application of identical single-qubit Clifford sequences and spatially resolved readout.

(a) Extracted pairwise correlated error $\varepsilon_{\text{corr}}^{(i,j)}$ for all ion pairs in an 8-ion chain. (b) Sum of local depolarising errors $(\varepsilon_L + \varepsilon_R)^{(i,j)}$. (c) Ratio of excess correlated noise to local errors. Correlated and local error per Clifford measured for all qubit pairs in an 8-ion chain (adapted from manuscript Fig. 4). **Redraw for thesis.**

This scaling is demonstrated on an eight-ion chain, with parallel single-qubit rotations driven by an elliptical 729 nm beam of major-axis waist radius $52(3) \mu\text{m}$ aligned to the ion chain. Addressing multiple qubits in parallel with a single shared drive is common to both ion-trap and neutral-atom platforms owing to its experimental simplicity, but unequal illumination across the register leads to position-dependent over- and under-rotation errors. With axial confinement of 0.5 MHz, the chain extent is $35 \mu\text{m}$, giving an expected Rabi-frequency difference of 11(1)% between the outer and inner ions (single-qubit gates were optimised for the inner ions, $i = 4$). The extracted pairwise correlations show the resulting spatially dependent error structure: both neighbouring and distant outer-ion pairs exhibit large correlated errors, and in particular the maximally separated ions

$i = 1$ and $j = 8$ experience similar under-rotations, producing a strong correlated error despite their spatial separation.

To improve sensitivity to both correlated and local contributions simultaneously, a common error offset was intentionally added by randomising the phase of each implemented $R_x(\pi/2)$ pulse (drawn from a normal distribution with standard deviation $s_\phi = 0.19$ rad, producing $\varepsilon_{\text{corr}}^{(0)} = 0.025$). This separates the relevant exponential decay rates and improves the stability of the correlated-population fits.

Add the noise-injection methodology in more detail here, including the complementary Stark-shift-beam injection experiments from the manuscript SM (App. “Noise injection methods”, Fig. S2): two addressing configurations (correlated two-spot vs. independent single-spot Stark shifts) used to independently calibrate and validate the $\varepsilon_{\text{corr}}$ vs. $\varepsilon_L, \varepsilon_R$ separation against a known injected error rate.

5.6 Conclusions and outlook

k -copy RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. Because of the symmetry enforced by twirling over the diagonal Clifford representation $Cl^{\otimes k}$ (Section 2.7.2), structured errors drive the decay of correlated populations into invariant subspaces in a way that is directly observable without full process reconstruction. Given a specified error model, these correlated decays can be used to directly quantify the magnitude of the individual error channels contributing to the total error per Clifford.

This method has been proposed and experimentally demonstrated, and its application extended to an eight-qubit register through pairwise comparisons (Section 5.5.3). This approach permits highly efficient parallel data collection: the time required on the device is independent of qubit number, and the protocol requires exclusively local operations followed by classical post-processing of correlated survival rates.

Expand this Outlook section substantially for the thesis, drawing together the forward-looking threads seeded earlier in this chapter and in Chapter ??:

- Extension to genuine ≥ 3 -body correlated error terms, beyond the pairwise diagnostic of Section 5.5.3.
- Experimental characterisation of ZZ -type entangling errors (Section 2.8), not demonstrated here.
- Sample-cost / Fisher-information analysis for multi-exponential fitting — flagged as an open numerics question in the manuscript SM and not yet resolved.
- Non-Markovian (temporally correlated) noise, via repeated identical Clifford RB sequences applied at different times on a single qubit, exploiting the same correlated-outcome analysis used here for spatial correlations.
- Character-RB-style initial-state engineering (Section 5.3) to isolate single exponential decays and reduce sample cost.

6

Motional Interactions

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Perhaps this should be chapter 4, as this is more like characterisation?

Presentation of the experimental results demonstrating motional control

Move from characterisation: sideband cooling ? Thermometry ? Heating rates ? Motional mode stability ? Motional mode coherence ? SDF Two-qubit entangling gates

And add: Description of CVQC and hybrid QC. $\pi/4$ gates Two-SDF toolkit Tomography method Squeezing Fock state preparation Planned Fredkin gate (would be amazing to show at least parity operation!)

- Describe Two-SDF toolkit

- Describe tomography method for motional states. Could show Fock state here.
- Results:
 - Squeezed state
 - Beam splitter
 - Parity operation
 - Fredkin gate

6.1 Computing on a Hybrid System

6.2 Gate Implementation

6.3 System Characterisation

6.3.1 Mode temp

6.3.2 Heating

6.3.3 Phase Noise

6.3.4 SDF

6.3.5 MS Gate

6.3.6 Squeezing

7

Virtual Distillation on an Ion Trap QC

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This chapter presents the theory of virtual distillation, a purification based error mitigation technique. We begin with the background and uses of error mitigation, and detail the virtual distillation protocol. Following this, we analyse the performance of virtual distillation under different physically motivated noise models, with experimental results presented for two-copies in the presence of both spatially correlated and uncorrelated noise. We close this chapter with a discussion on extending this method to more than two-copies, and to hybrid systems.

7.1 Error Mitigation

- Error mitigation motivation. Mainly on contrasting to quantum error correction.
- QEM is framed as finding expectation value of observable, rather than maintaining coherence for a long computation.
- How QEM reduces bias (at the expense of required samples). Look at Cai paper for structured argument.

7.2 Virtual Distillation Theory

- Introduce virtual distillation, a purification technique. Discuss scaling with copy number, and the error models it deals with.

7.2.1 Protocol

Either Hadamard test or B-matrix.

7.2.2 Mechanism

- It can be used for both spin and bosonic objects

7.2.3 Noise Floor

- Discuss noisefloor (coherent mismatch)
- Go into detail on how spatially correlated noise can be an issue here, discuss network paper as opposite case

7.3 Experimental Implementation

7.3.1 Calibrations

π -4 gates, single addressing

7.3.2 Injected Spatial Correlations

- two copy, B matrix demonstration

7.4 Extending to >2 copies

- plan for c-swap implementation.

7.5 Error Mitigation on Hybrid Systems

- What errors does this mitigate?
- How to implement it?
- Results?

7.6 Conclusions

8

Conclusion and Outlook

Appendices

A

Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Detuning errors

To exemplify the method for calculating the effect of an error channel on a single qubit gate, we consider the simple case of a coherent detuning error. We define the Fidelity of a quantum channel $\mathcal{E}$ with respect to a unitary U as

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \quad (\text{A.1})$$

where the integral is taken over the uniform Haar measure on the space of pure states. We shall see that the language used for this analysis of single qubit gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this

can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(W)|^2 + d \right), \quad (\text{A.2})$$

where V is the error unitary, $W = U^\dagger V$ and d is the dimension of the Hilbert space. Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.1 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2. \end{aligned} \quad (\text{A.3})$$

Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.4})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.5})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.6})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)} (\mathbb{I} + S). \quad (\text{A.7})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.8})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter 7. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.2.

Considering the explicit case of a $R_x(\pi/2)$ gate with a coherent detuning error, we can write the error unitary as

$$V = \exp \left(-i \frac{\pi}{4} \left(\hat{\sigma}_x + \frac{\delta}{\Omega} \hat{\sigma}_z \right) \right), \quad (\text{A.9})$$

where δ is the detuning from resonance, and Ω is the Rabi frequency. The ideal unitary is given by $U = \exp(-i \frac{\pi}{4} \hat{\sigma}_x)$, yielding trace of the product $W = U^\dagger V$ as

$$\text{Tr}(W) = 2 - \frac{1}{2} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4, \quad (\text{A.10})$$

where we have assumed the detuning is small compared to the Rabi frequency. Inserting this into equation A.2, we find the fidelity of the detuned gate is given by

$$F(V, U) = 1 - \frac{1}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (\text{A.11})$$

To convert this to an error contribution to a single qubit Clifford gate, we introduce the scalar κ , which depends on the specific decomposition of the Clifford gate set into the native gate set of the system. The error contribution is then given by

$$\epsilon = \kappa(1 - F(V, U)) = \frac{\kappa}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (\text{A.12})$$

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